



化学测量学—人工智能化学分析

Chapter 9

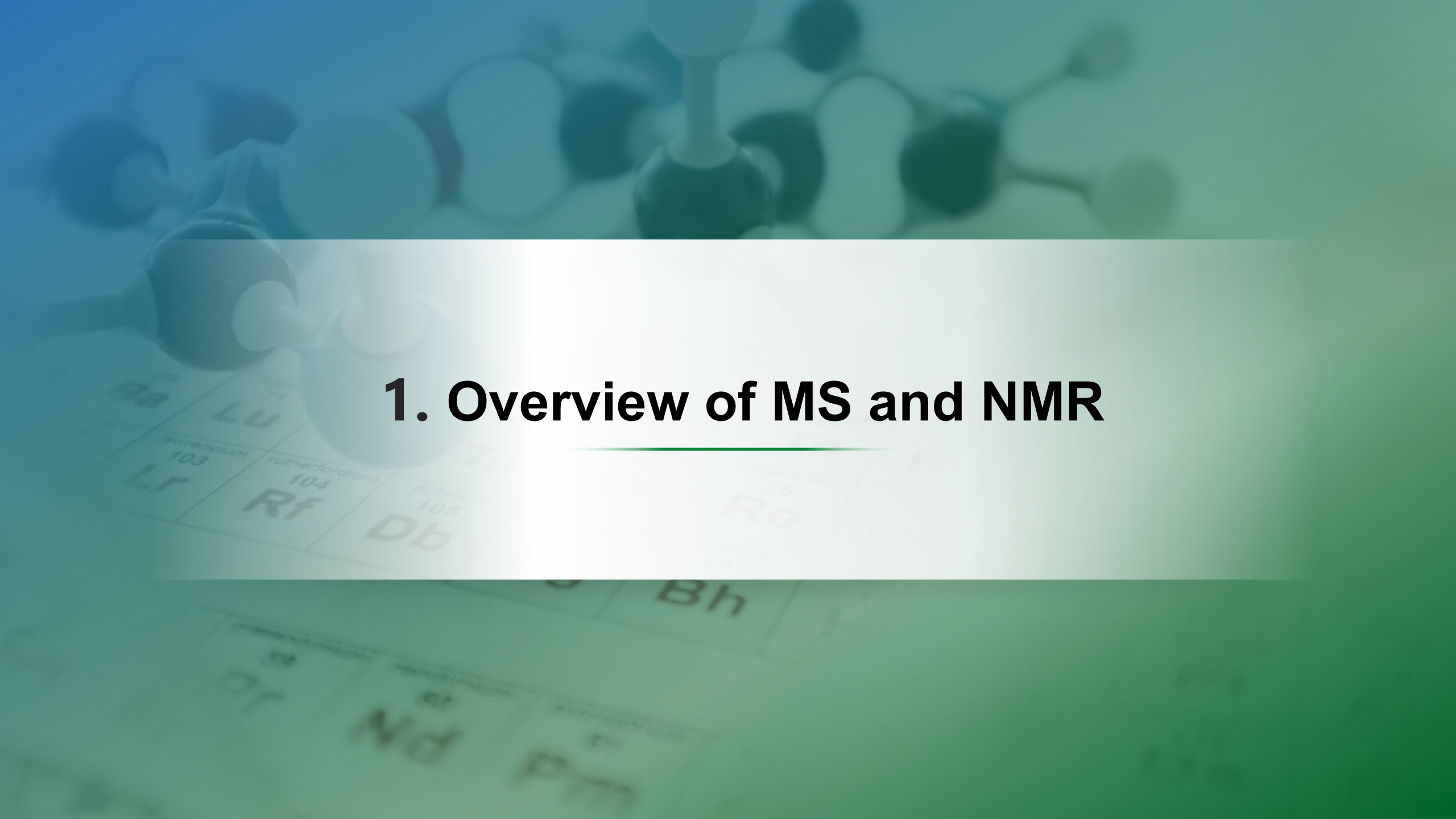
MS and NMR

人工智能化学分析

马昊 厦门大学

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- 2. Library matching**
- 3. Spectral prediction**
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- 5. Molecular generation**



1. Overview of MS and NMR

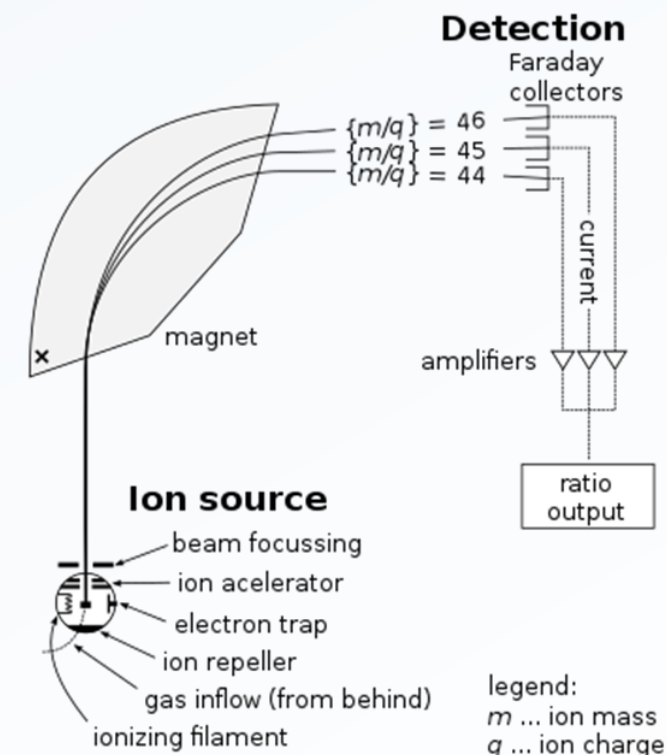
MS and NMR

■ MS

<https://masspec.scripps.edu/learn/ms/index.html>

Mass spectrometry (MS) is an analytical technique that is used to measure the mass-to-charge ratio of ions. The results are presented as a mass spectrum, a plot of intensity as a function of the mass-to-charge ratio. Mass spectrometry is used in many different fields and is applied to pure samples as well as complex mixtures.

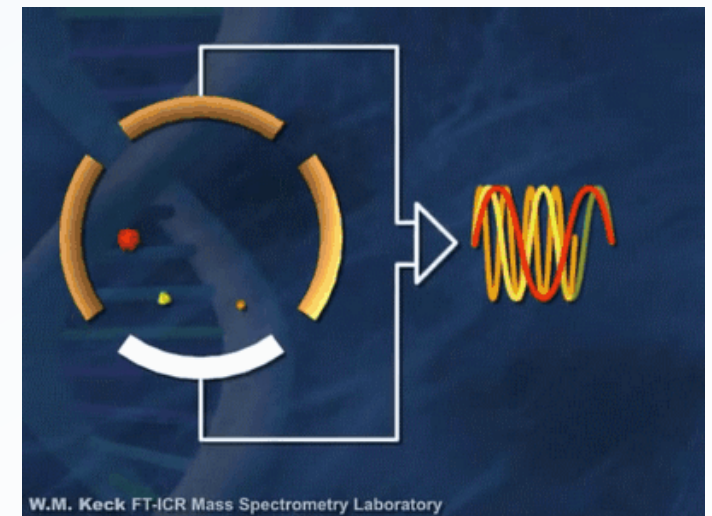
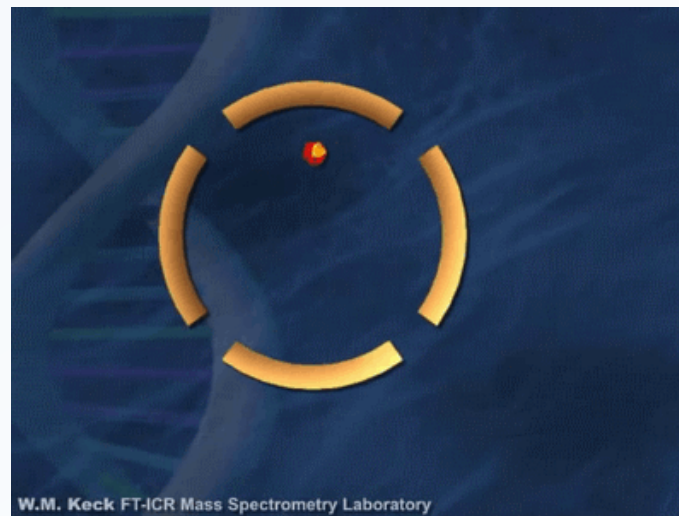
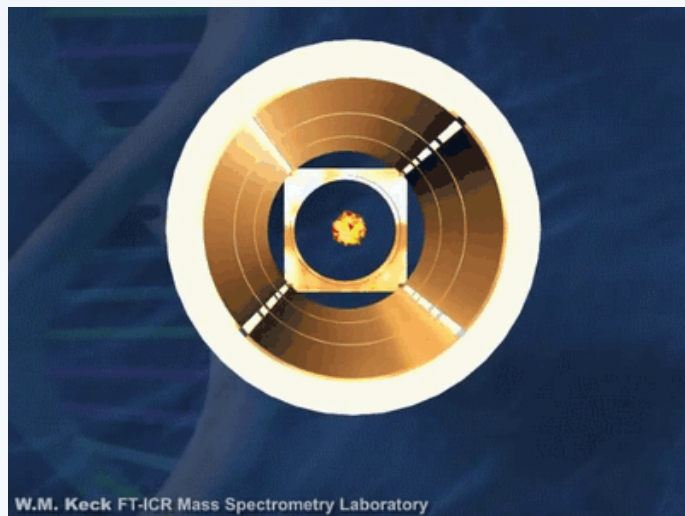
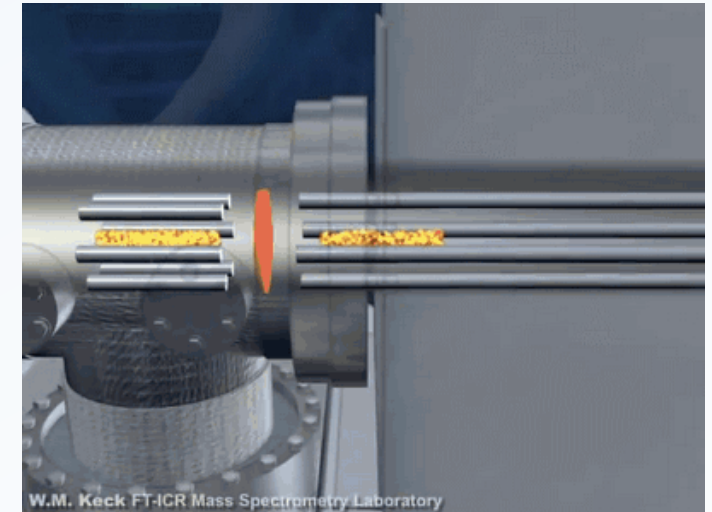
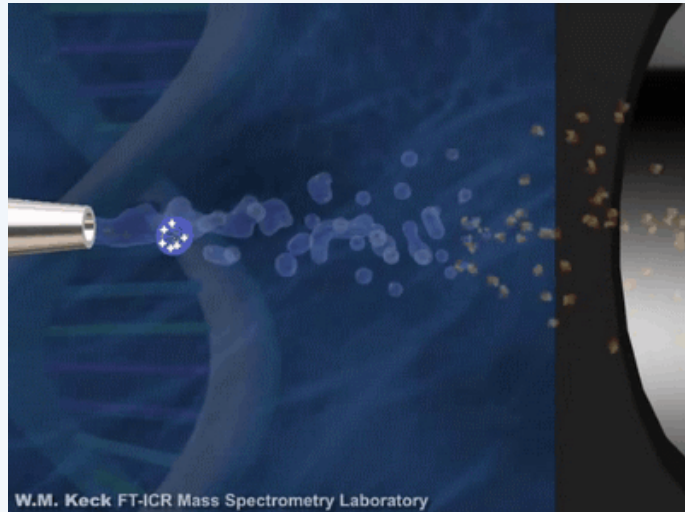
A mass spectrum is a type of plot of the ion signal as a function of the **mass-to-charge ratio**. These spectra are used to determine the elemental or isotopic signature of a sample, the masses of particles and of molecules, and to elucidate the chemical identity or structure of molecules and other chemical compounds.



MS and NMR

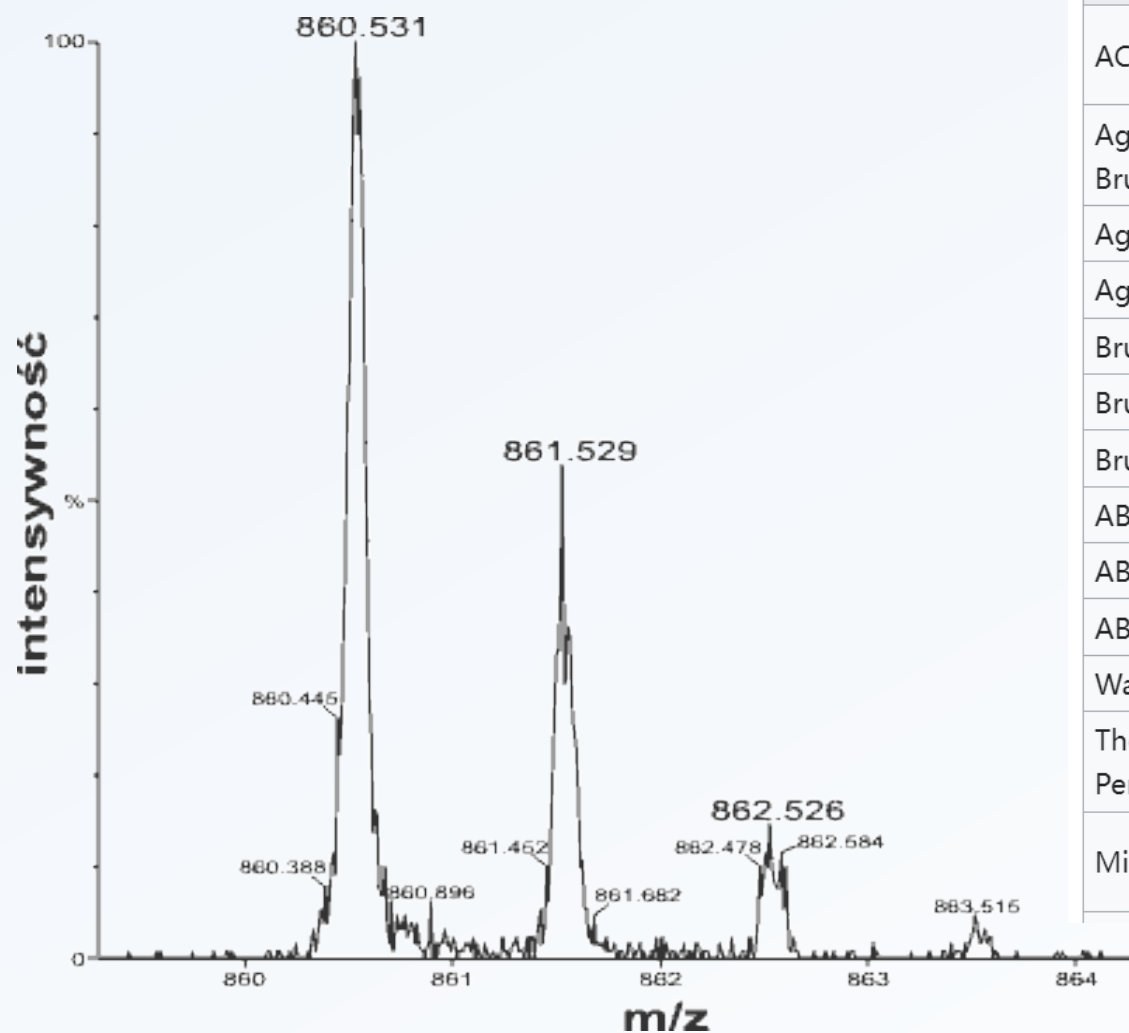
■ MS

Fourier transform ion cyclotron resonance mass spectrometry



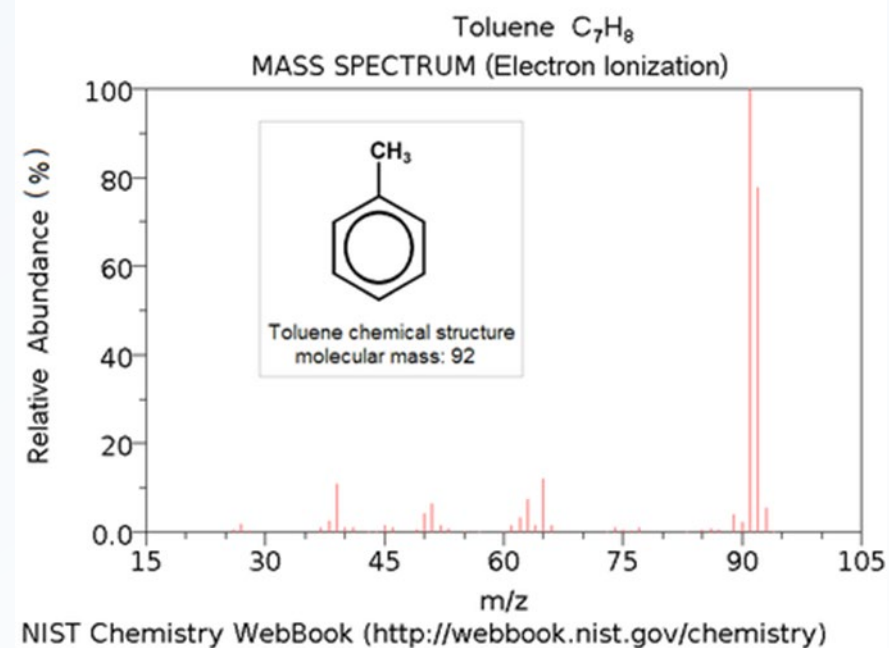
MS and NMR

■ MS



Company	Extension	File type
ACD/Labs	*.spectrum	Imports LC/MS and GC/MS data from most major instrument vendors listed here
Agilent Bruker	.D (folder)	Agilent MassHunter, Agilent ChemStation, or Bruker BAF/YEP/TDF data format
Agilent/Bruker	.YEP	instrument data format
Agilent	.AEV, .ASR	ASCII Report format (for Analytical Studio Reviewer)
Bruker	.BAF	instrument data format
Bruker	.FID	instrument data format
Bruker	.TDF	timsTOF instrument data format
ABI/Sciex	.WIFF, .WIFF2	instrument data format
ABI/Sciex	.t2d	4700 and 4800 file format
ABI/Sciex	.dat	Voyager-DE series file format
Waters	.PKL	MassLynx peak list format
Thermo PerkinElmer	.RAW*	Thermo Xcalibur PerkinElmer TurboMass
Micromass**/Waters	.RAW* (folder)	Waters MassLynx

Since the precise [structure](#) or [peptide sequence](#) of a molecule is deciphered through the set of fragment masses, the interpretation of [mass spectra](#) requires combined use of various techniques. Usually the first strategy for identifying an unknown compound is to compare its experimental mass spectrum against a library of mass spectra. If no matches result from the search, then manual interpretation or [software assisted interpretation of mass spectra](#) must be performed. Computer simulation of [ionization](#) and fragmentation processes occurring in mass spectrometer is the primary tool for assigning structure or peptide sequence to a molecule. An *a priori* structural information is fragmented *in silico* and the resulting pattern is compared with observed spectrum. Such simulation is often supported by a fragmentation library that contains published patterns of known decomposition reactions. [Software](#) taking advantage of this idea has been developed for both small molecules and [proteins](#).



MS and NMR

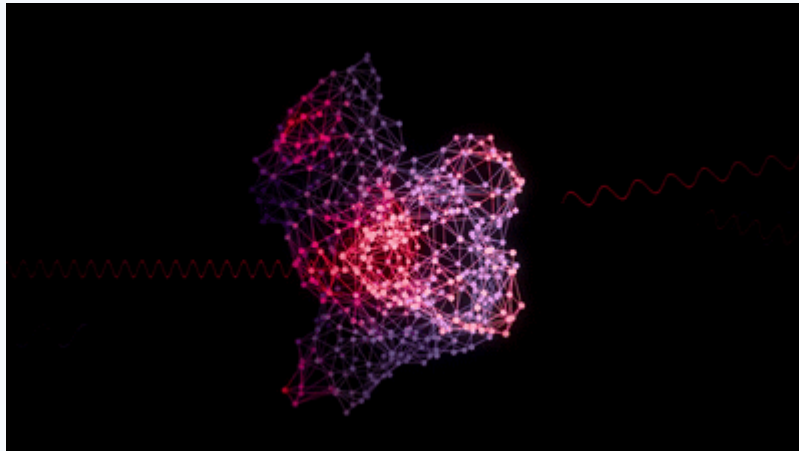
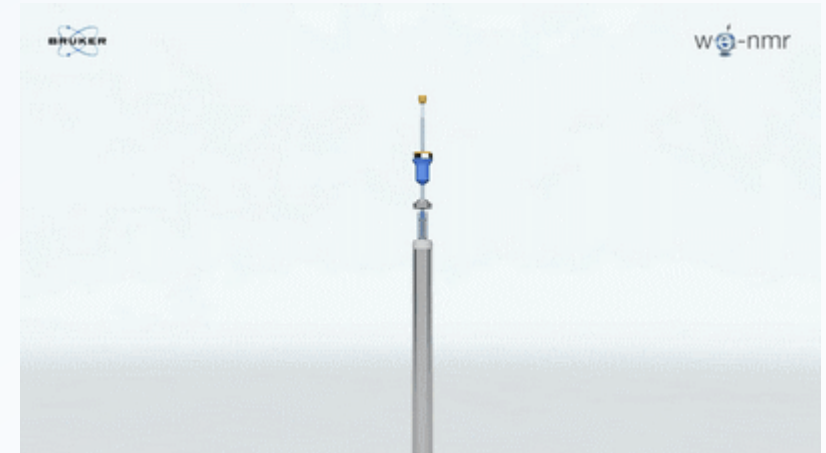
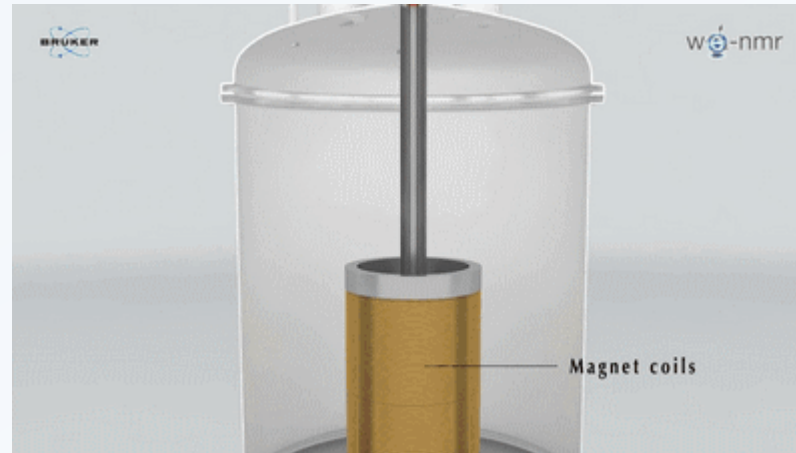
■ NMR

Nuclear magnetic resonance (NMR) is a [physical phenomenon](#) in which [nuclei](#) in a strong constant [magnetic field](#) are perturbed by a weak oscillating magnetic field (in the [near field](#)^[1]) and respond by producing an electromagnetic signal with a frequency characteristic of the magnetic field at the nucleus. This process occurs near [resonance](#), when the oscillation frequency matches the intrinsic frequency of the nuclei, which depends on the strength of the static magnetic field, the chemical environment, and the magnetic properties of the [isotope](#) involved; in practical applications with static magnetic fields up to ca. 20 [tesla](#), the frequency is similar to [VHF](#) and [UHF](#) television broadcasts (60–1000 MHz). NMR results from specific [magnetic](#) properties of certain atomic nuclei. [Nuclear magnetic resonance spectroscopy](#) is widely used to determine the structure of organic molecules in solution and study [molecular physics](#) and [crystals](#) as well as non-crystalline materials. NMR is also routinely used in advanced [medical imaging](#) techniques, such as in [magnetic resonance imaging](#) (MRI). The original application of NMR to condensed matter physics is nowadays mostly devoted to strongly correlated electron systems. It reveals large many-body couplings by fast broadband detection and it should not to be confused with solid state NMR, which aims at removing the effect of the same couplings by [Magic Angle Spinning](#) techniques.

MS and NMR

■ NMR

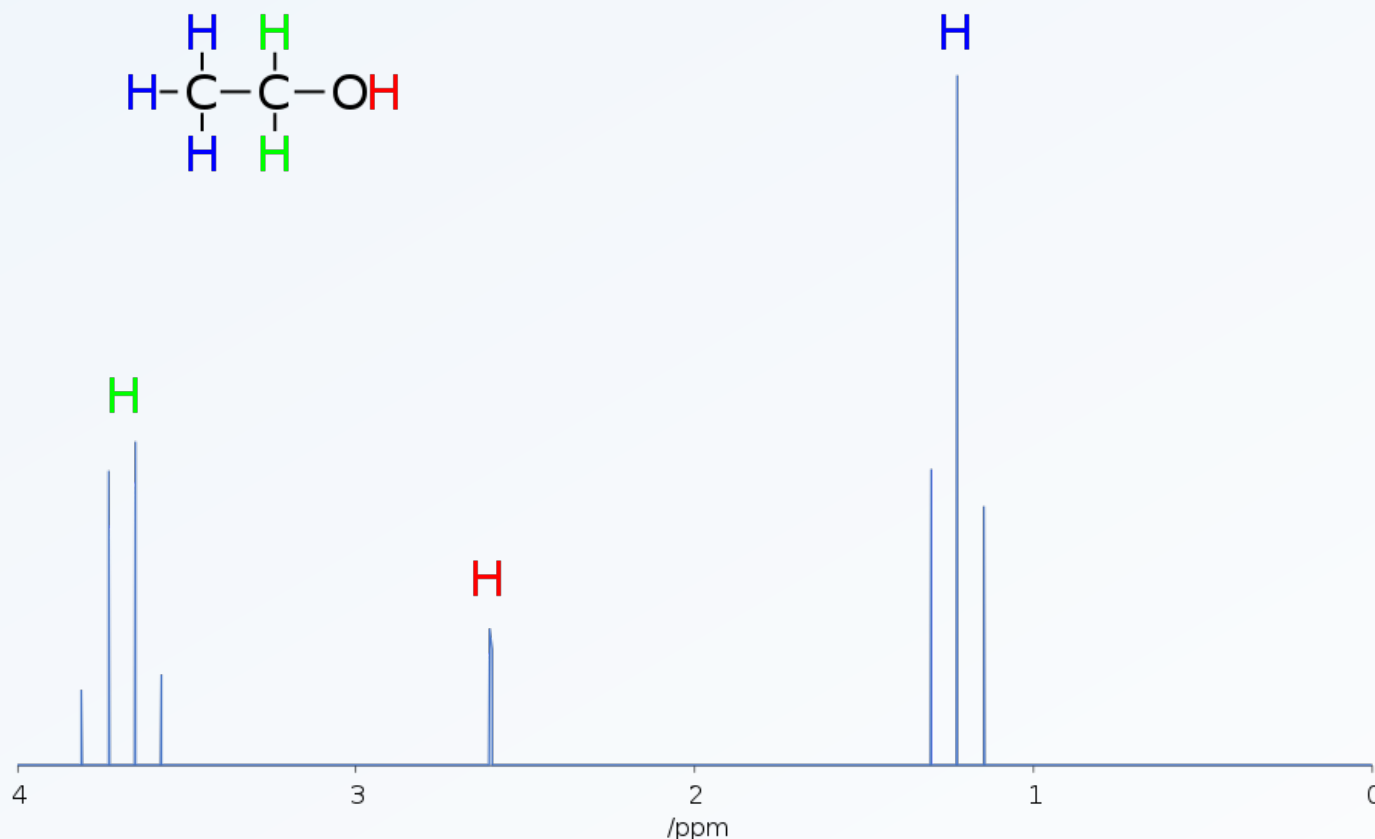
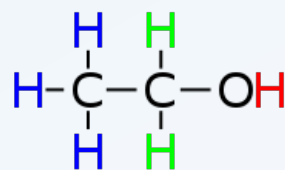
In Lab



MS and NMR

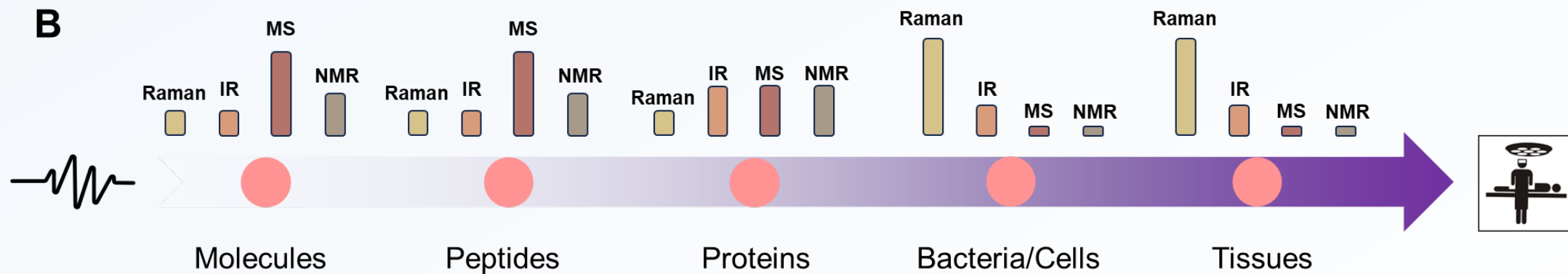
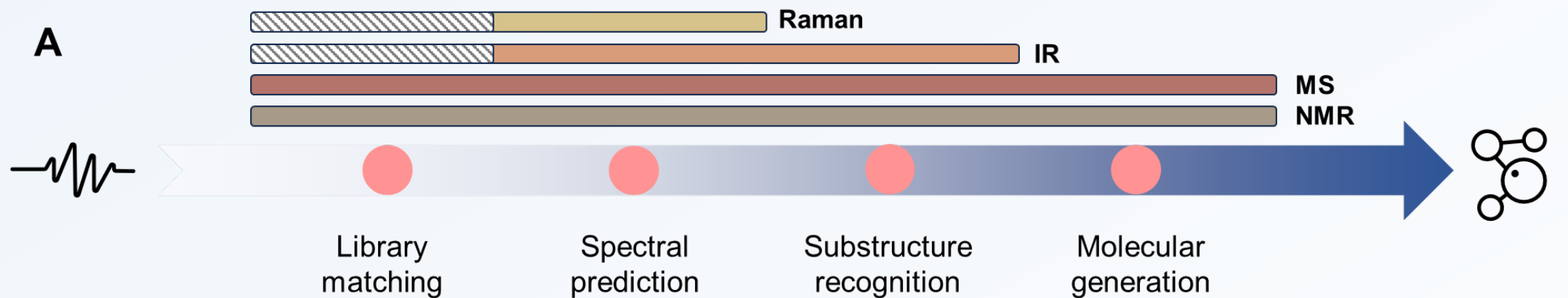
■ NMR

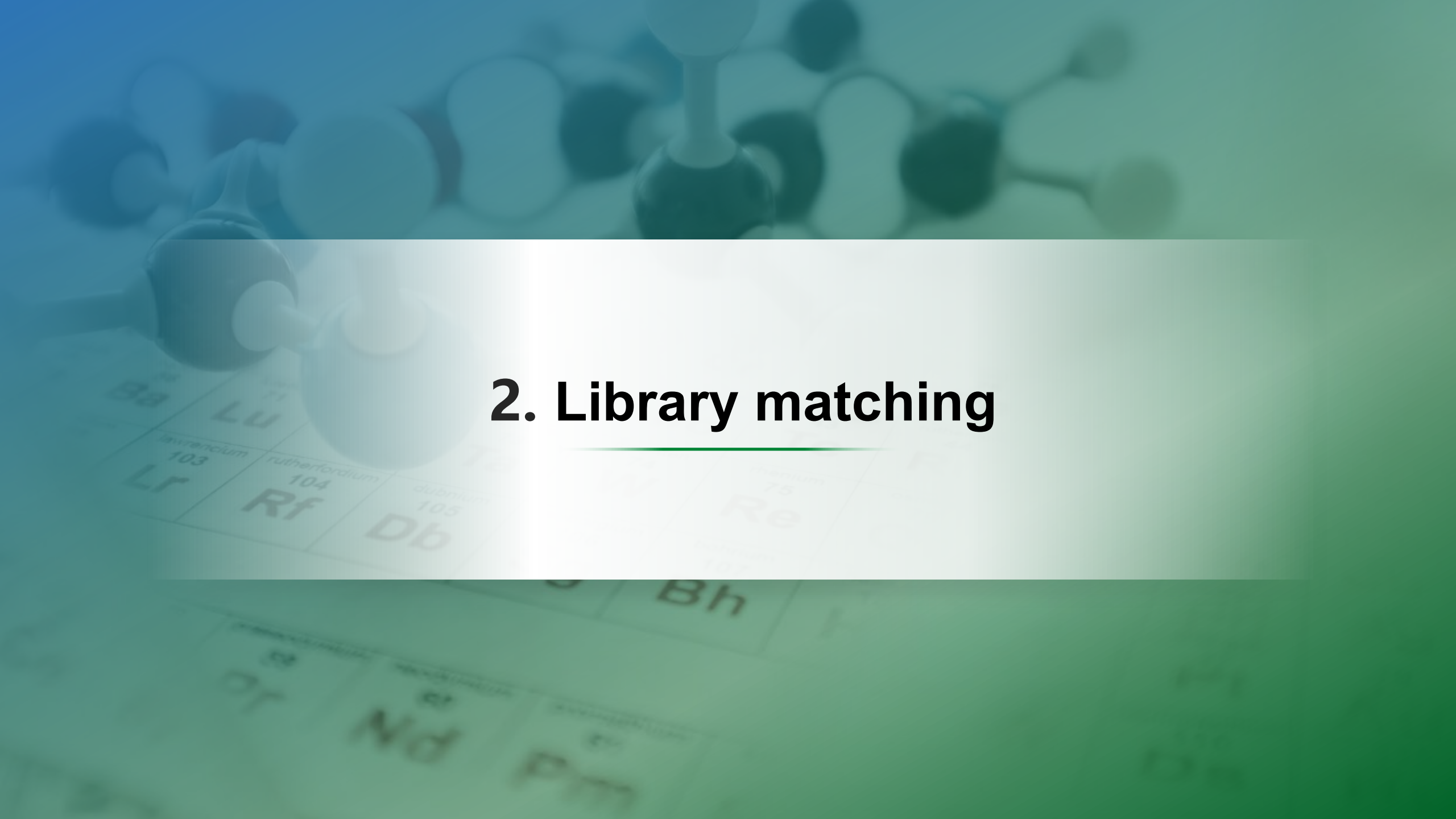
Ethanol



醛氢9-10.5 ppm
 芳环及苯环6-9.5 ppm
 烯氢4.5-7.5 ppm
 与氧原子相连的氢3.0-5.5ppm
 与氮原子相连的氢2.0-3.5ppm
 炔氢1.6-3.4 ppm
 脂肪氢0-2.5 ppm
 活泼氢：醇类0.5-5.5ppm
 酚类4.0-12.0 ppm
 酸类：9-13.0 ppm
 氨活泼氢：酰胺5-8.5ppm
 芳香氨 3.0-5.0ppm
 脂肪氨0.6-3.5 ppm。

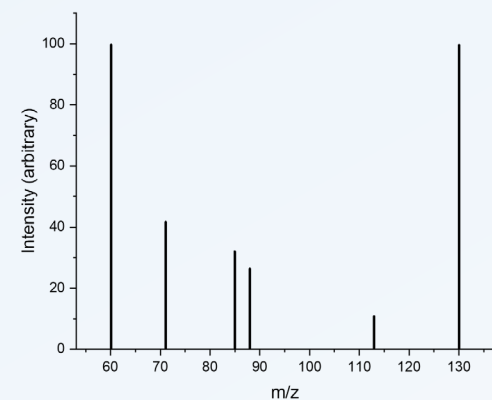
MS and NMR





2. Library matching

Library matching

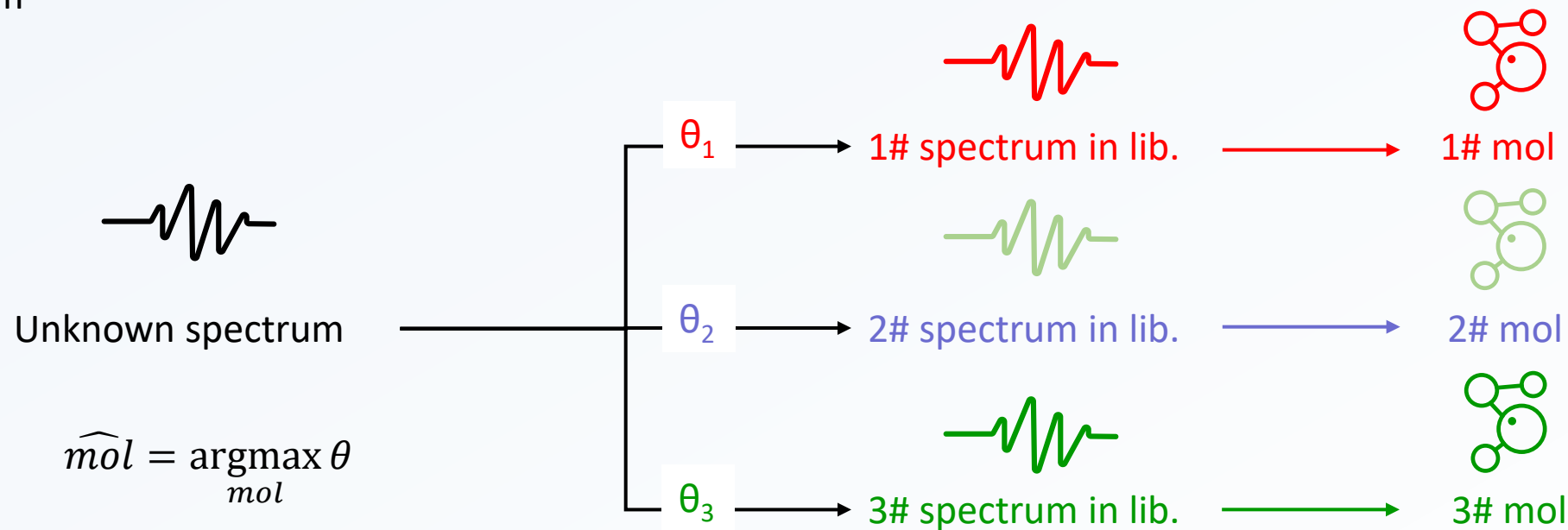
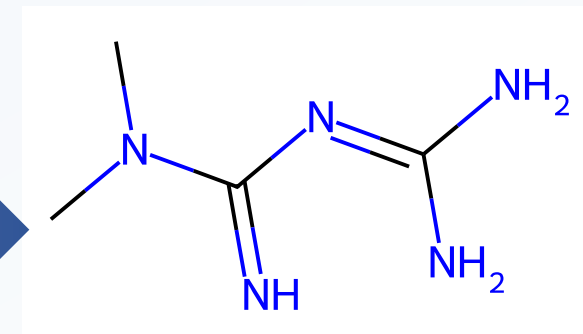


Unknown spectrum

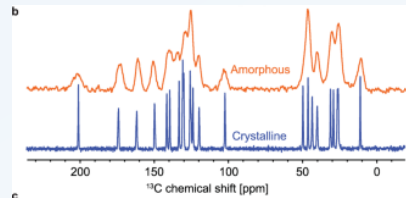
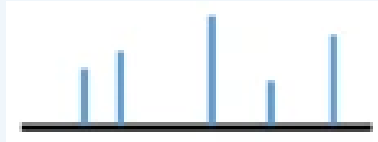
Library matching

Spectral prediction

De novo generation



Library matching

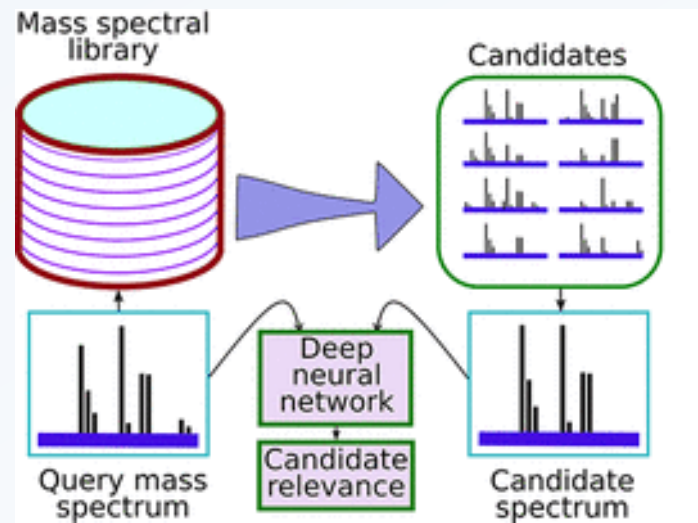


AI+MS\NMR

Spectra analysis



Searching



Spectral similarity

Structural similarity

Library matching

Spectral similarity measures:

$$S_{dp}(a, b) = \frac{\left(\sum_{n=1}^M n^a q_n^b n^a r_n^b \right)^2}{\sum_{n=1}^M (n^a q_n^b)^2 \sum_{n=1}^M (n^a r_n^b)^2}$$

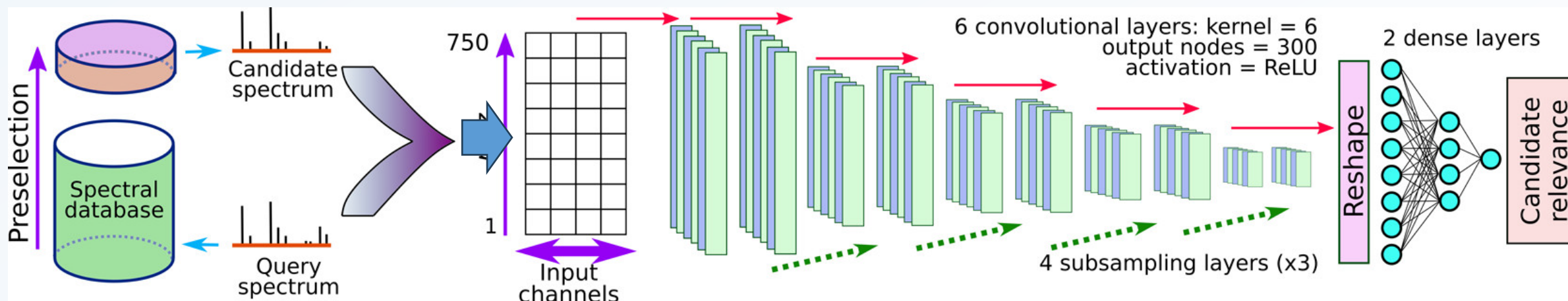
Dot product

$$S_c = \frac{N_Q S_{dp}(a, b) + \left(\frac{N_{QR}}{N_{QR} - 1} \right) \sum_{i=2}^{i=N_{QR}} \left(\frac{q_{n_i}^d}{q_{n_{i-1}}^d} \frac{r_{n_{i-1}}^d}{r_{n_i}^d} \right)^\sigma}{N_Q + N_{QR}}$$

Composite spectral similarity

Manhattan distance

Euclidean distance



Using deep convolutional neural network for the EI-MS library search.

Library matching

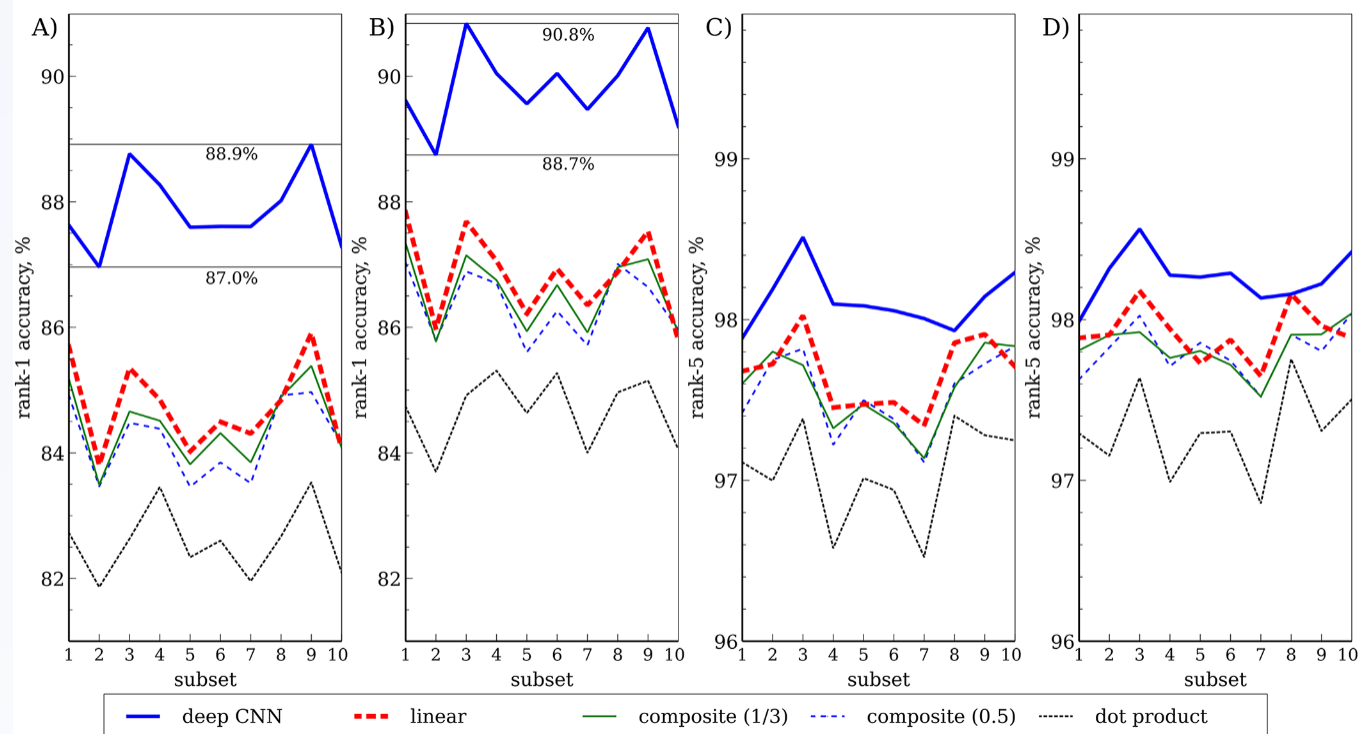
Feature Engineering

Table 1. Input Features for Deep CNN^a

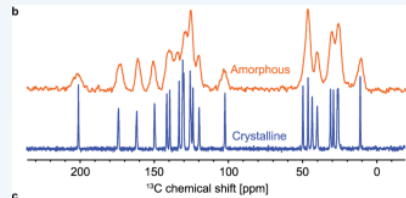
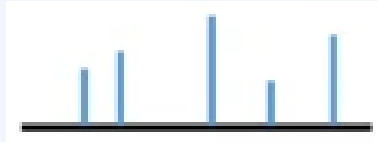
j	$f_j(q, r, n)$
1	If $q = 0, r \neq 0$, then -1 If $q = r = 0$, then 0 If $q \neq 0, r \neq 0$, then $(1 - x^2)/(1 + x^2)$, where $x = r/q$
2	$1 - f_1(q, r, n) $
3	r
4	q
5	If $r = 0, q \neq 0$, then q , else 0
6	If $q = 0, r \neq 0$, then r , else 0
7–12	The same as 1–6, but $q^{0.5}, r^{0.5}$ were used instead of q, r
13–16	The same as 3–6, but nq, nr were used instead of q, r
17–20	The same as 3–6, but $nq^{0.5}, nr^{0.5}$ were used instead of q, r
21–26	The same as 1–6, but $q^{1/3}, r^{1/3}$ were used instead of q, r
27–30	The same as 3–6, but $q/S_q, r/S_r$ were used instead of q, r . S_q, S_r —sums of all intensities in reference and query peaks
31	If $r = 0, q \neq 0$, then 1 , else 0
32	If $q = 0, r \neq 0$, then 1 , else 0
33	If $q \neq 0$, then 1 , else 0
34	If $r \neq 0$, then 1 , else 0
35	If $q \neq 0, r \neq 0$, then 1 , else 0
36	n

^aIf q and r are intensities of query and reference spectra corresponding to $m/z = n$, the function $f_j(q, r, n)$ describes the element at the j -th row and n -th column of an input matrix.

CNN performs better than all the spectral similarity measurements.



Library matching

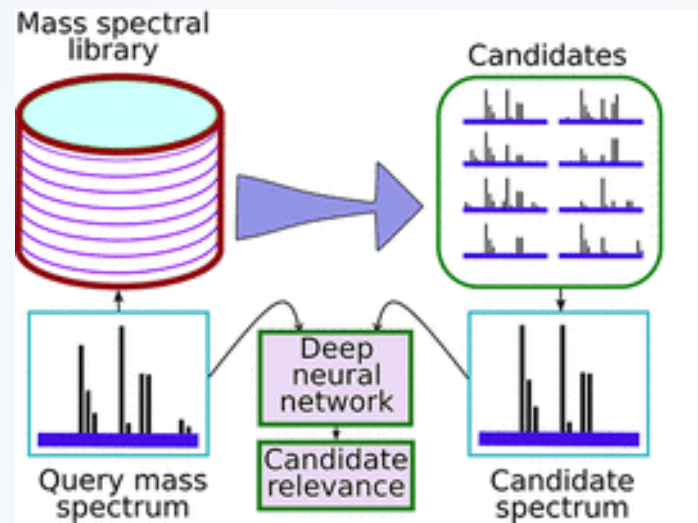


AI+MS\NMR

Spectra analysis



Searching



{
Spectral similarity
Structural similarity

Library matching

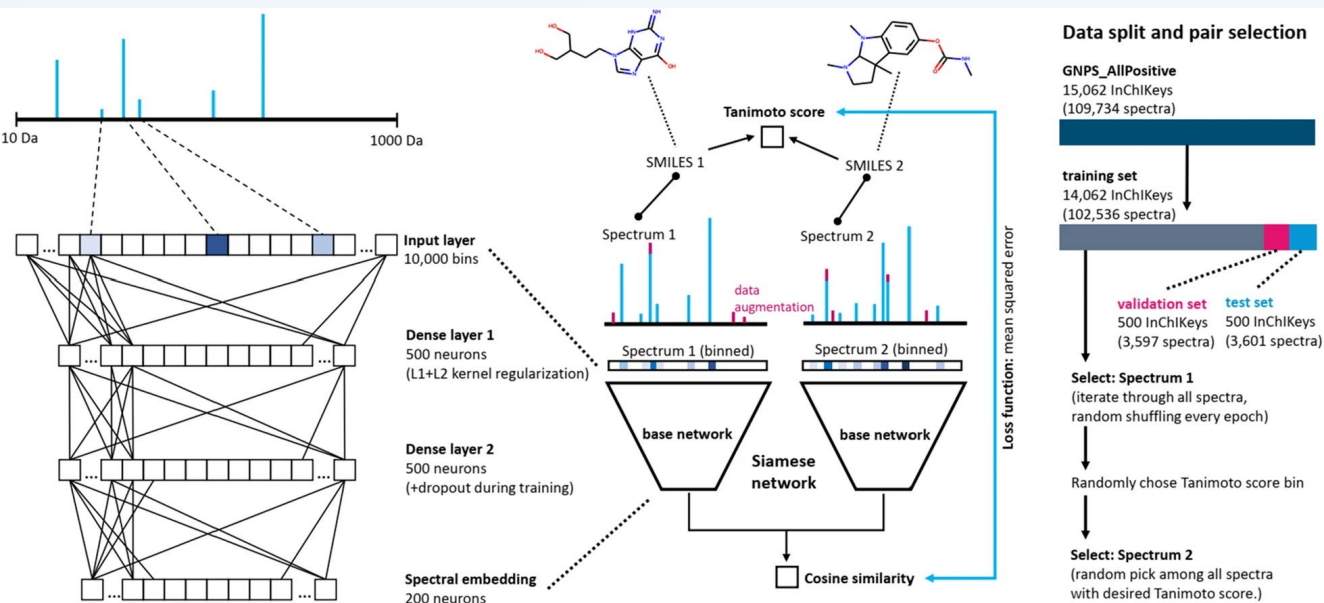
训练MS数据分子
指纹预测因子



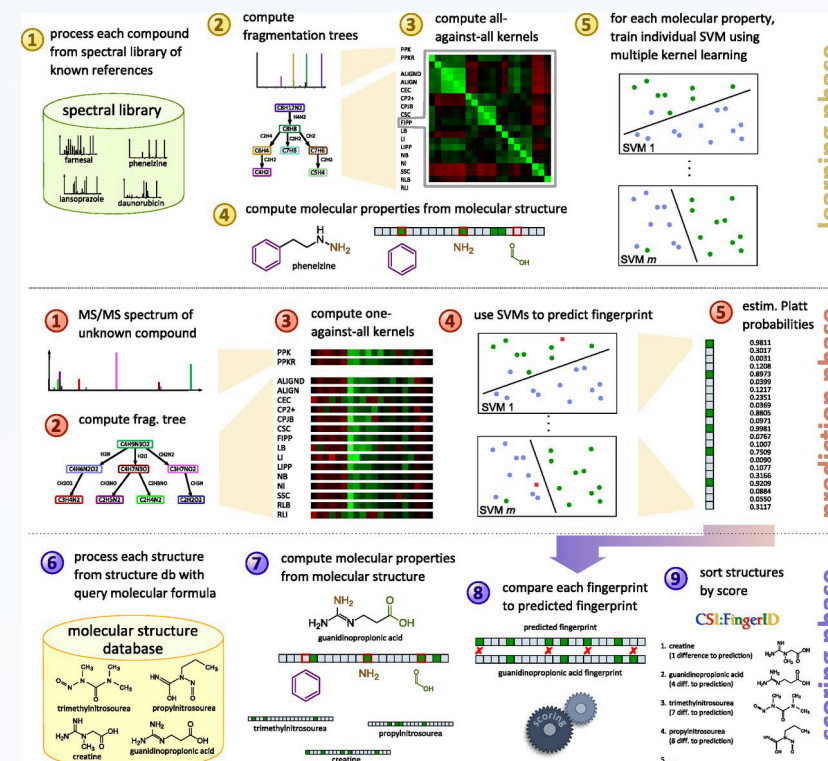
未知化合物的MS
数据来预测指纹



与结构数据库
中的分子结构
指纹进行比较
匹配



Spectrum → SMILES → Structure similarity → Matching

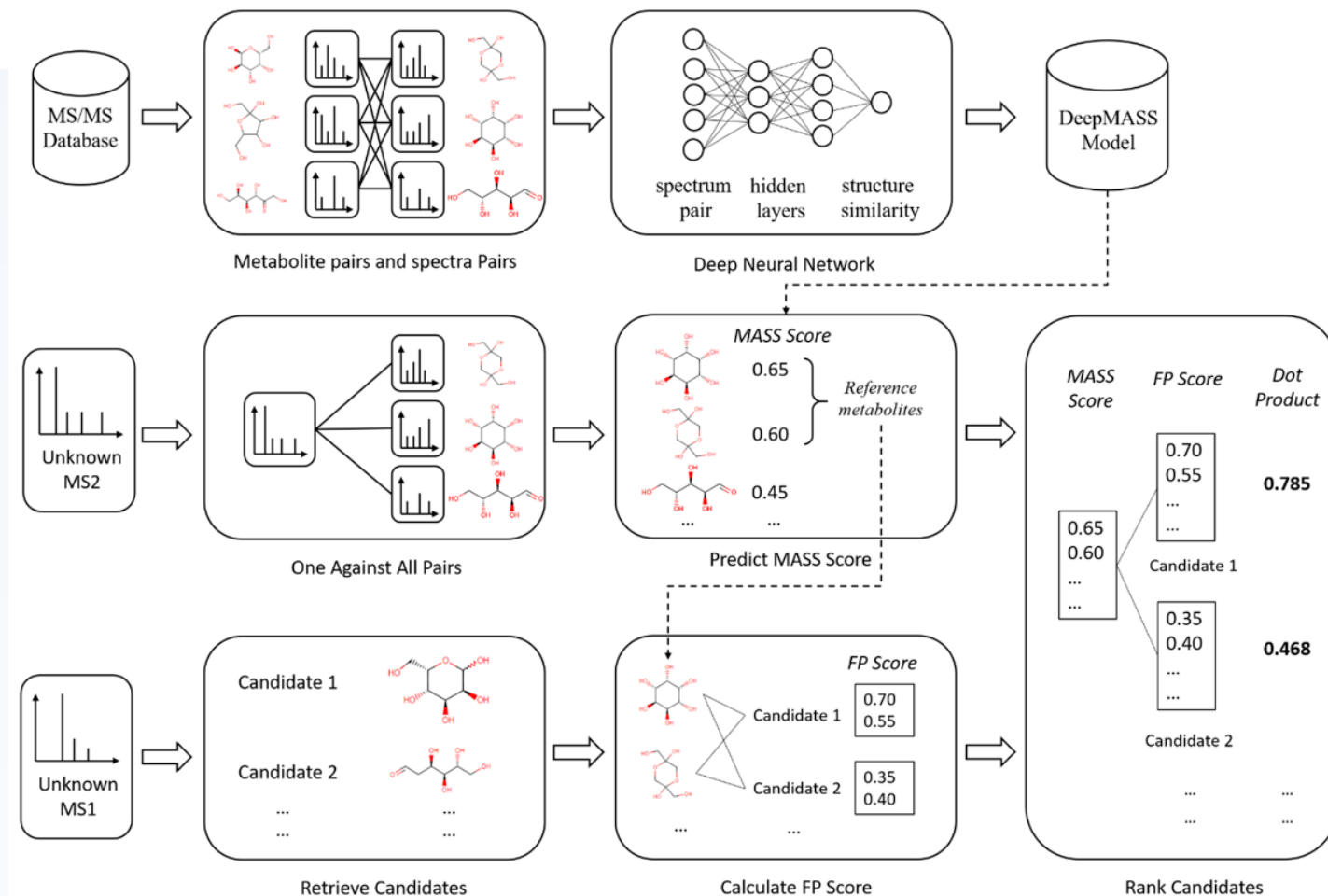


Deep MS/MS-Aided Structural-Similarity Scoring for Unknown Metabolite Identification

Hongchao Ji, Yamei Xu, Hongmei Lu,* and Zhimin Zhang*

- in silico fragmentation methods
- fingerprint-based methods
- known-to-unknown methods

To improve the automation and generalization ability of the known-to-unknown methods, we need novel machine learning methods, which can learn features from raw data directly, has better generalization ability than rule-based approaches, and can be easily scaled to take advantage of increasing MS/MS spectral database. Recently, deep learning revolution has changed the machine learning community fundamentally.²⁷ Various architectures of deep neural networks emerged for different applications, including deep neural networks,²⁸ convolutional neural networks,²⁹ recurrent neural networks,³⁰ deep belief networks,³¹ and generative adversarial networks.³² Meanwhile, deep learning frameworks, such as Tensorflow³³ and Keras,³⁴ enable researchers to go from idea to result swiftly. Therefore,



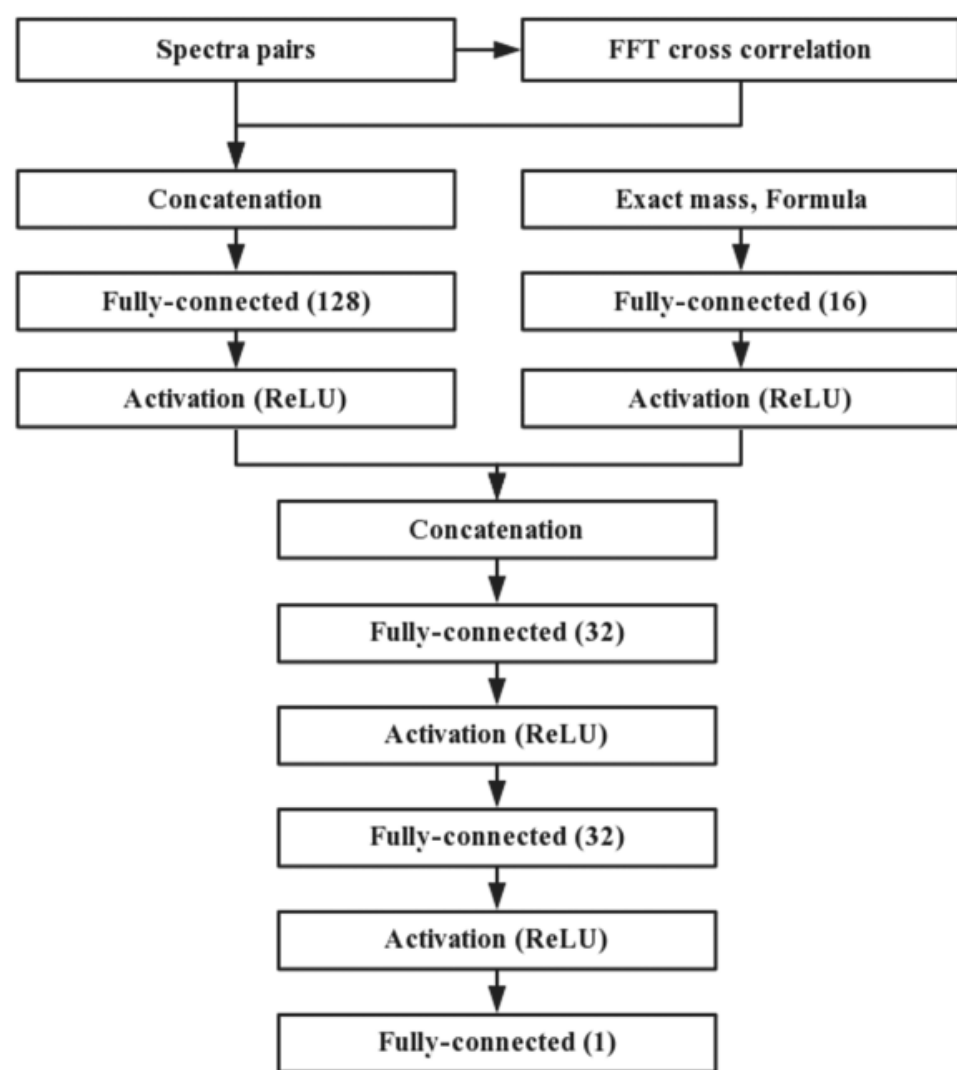


Figure 2. Architecture of the deep neural network of DeepMASS.

```

from keras.models import Model, model_from_json
from keras.layers import Dense, Input, Dropout, concatenate
from keras.optimizers import SGD
from sklearn.preprocessing import MinMaxScaler
from sklearn.externals import joblib

```

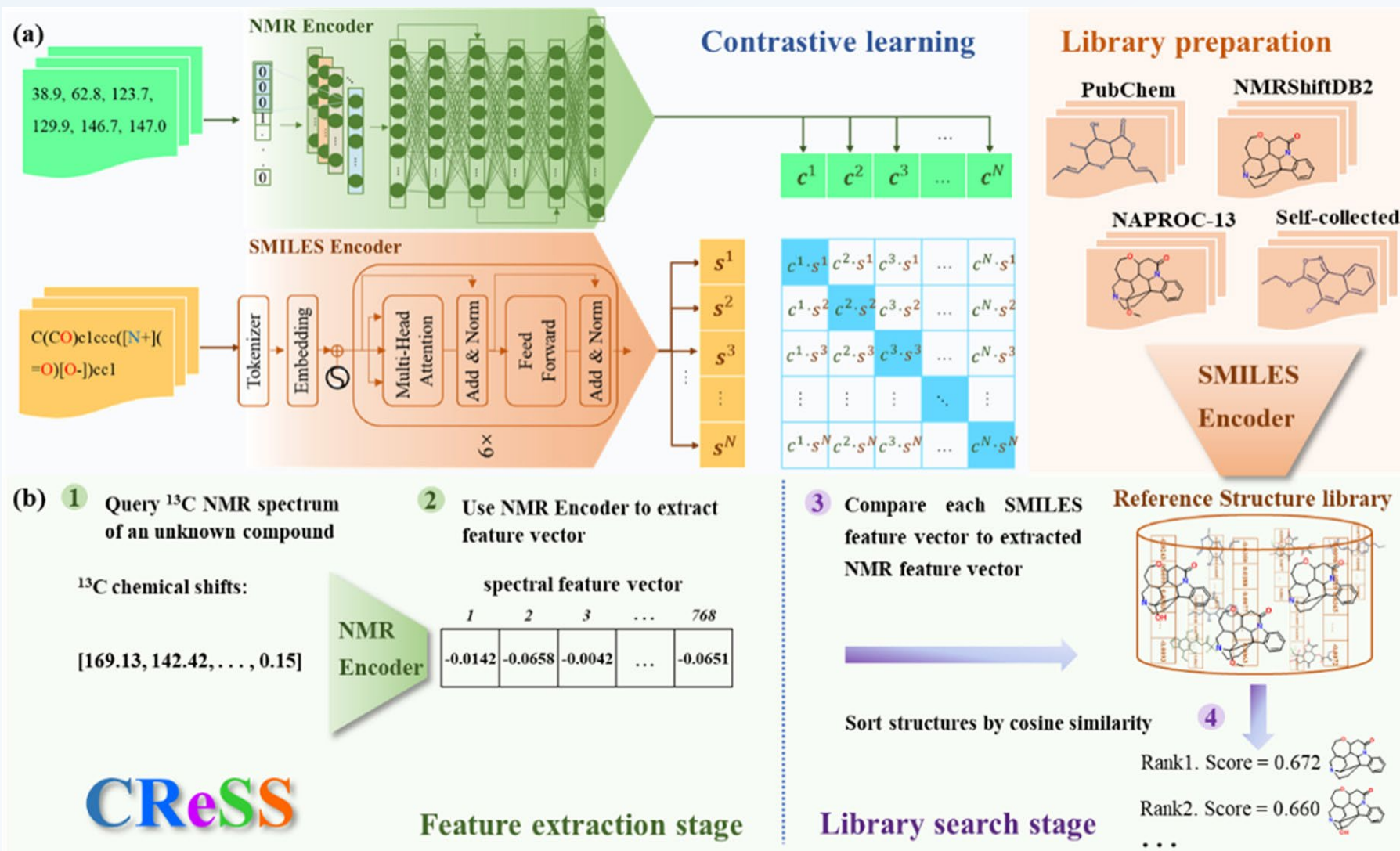
```

# build model
spec_input = Input(shape=(train_X1.shape[1],), sparse=True)
# spec_input = Input(shape=(train_X1.shape[1],))
formula_input = Input(shape=(train_X2.shape[1],))
spec_output = Dense(128, activation='relu')(spec_input)
formula_output = Dense(16, activation='relu')(formula_input)
hidden = concatenate([formula_output, spec_output])
hidden = Dense(32, activation='relu')(hidden)
hidden = Dropout(0.3)(hidden)
hidden = Dense(32, activation='relu')(hidden)
hidden = Dropout(0.2)(hidden)
predictions = Dense(1, activation='linear')(hidden)

model = Model(inputs=[spec_input, formula_input], outputs=predictions)
model.compile(optimizer=SGD(lr=0.05), loss='mse', metrics=['mae'])
model.fit([train_X1, train_X2], train_Y, epochs=20)

```

Library matching



原理：使用对比学习引入
NMR谱图信息+SMILE结
构信息，预测两条光谱的
相似程度，实现相似度与
结构的关联

Anal. Chem. 2021, 93, 50, 16947–16955

Library matching

Inspired by NLP

RESEARCH ARTICLE

Spec2Vec: Improved mass spectral similarity scoring through learning of structural relationships

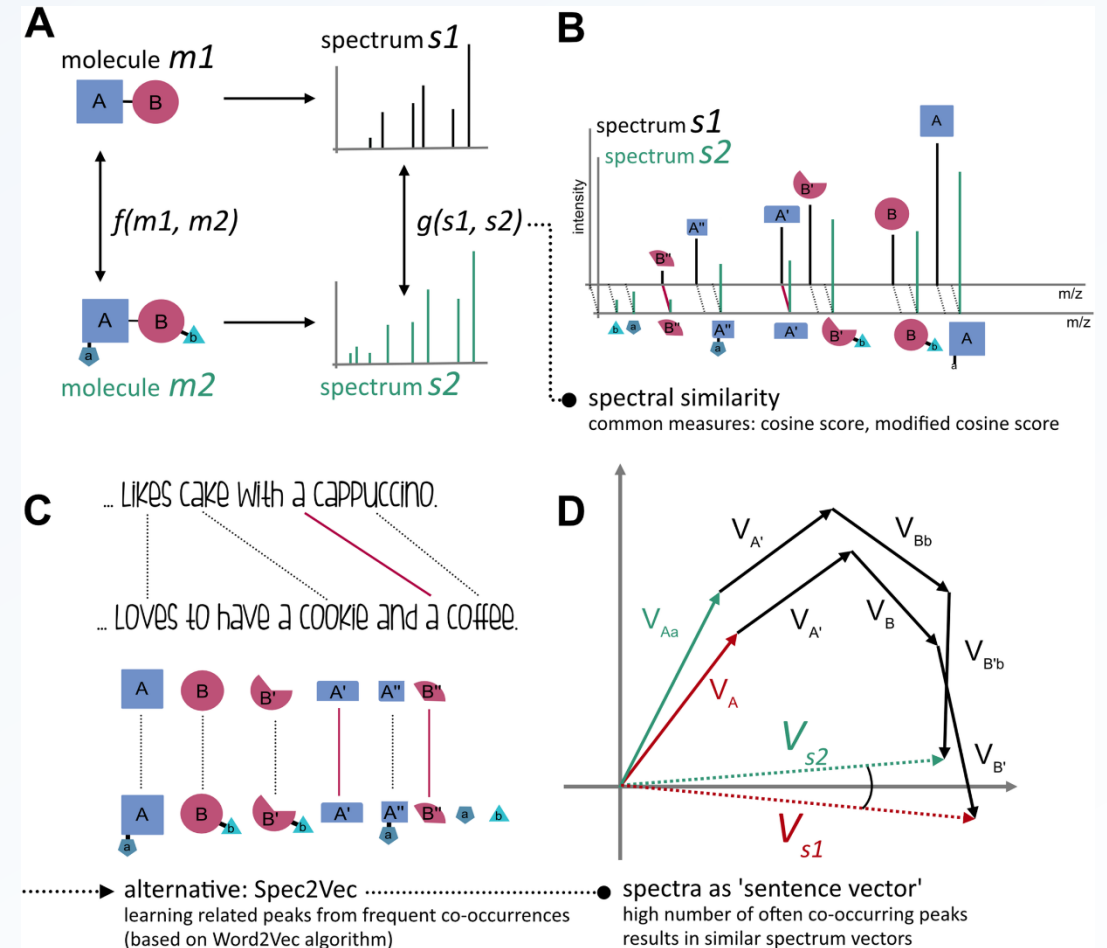
Florian Huber^{1*}, Lars Ridder¹, Stefan Verhoeven¹, Jurriaan H. Spaaks¹, Faruk Diblen¹, Simon Rogers², Justin J. J. van der Hooft^{3*}

Mechanism:

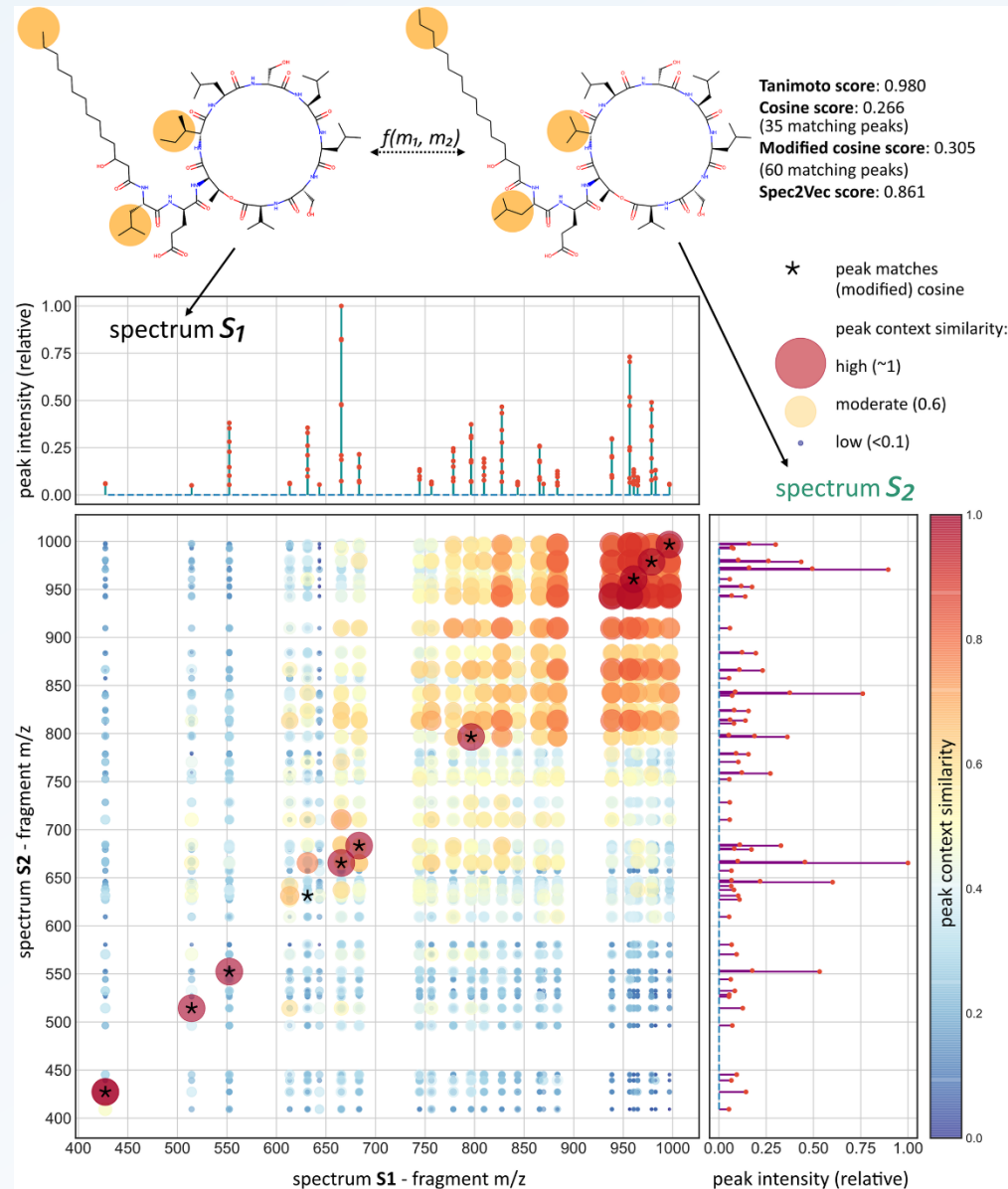
- Label is **structure similarity**
- Spec2Vec training is based on **frequency of each peaks**

MS-MS spectra can be considered as signatures of molecules: spectra are known to contain structural information of the original molecule, but without a straightforward way to translate mass spectral features into structural ones describing the fragmented molecule.

PLoS Comput Biol 2021, 17(2), e1008724

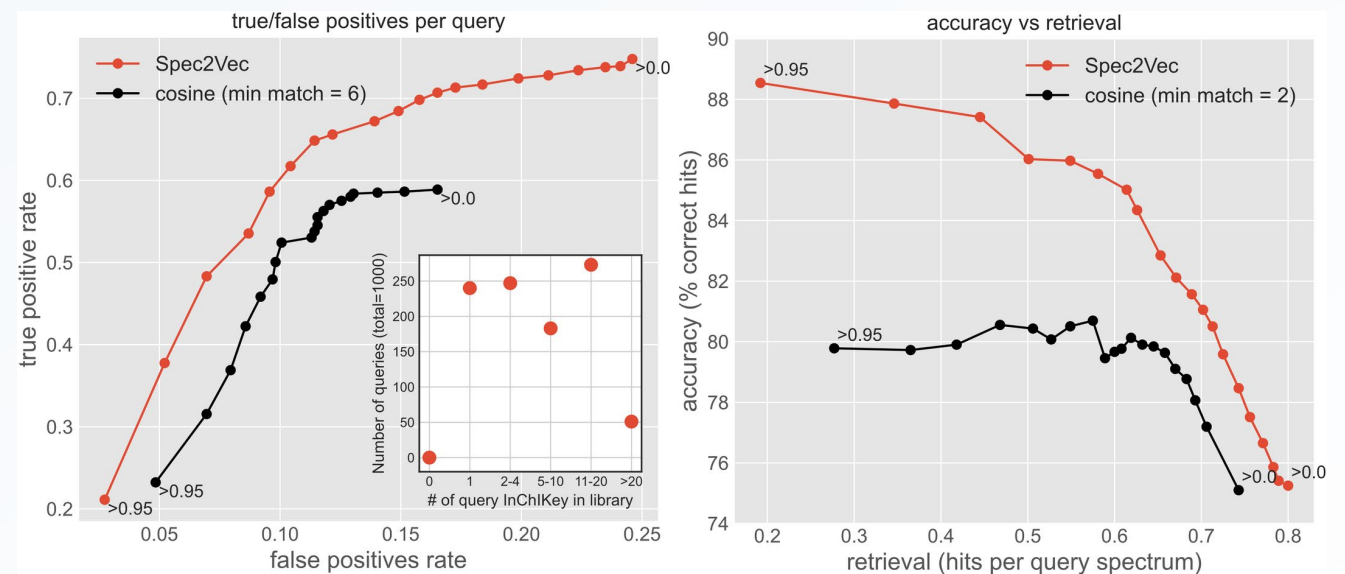


Library matching

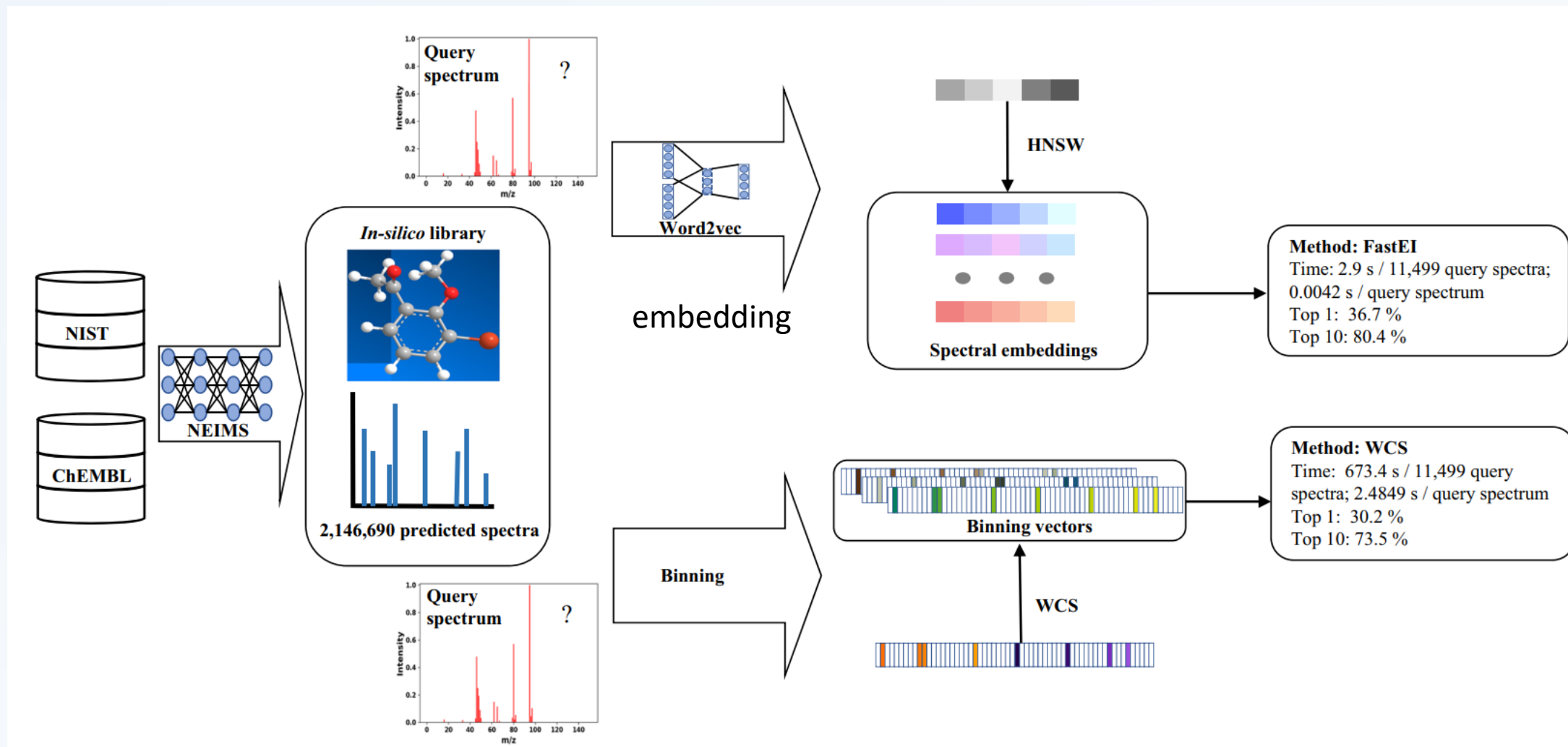


Advantages:

- Correlate the structure and similarity
- High accuracy



Library matching

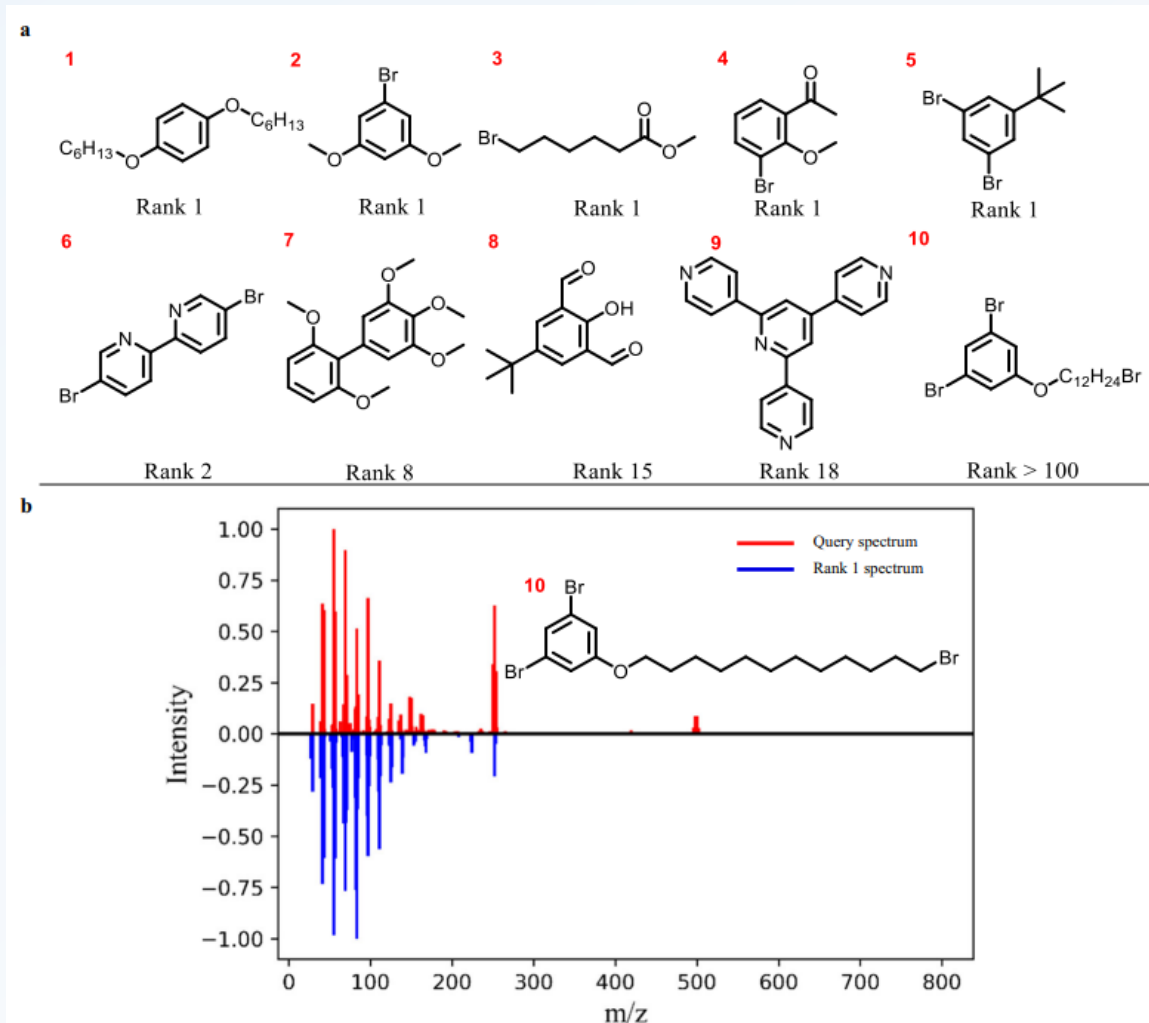


Hierarchical Navigable Small-world Graph (HNSW)

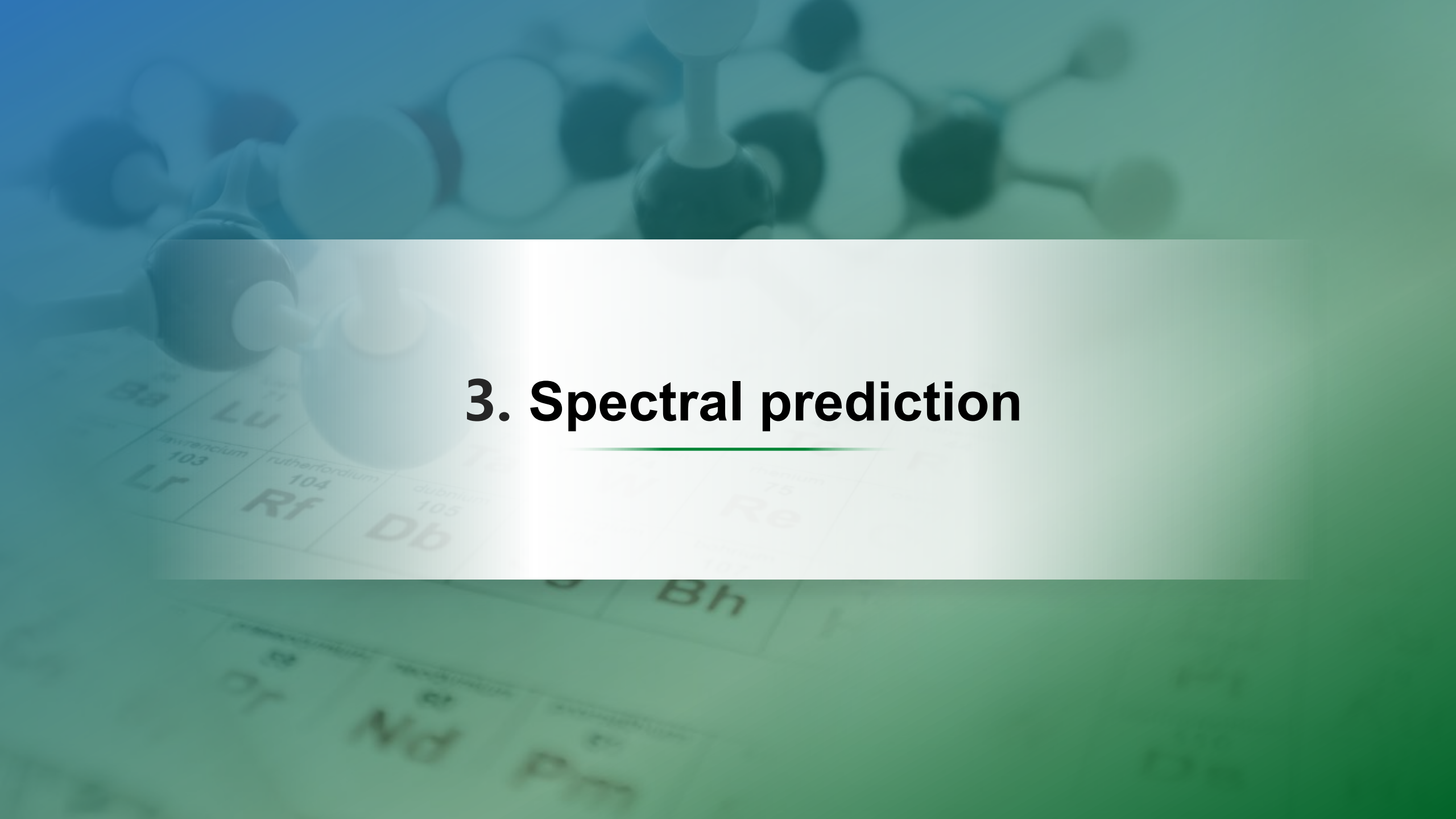
Nat Commun 14, 3722 (2023)

Library matching

Unknown compound identification



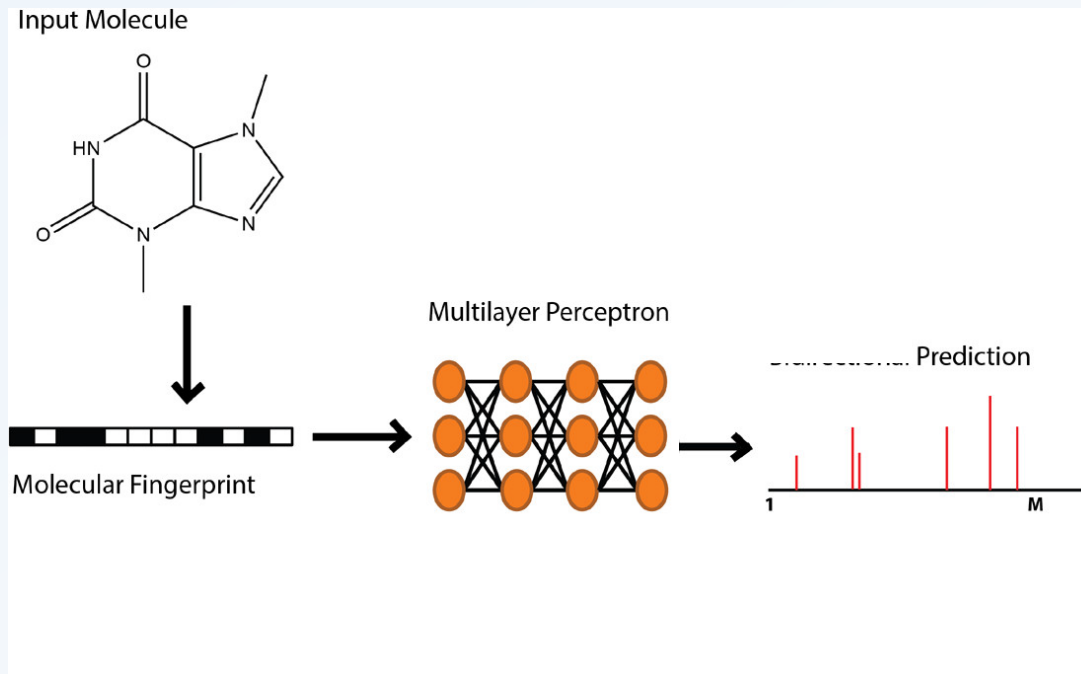
matching due to the limited compound coverage. To demonstrate the advantage of the million-scale *in-silico* library and the accuracy of FastEI, 10 compounds beyond the NIST 2017 library were collected from an organic laboratory. Their structures are shown in Fig. 3a. Compounds 1, 2, 4, and 5 are common aromatic compounds with different substituents on the benzene rings. If these 10 molecules are identified by matching the NIST 2017 library directly, the correct result cannot be achieved. Because there are no correct molecules in the NIST 2017 library, and all the candidates are false positives. With the aid of the large-scale *in-silico* library, these compounds can be mostly identified by FastEI. Their ranks are shown in Fig. 3a. It can be seen that their top 1 accuracy is 50%, and the top 10 accuracy can reach 70%. The compound with a rank >100 is only compound 10, which has a 12-carbon long-chain substituted on the benzene ring. Its MS spectrum is shown in Fig. 3b. It can be seen that most peaks are concentrated in $m/z < 200$. Many chain alkanes candidates are ranked before the target compound, possibly due to the existence of a long-chain structure (Fig. 3c). Overall, these results show that FastEI with the large-scale *in-silico* library can achieve good identification results for molecules beyond the NIST 2017 library. Expanding the spectra library by machine learning-based methods is beneficial to compound identification through reducing the proportion of false negatives.



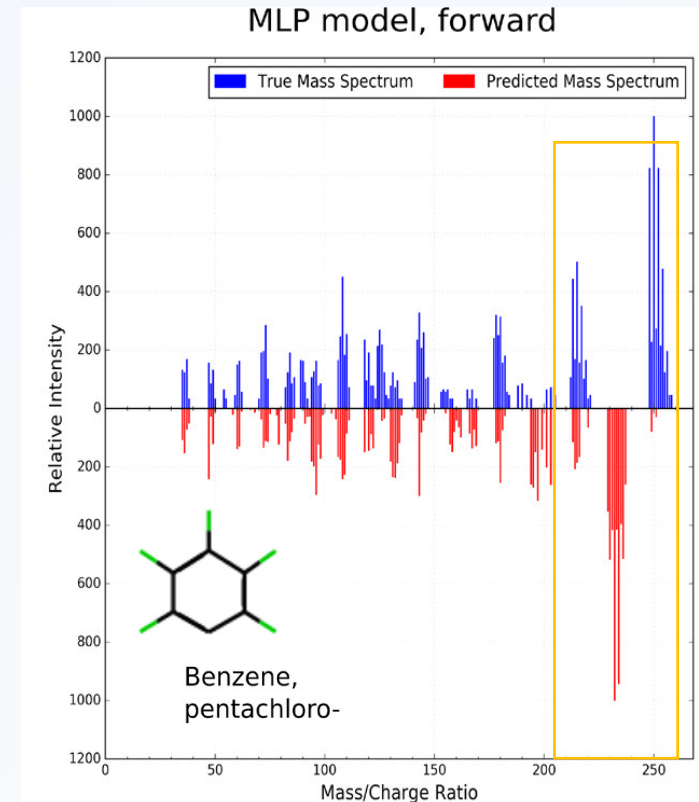
3. Spectral prediction

The first application of DL in predicting EI-MS

J. N. Wei et al. ACS Cent. Sci. 2019, 5, 700



MLP was employed to realize (fingerprint)
vector to (intensity) vector



Inaccurate prediction in
high m/z region
(neutral losses)

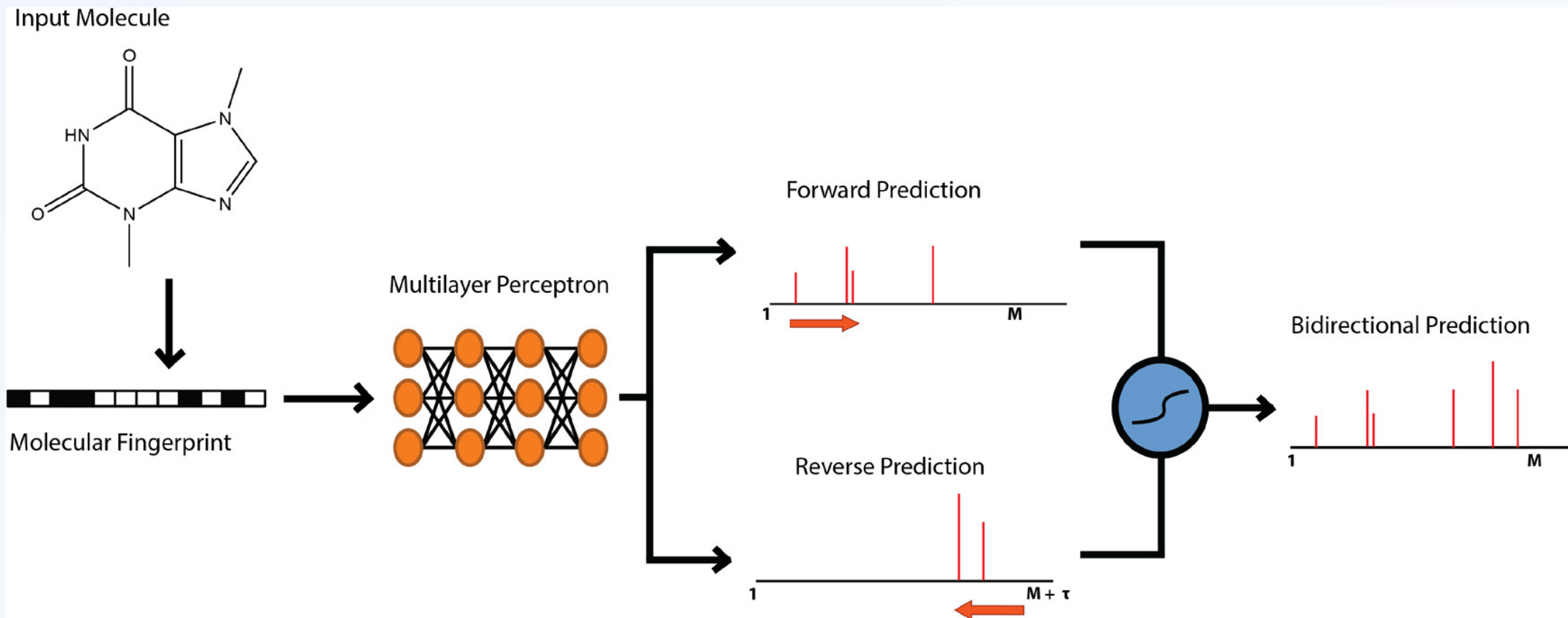
1. Molecular fingerprint vector can hardly capture features of big fragments
2. Direct prediction did not bridge the gap between fragments with low and high m/z

Key novelty

Forward: $peak_i = \sum w^{forward} feature_i + b$

Reverse: $peak_{M-i} = \sum w^{reverse} feature_i + b$

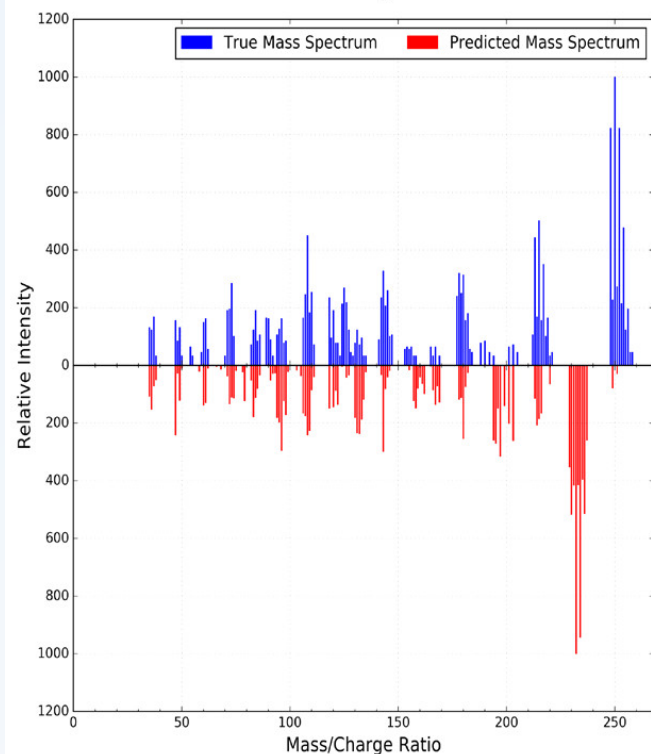
Final: $peak_i = \sigma \cdot peak_i + (1 - \sigma) \cdot peak_{M-i}$



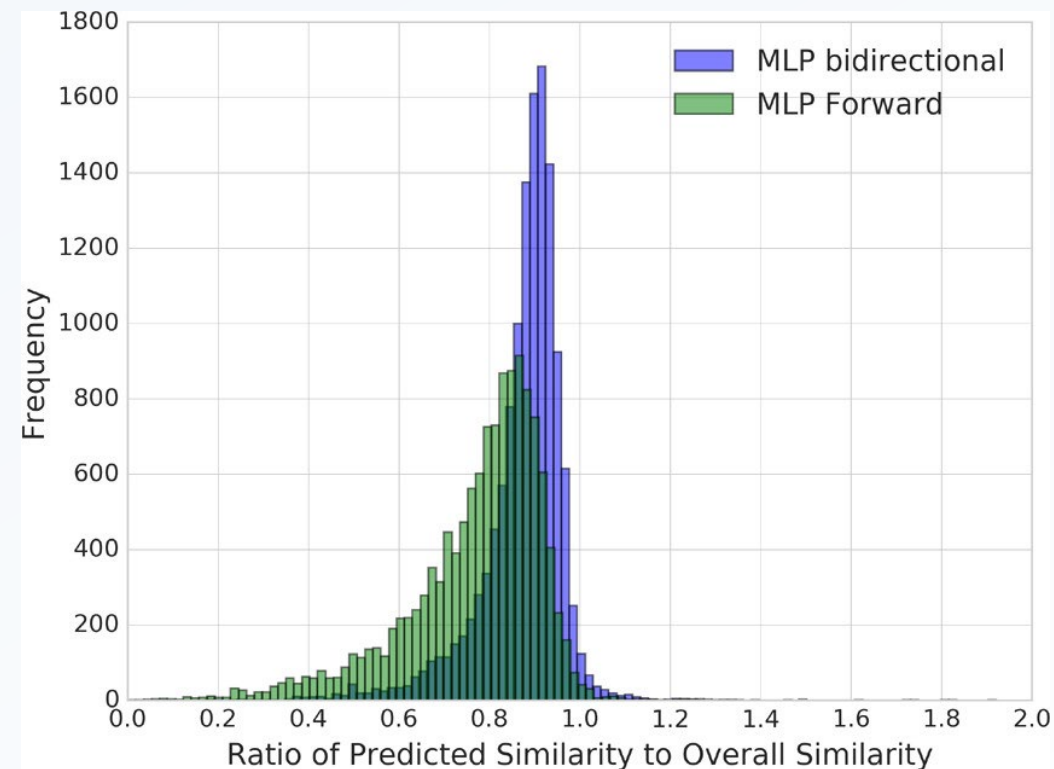
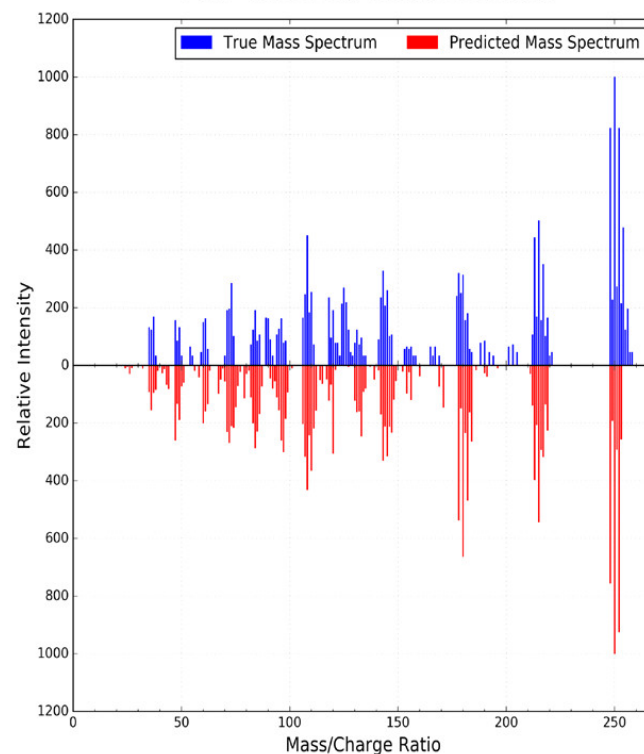
Peak at $m/z=i$ is contributed by features from i and $M-i$

Results

MLP model, forward



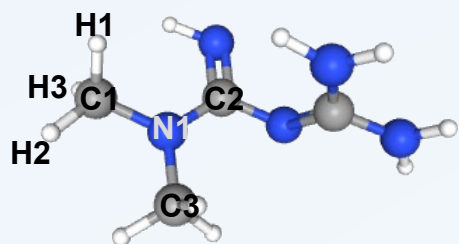
MLP model, bidirectional



Only 50% of the test set achieved a similarity of 0.9 or higher,
indicating a further improvement is highly demanded in this field.

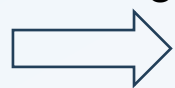
$$\text{Similarity}(I_q, I_l) = \frac{\sum_{k=1}^{M_{\max}} m_k I_{qk}^{0.5} \cdot m_k I_{lk}^{0.5}}{\left\| \sum_{k=1}^{M_q} (m_k I_{qk}^{0.5})^2 \right\| \left\| \sum_{k=1}^{M_l} (m_k I_{lk}^{0.5})^2 \right\|}$$

Limitations



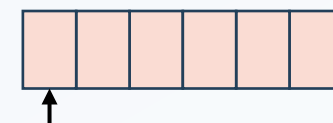
Metformin

labeling
& listing

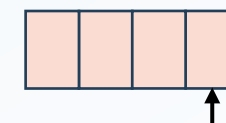


C1	N1	...
H1, H2, H3, N1	C1, C2, C3	...
C3	H1, H2, H3, ...	...
...	...	...

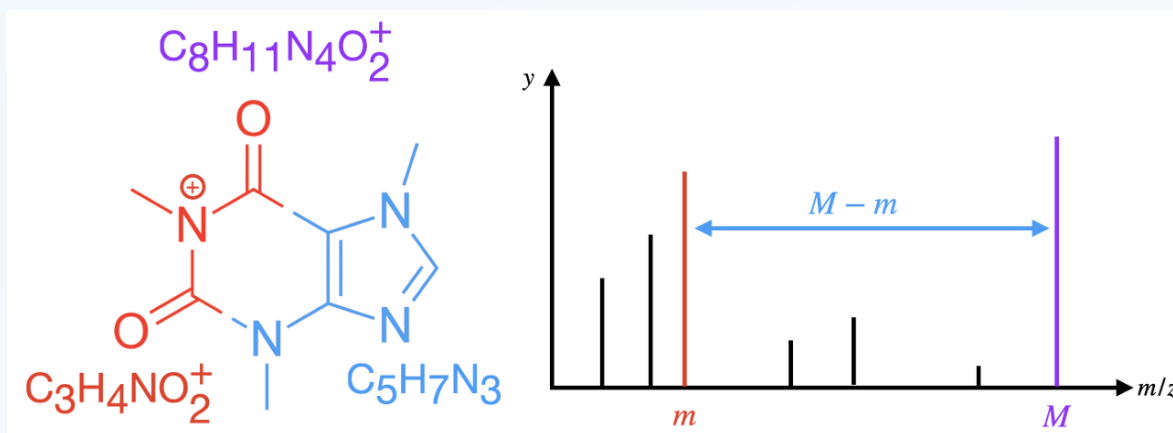
Hashing



1st feature

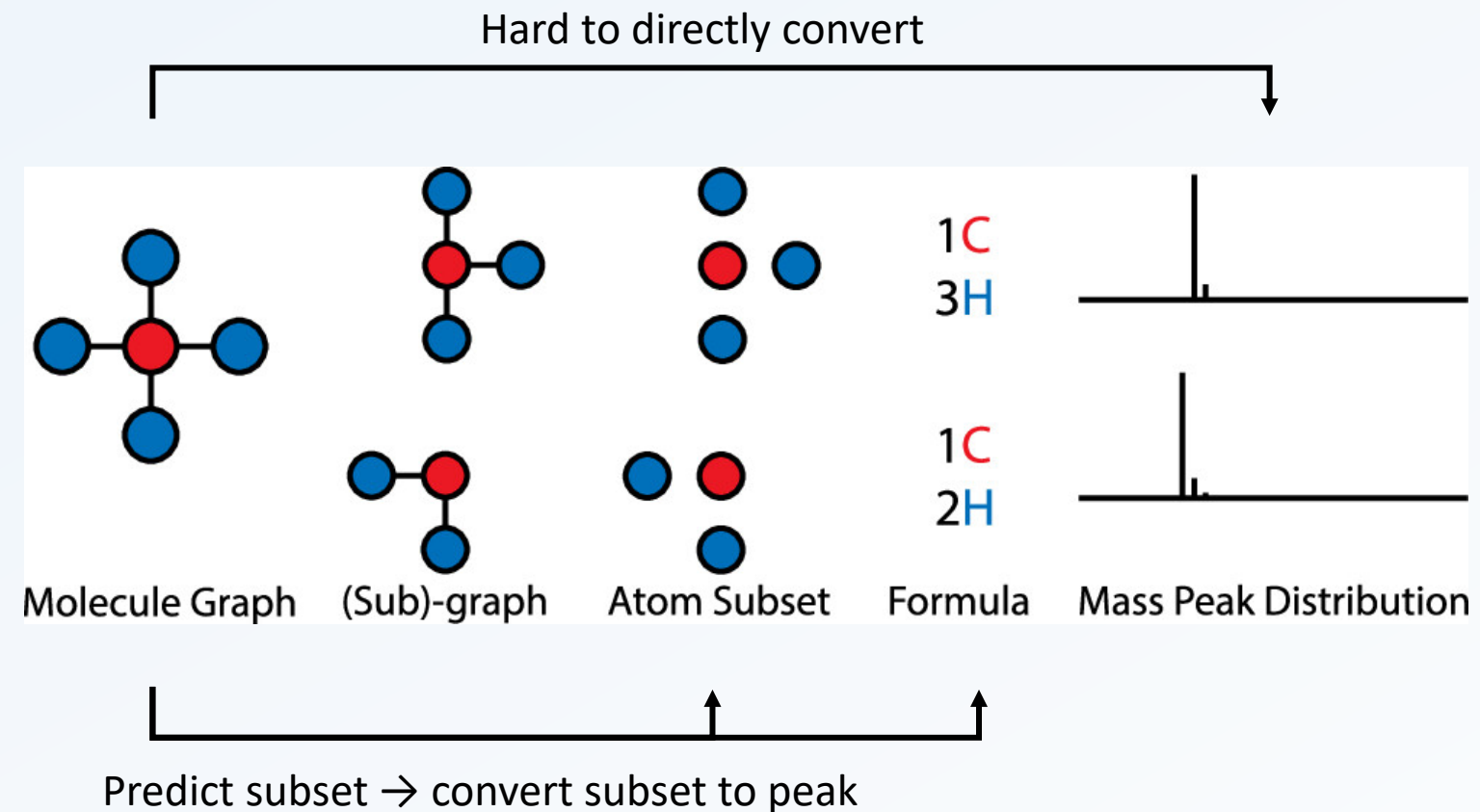


nth feature

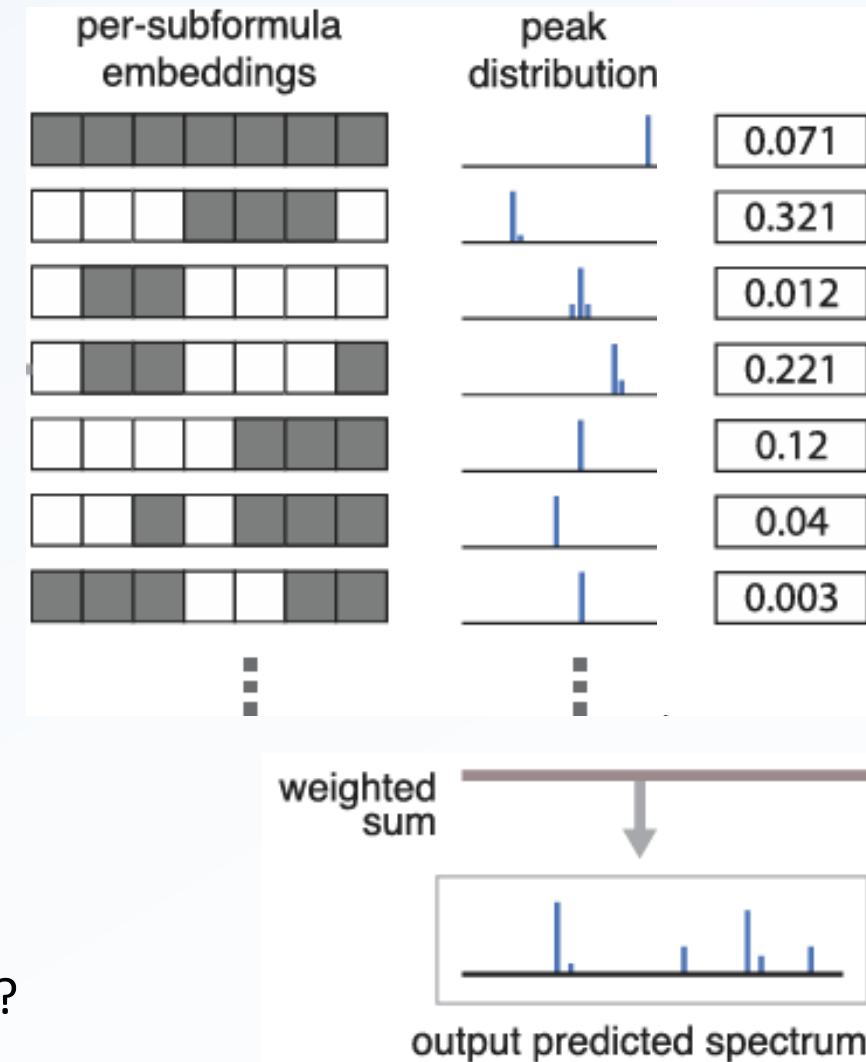


1. Molecular fingerprint vector can hardly capture features of big fragments
2. Spectra are not only intensity vector but collection of peaks of fragments

Rapid Approximate Subset-Based Spectra Prediction for EI-MS

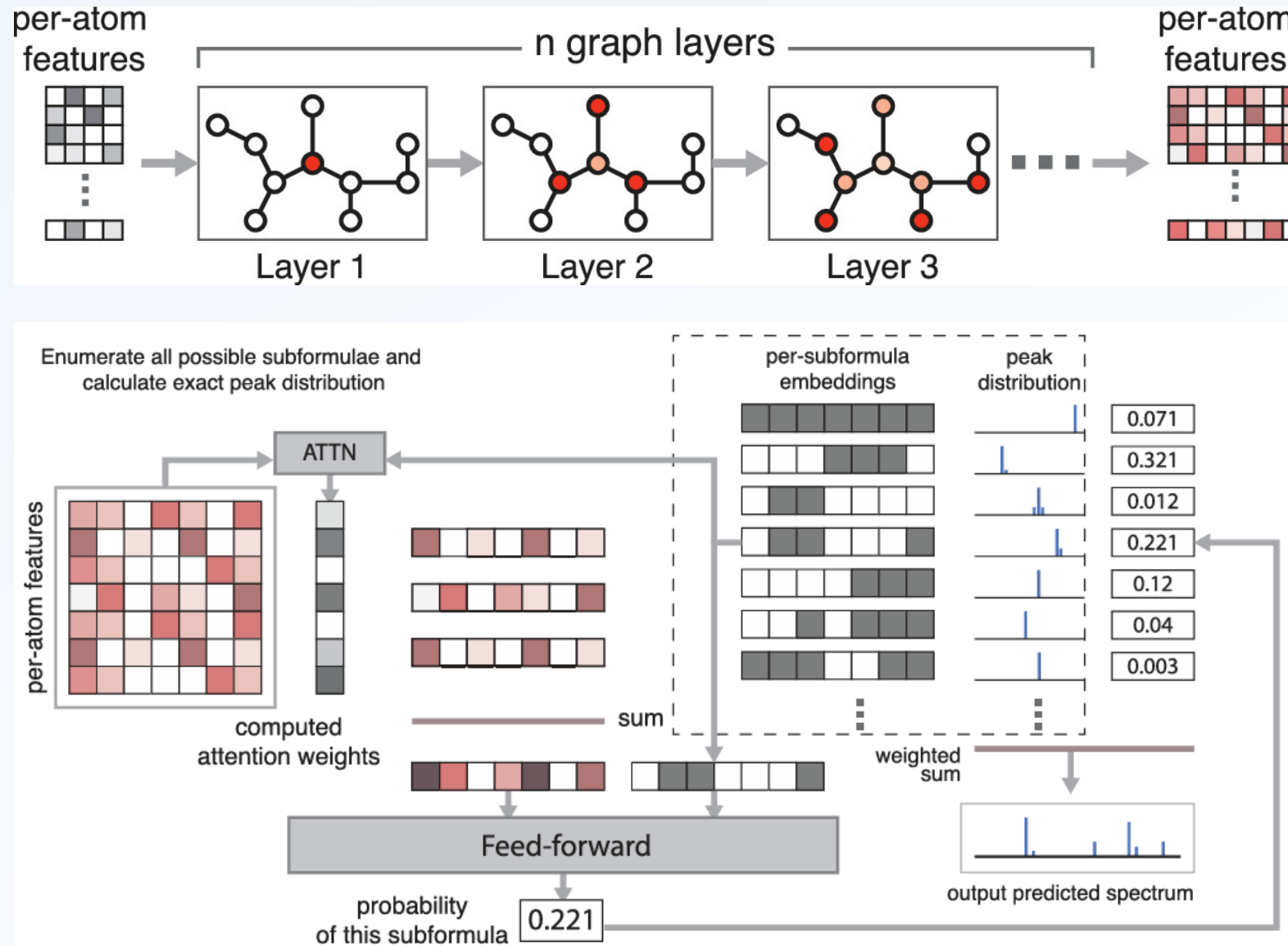


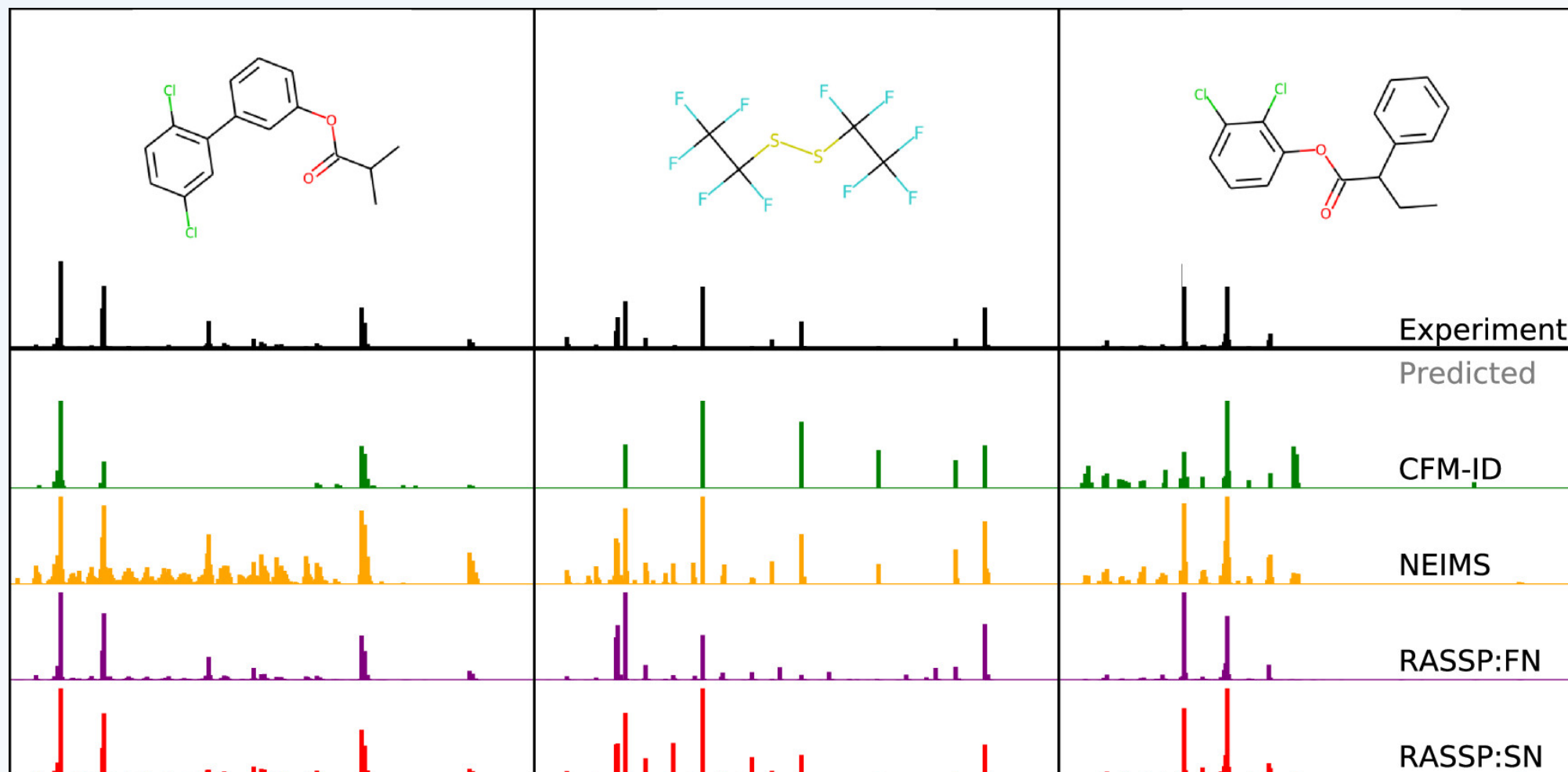
How to determine subsets in the molecule?



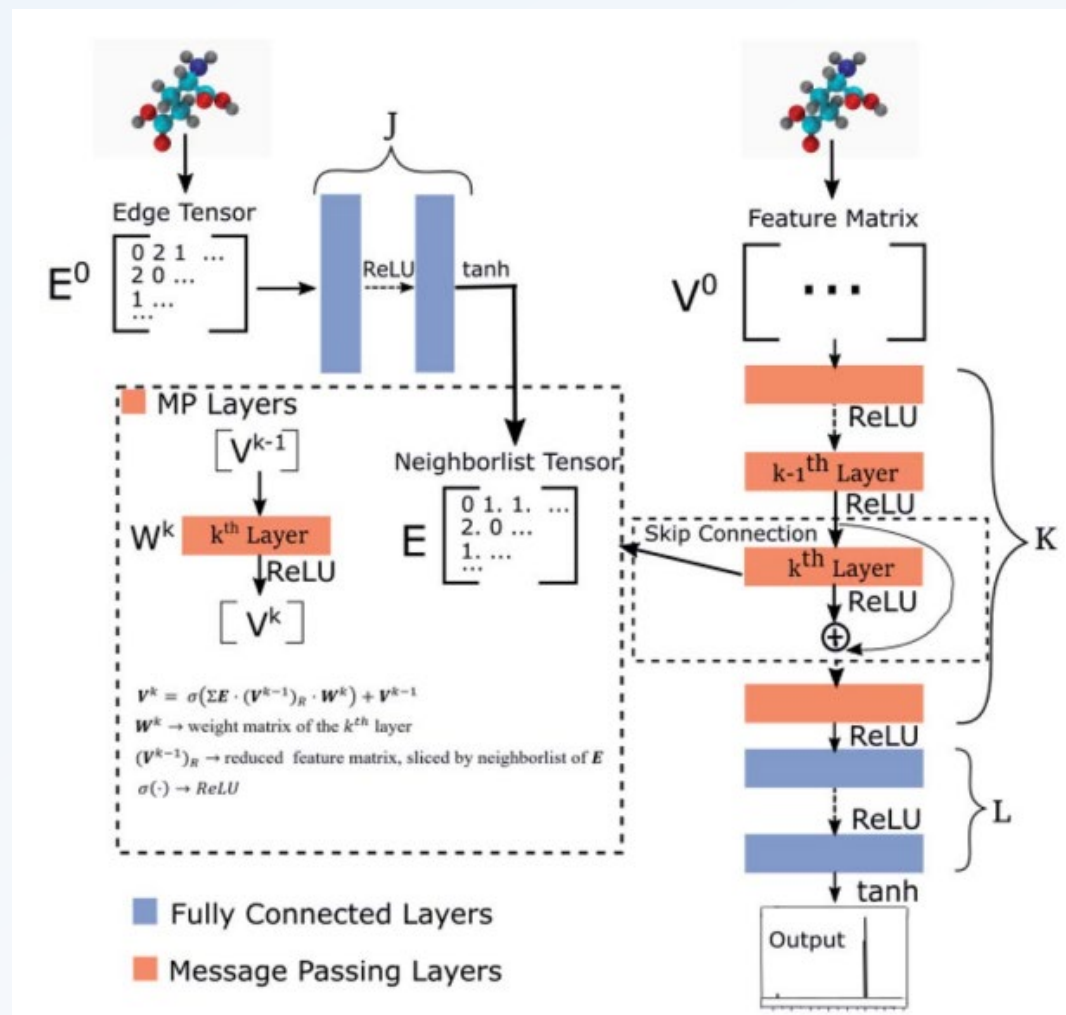
Key novelty

- ① extract atomic features by GNN
- ② find the most related atoms with subformula
- ③ calculate the peak distribution

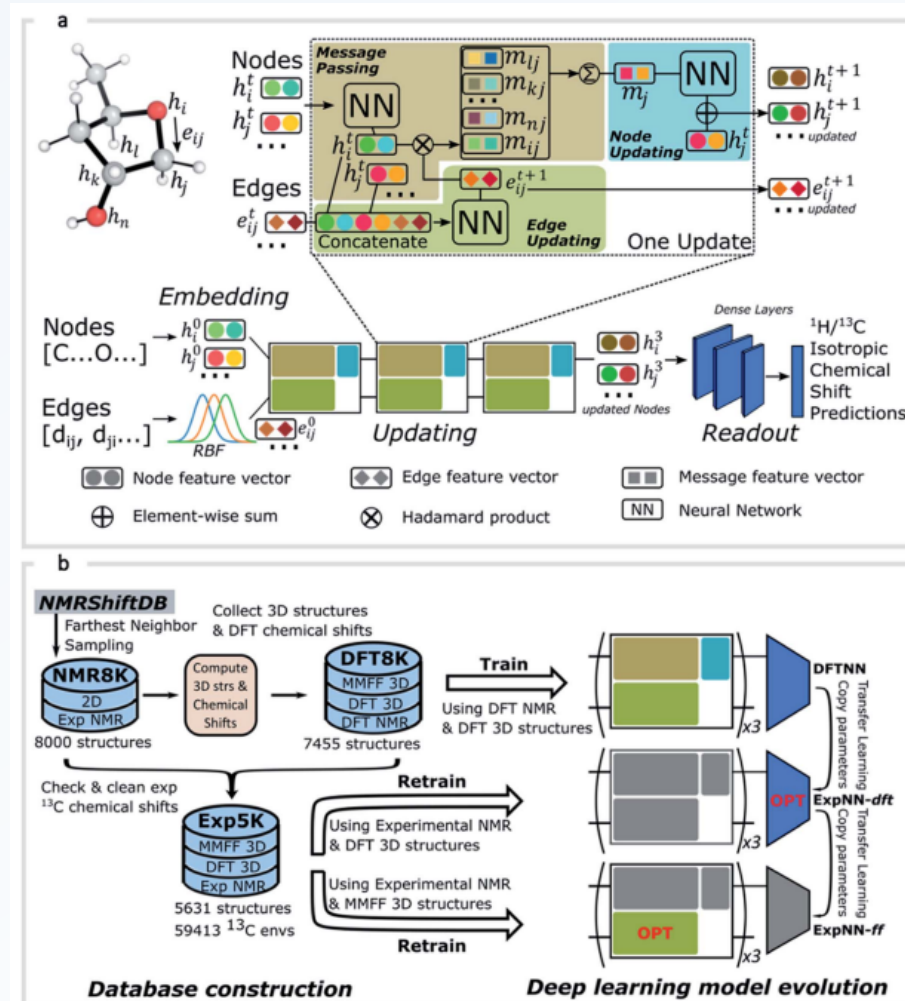




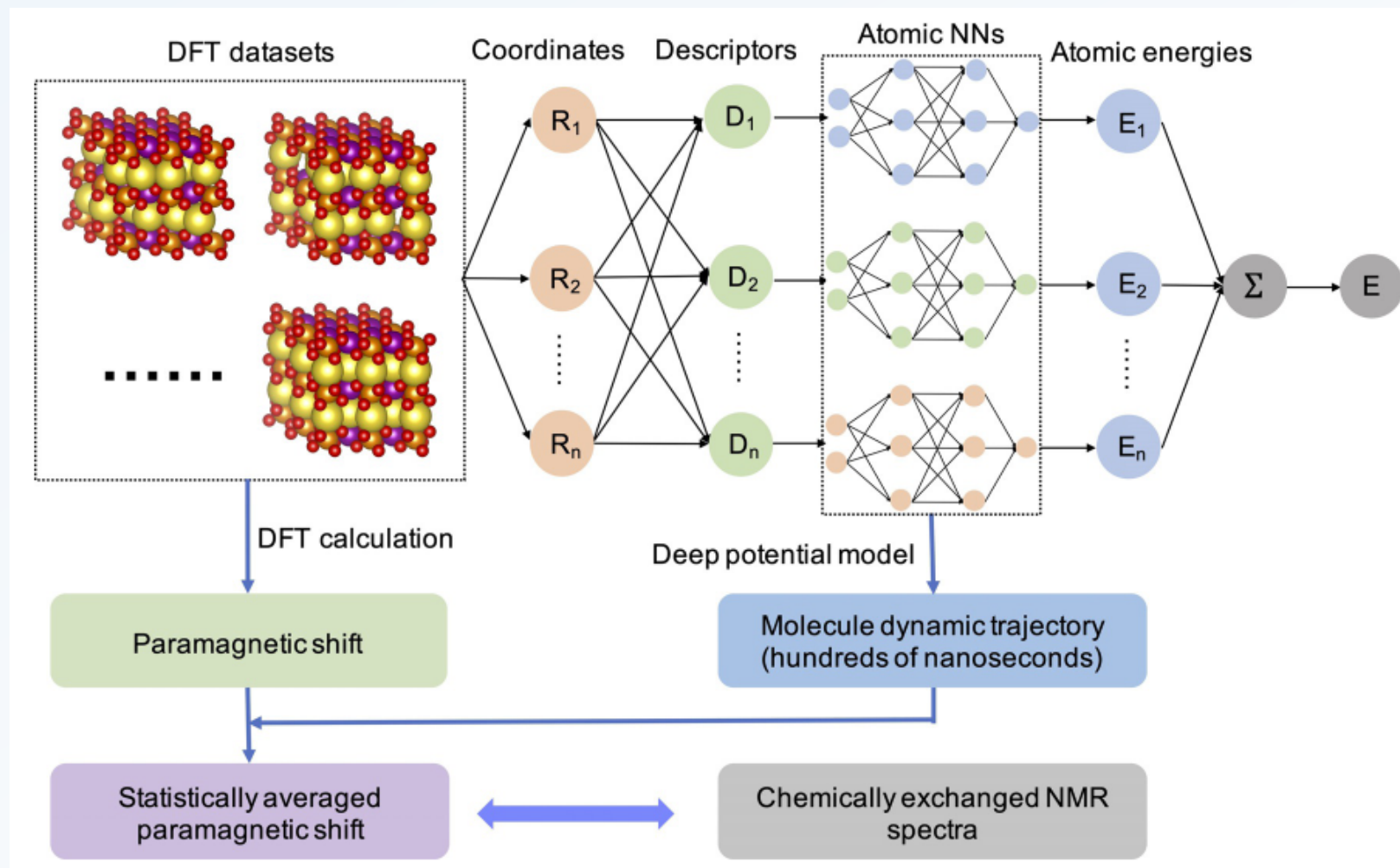
Developed methods (RASSP) performed better than previous ones



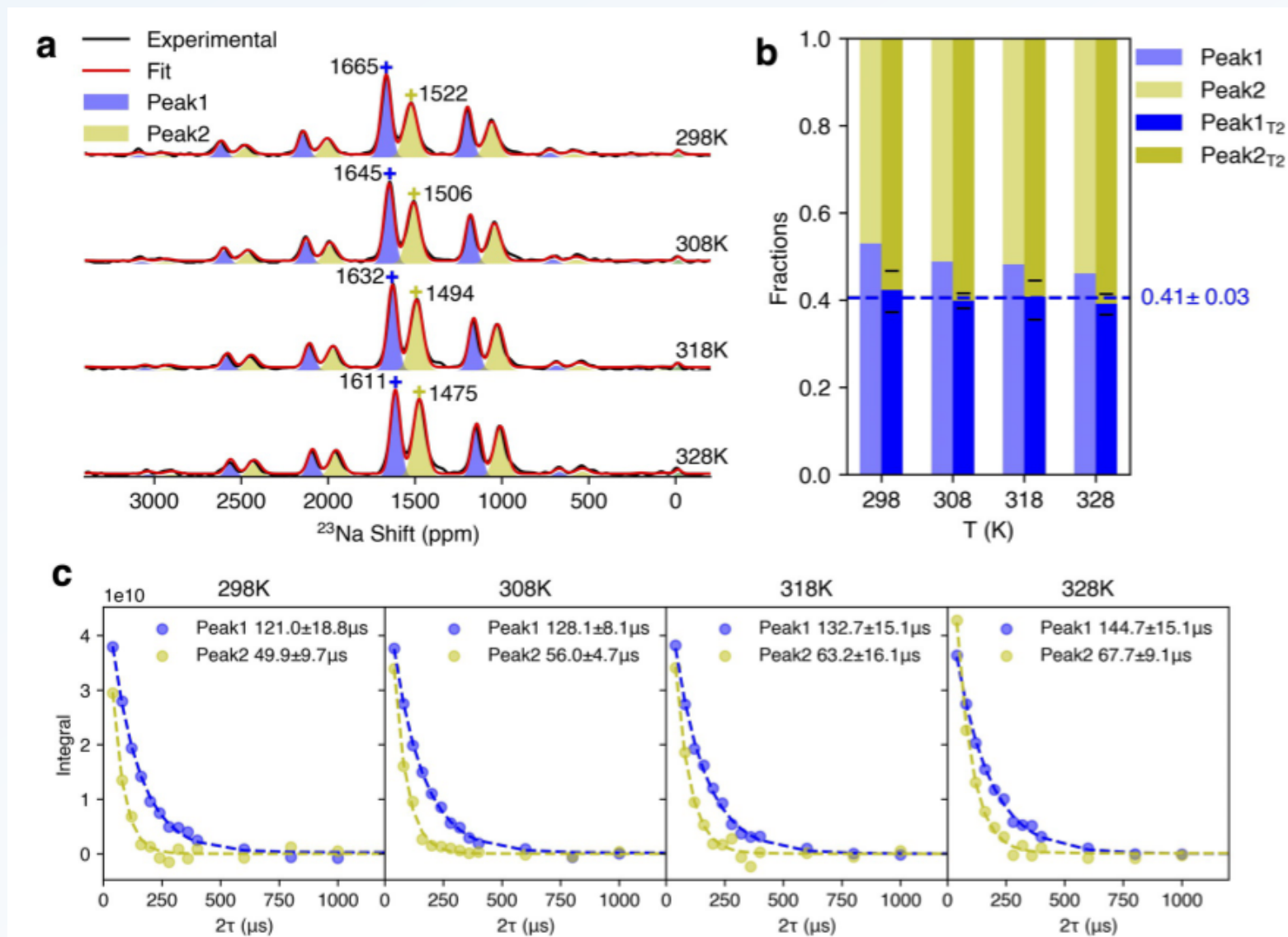
Chem. Sci., 2021,12, 10802-10809



Chem. Sci., 2021,12, 12012-12026

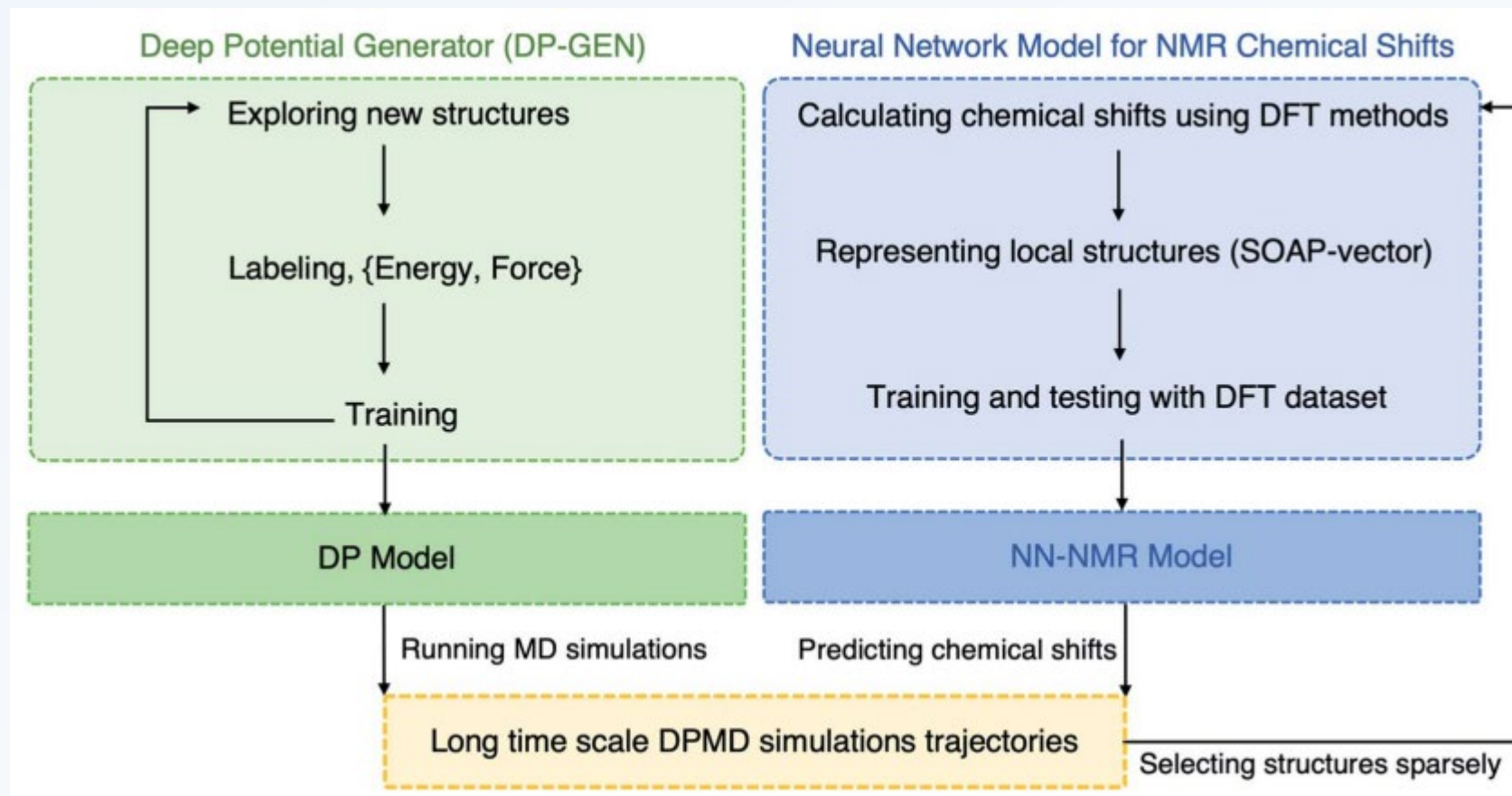


Angew. Chem. Int. Ed. 2021, 60, 12547 – 12553

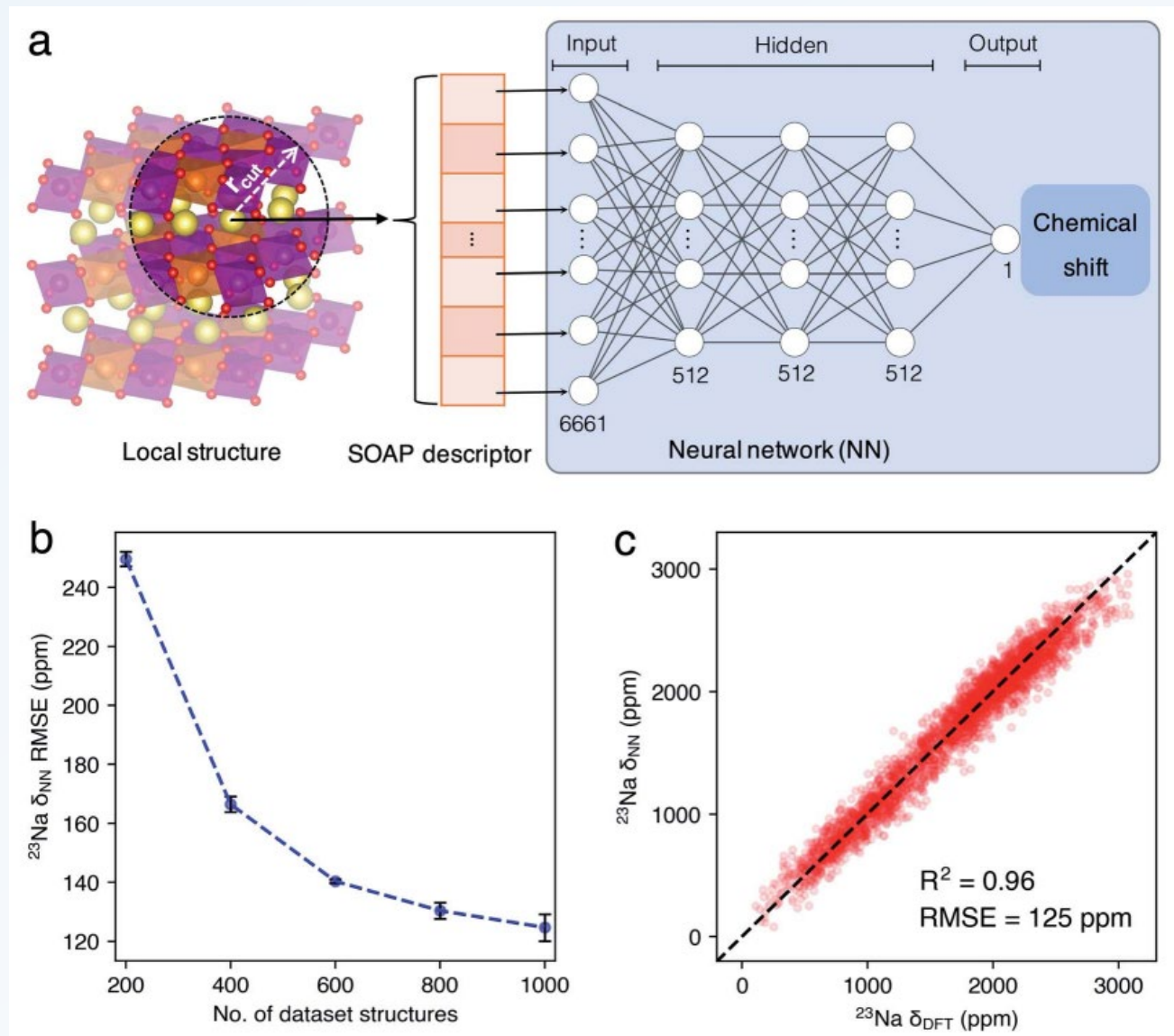


An MLP model for accelerating configuration sampling

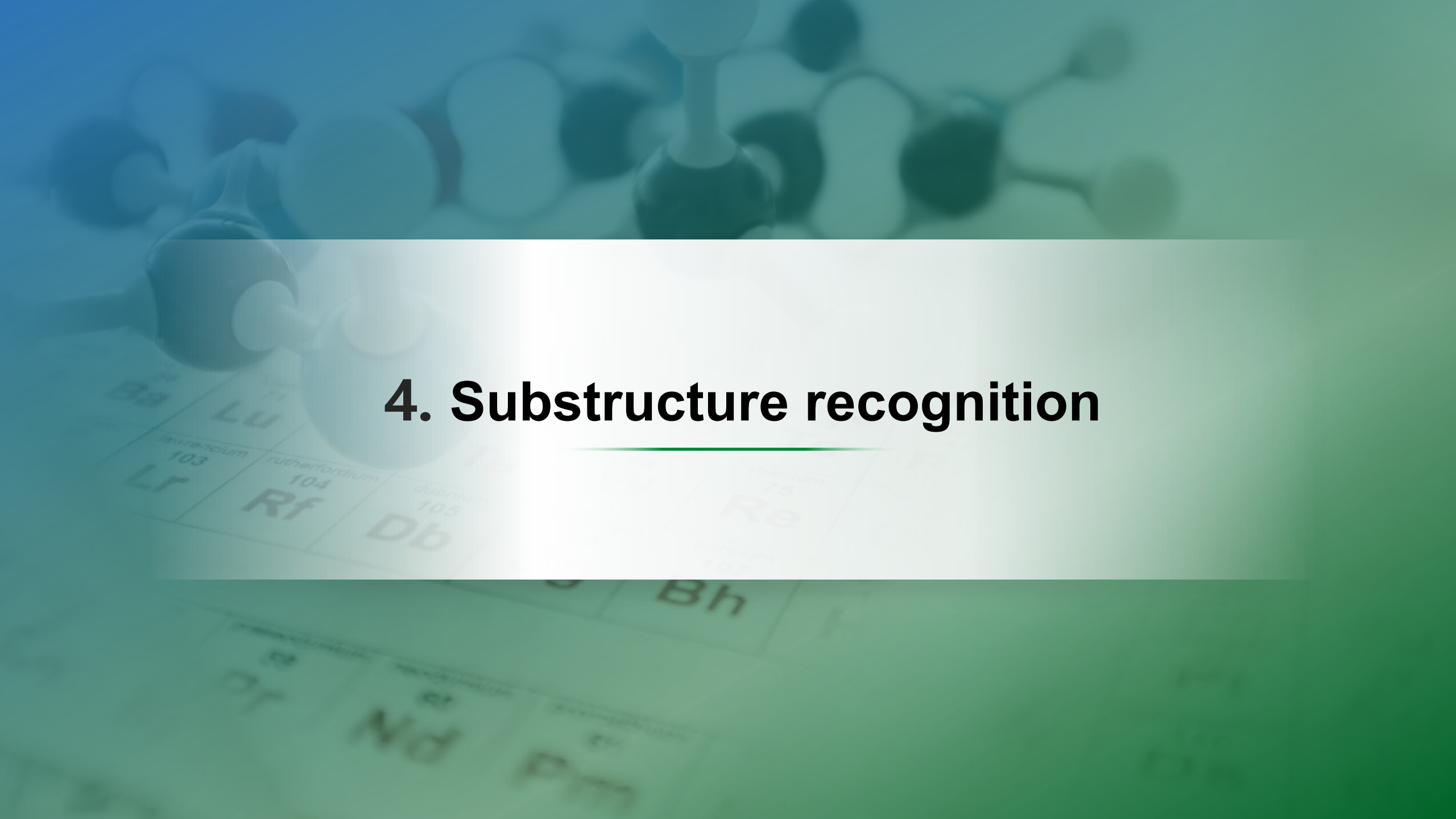
A Neural Network (NN) model for fast chemical shift prediction, named the NN-NMR model



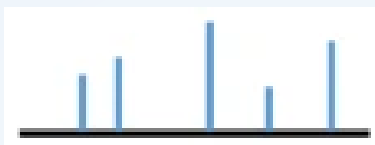
Chem. Sci., 2022,**13**, 7863-7872



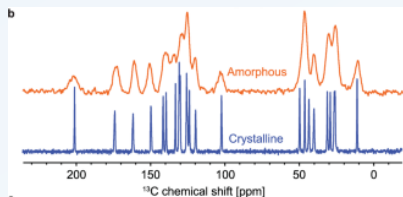
code



4. Substructure recognition

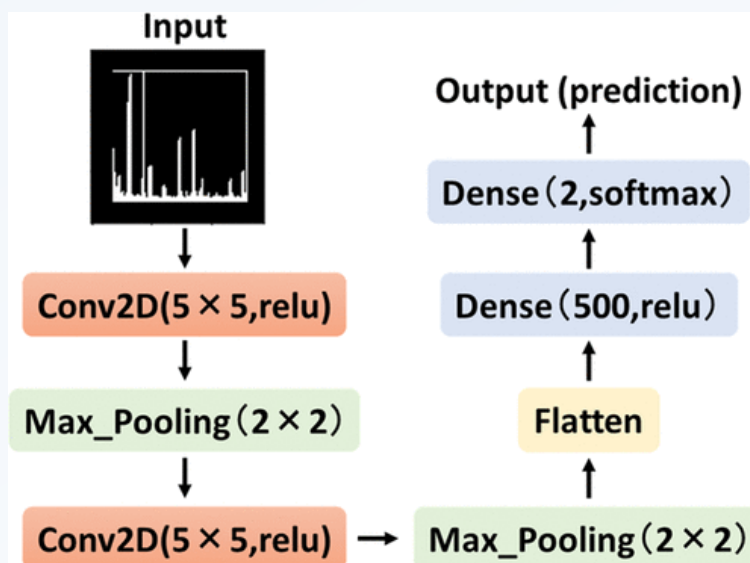


Pattern recognition



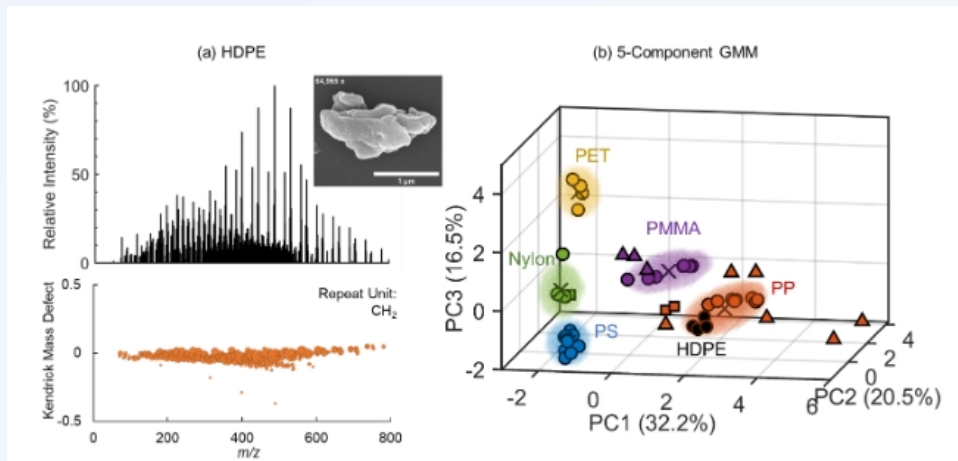
AI+MS\NMR

谱学分析



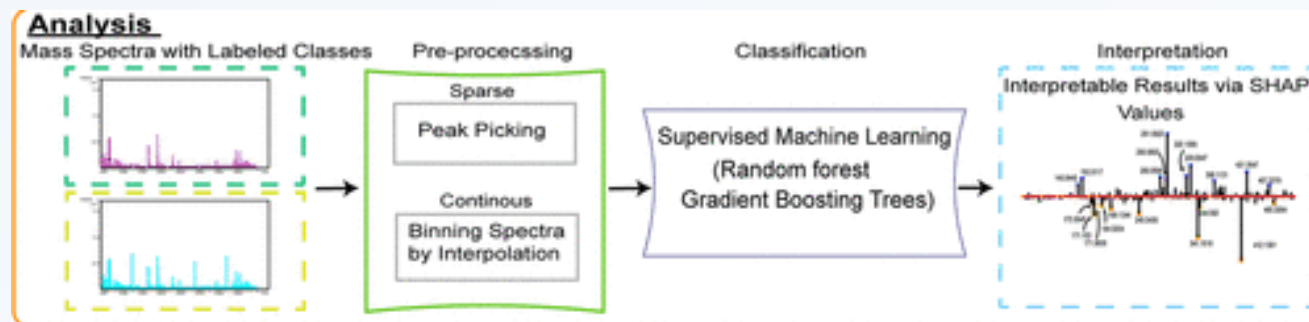
Traditional ML for classification
DL for classification

Traditional ML for classification



质谱+PCA快速筛选微塑料和纳米塑料

Anal. Chem. 2023, 95, 33, 12373–12382



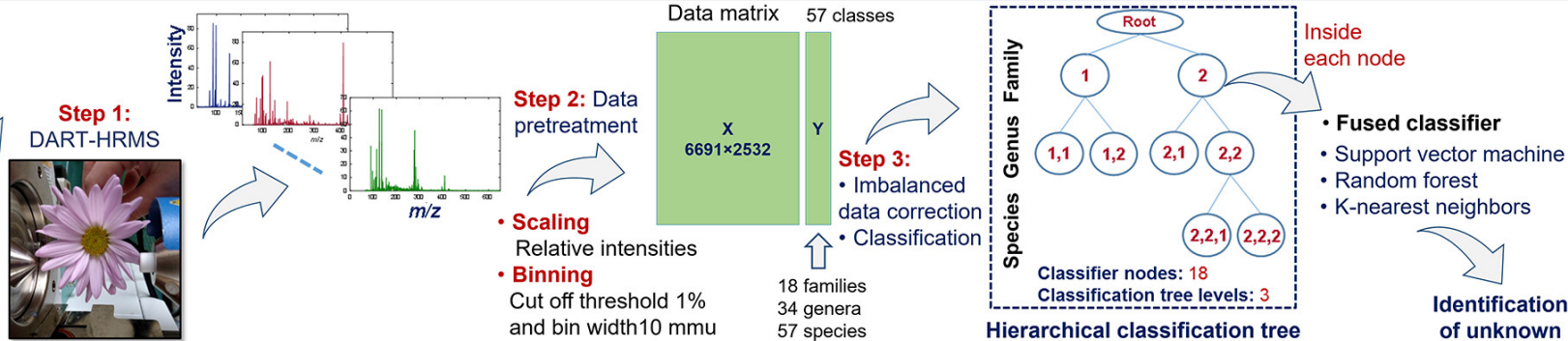
树模型的细胞分类

Anal. Chem. 2020, 92, 13, 9338–9347

A: Identification

Plant Materials

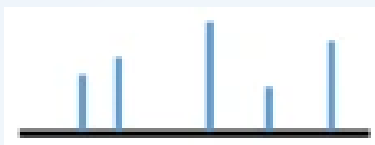
- Seed
- Flower
- Leaf
- Powder
- Liquid
- Stem
- Capsule
- Oil/extract



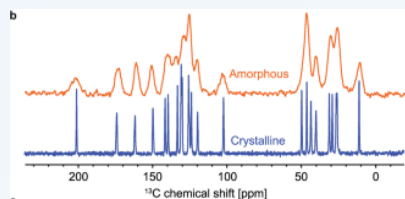
质谱+机器学习快速识别精神活性植物材料的物种

Anal. Chem. 2022, 94, 48, 16570–16578

Problems:
需要特征峰提取
对于大数据集效果不好

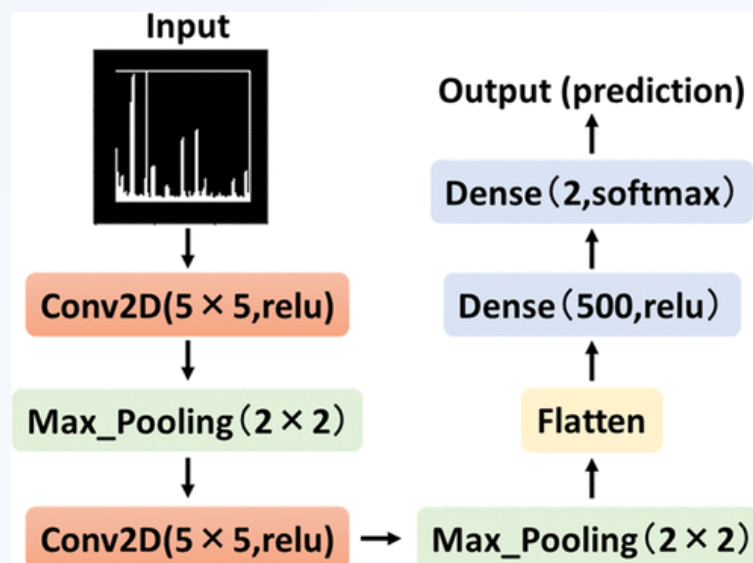


Pattern recognition



AI+MS\NMR

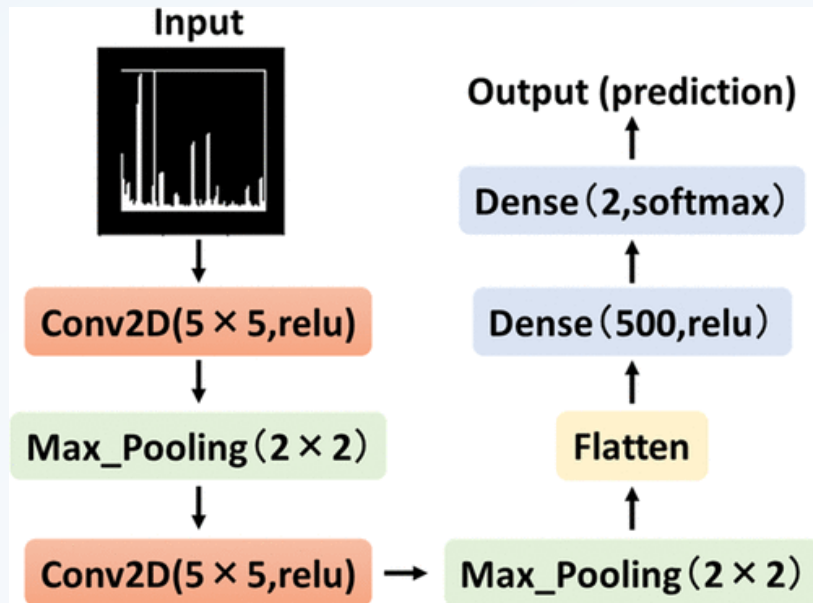
谱学分析



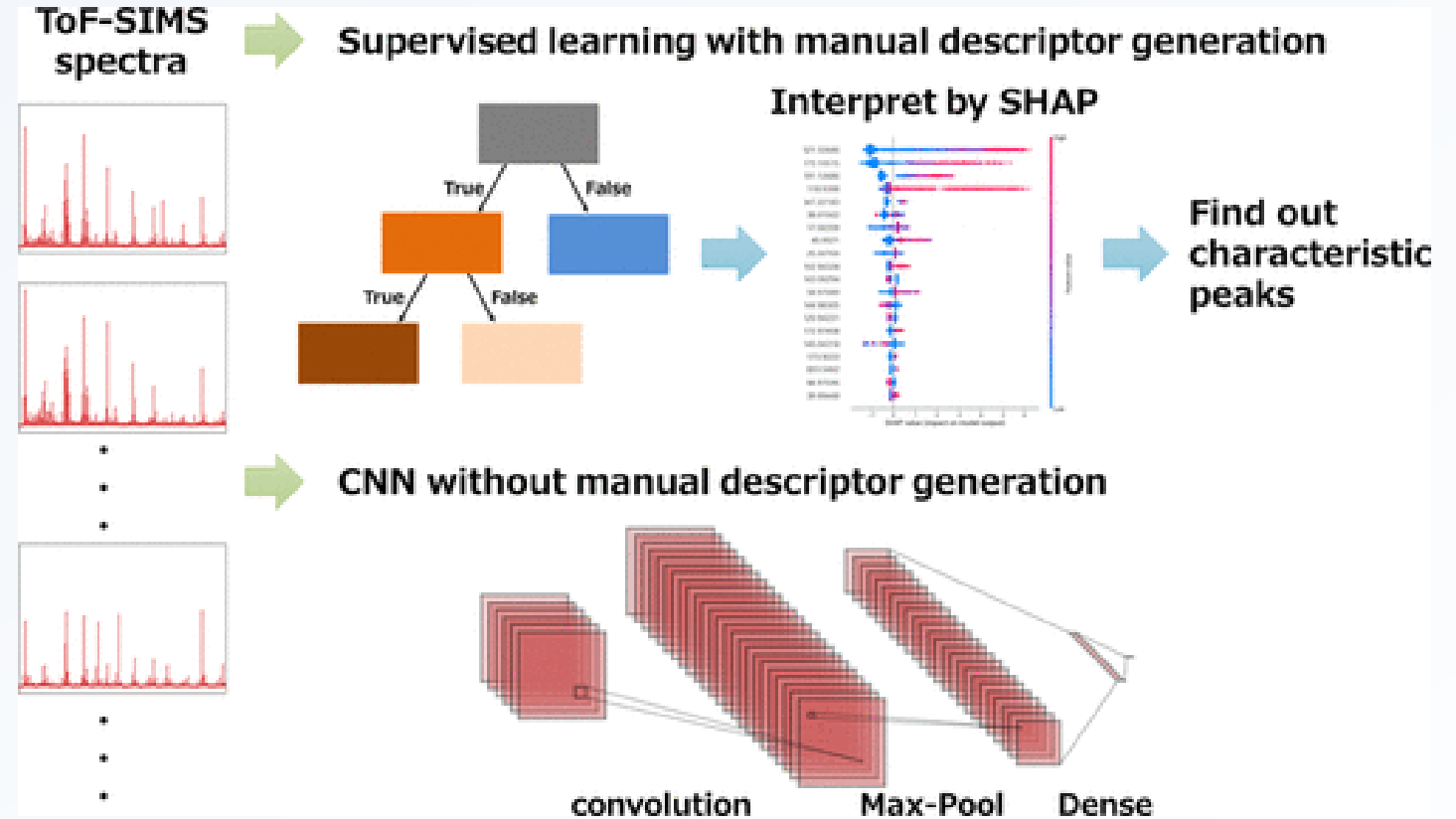
Traditional ML for classification
DL for classification

Advantages:

- Capability for bigdata
- No need to extract features



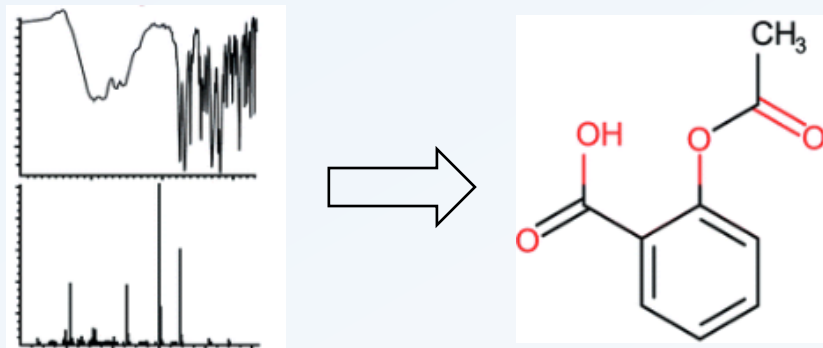
Identification of Silane Coupling Agents in Multicomponent Films



Anal. Chem. 2022, 94, 5, 2546–2553

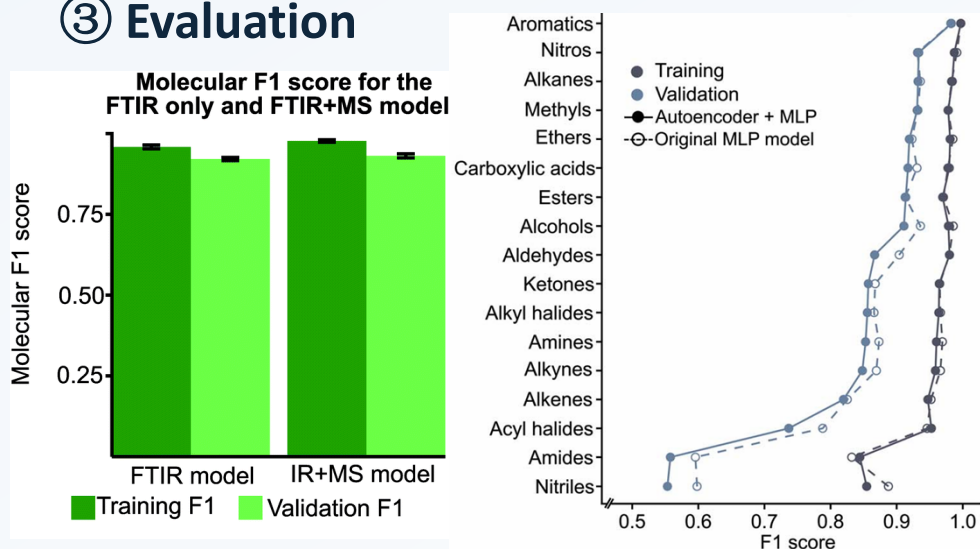
MLP for identifying functional groups (多光谱协同)

① Background

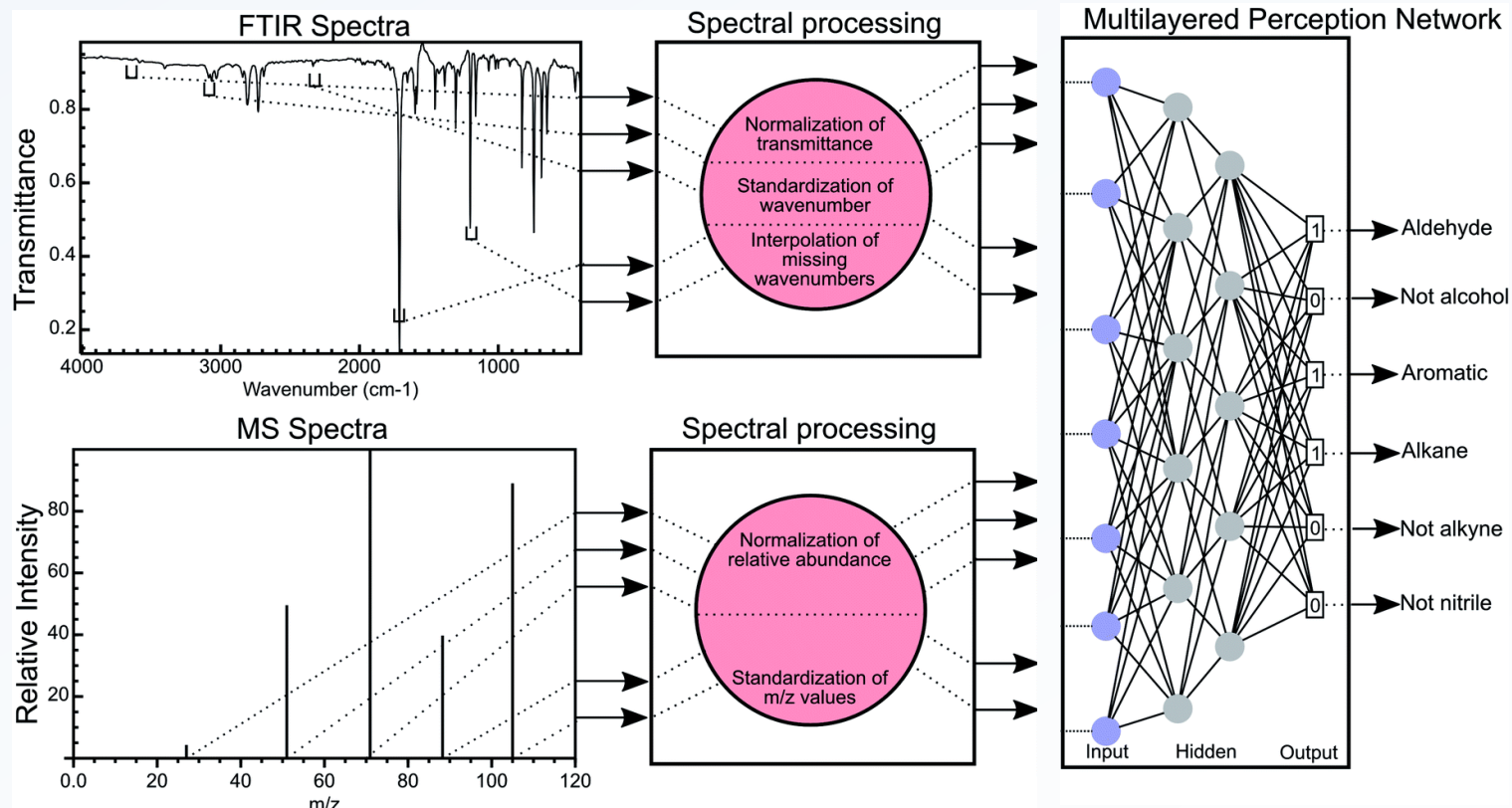


Given the IR/mass spectrum, how to inverse the structure (functional groups)?

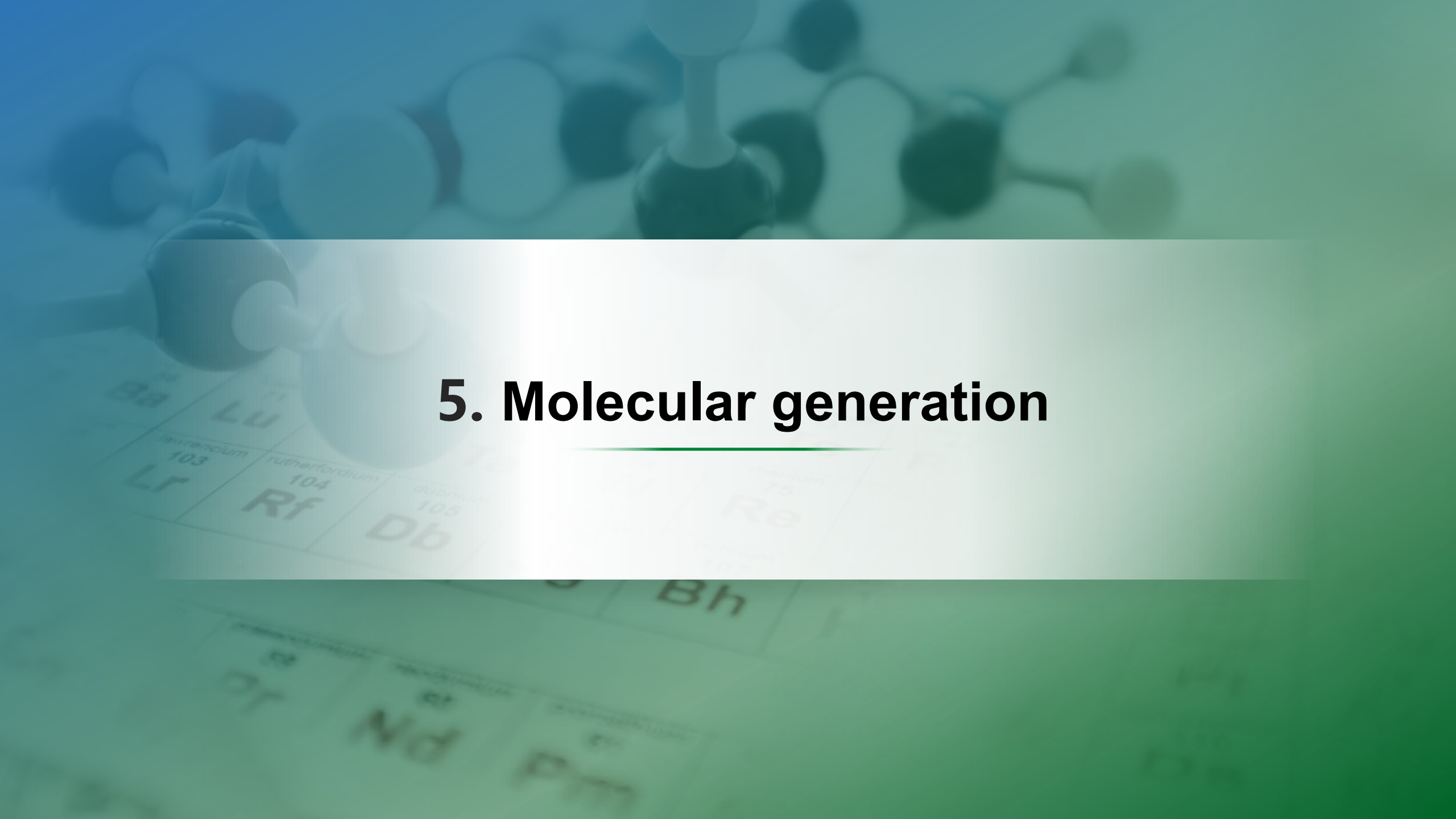
③ Evaluation



② Method

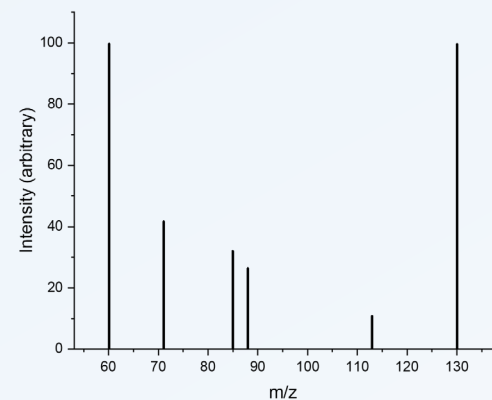


MLP was used to **encode(represent) spectra** and predict potential functional groups

The background of the slide features a blurred molecular model with white and grey spheres connected by rods, overlaid on a periodic table. The periodic table is partially visible, showing elements like Lu, Rf, Db, Re, Bh, Nd, and Pt. A semi-transparent white rectangular box is centered over the image, containing the section header.

5. Molecular generation

De novo generation

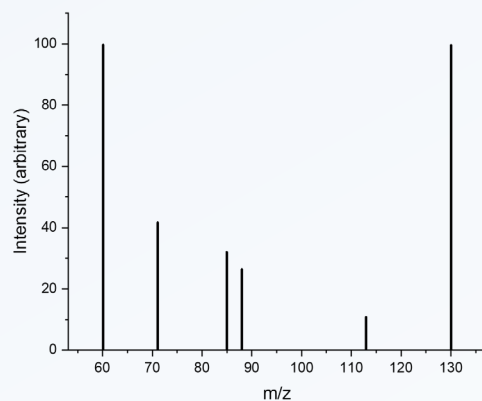
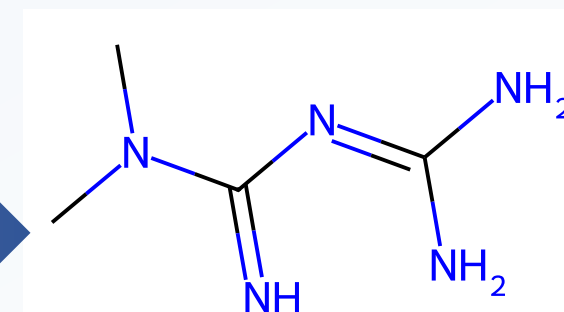


Unknown spectrum

Library matching

Spectral prediction

De novo generation



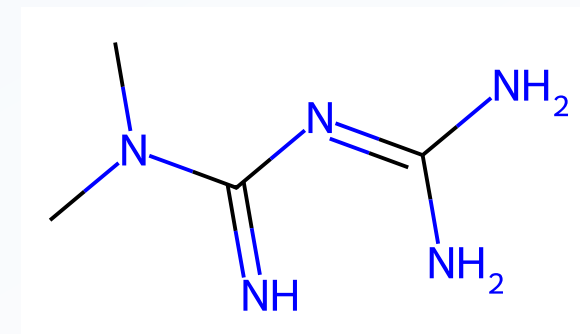
Unknown spectrum

Input



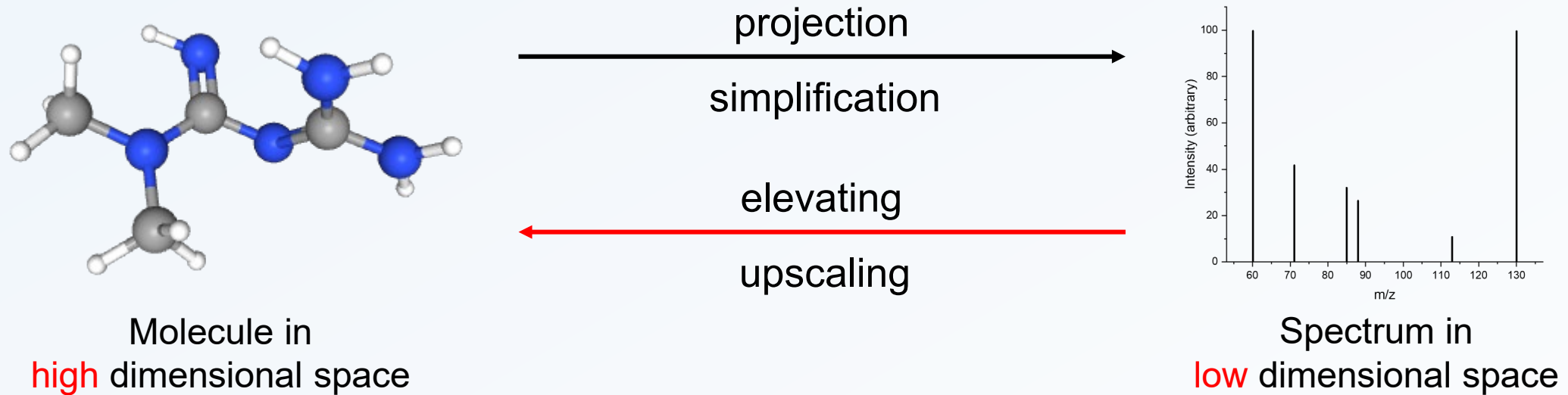
Algorithms

Output



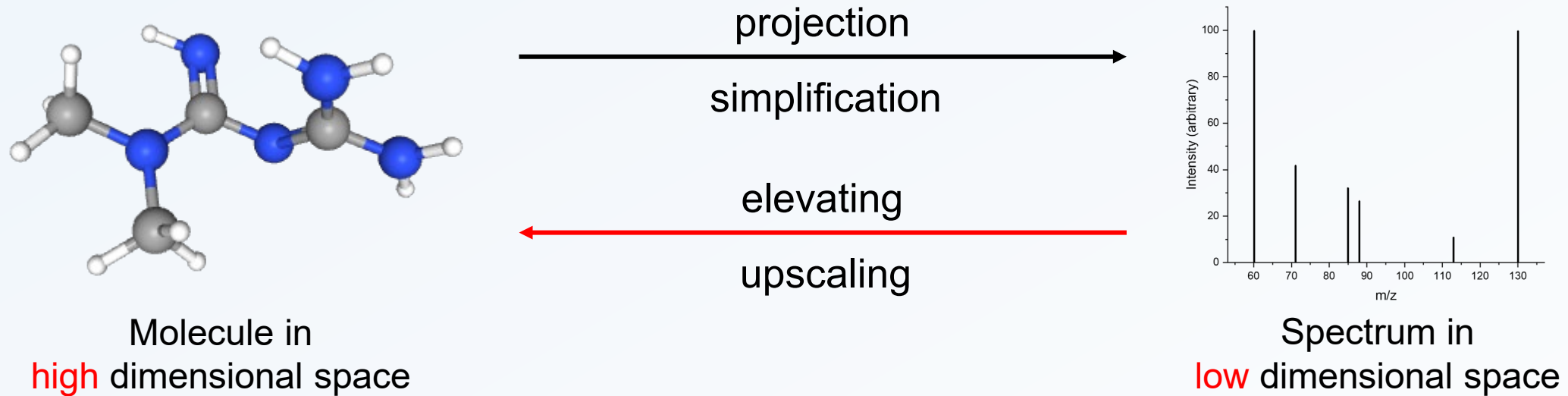
Predicted molecule

Challenges of *de novo* generation



1. How to make it possible to convert spectra to molecules?

Challenges of *de novo* generation



1. Find pattern within massive data
2. Combine the prior knowledge with spectra

De novo generation from MS by basic Transformer



21million molecules

$\xrightarrow[\text{fragmentation method}]{\text{In-silico}}$

21 million *in-silico* spectrum-molecule pairs

$[m/z_1, m/z_2, \dots, m/z_n]$

$[68.31, 53.02, \dots, 0.11]$

	0	0	...	0
	0	0		0
	⋮	⋮		⋮
	⋮	⋮		1
6831 th	1	1	5302 th	11 th
	⋮	⋮		⋮



Top-10 recall: 53% (93 test samples)

1. More data

2. More prior information

Improve performance with more data

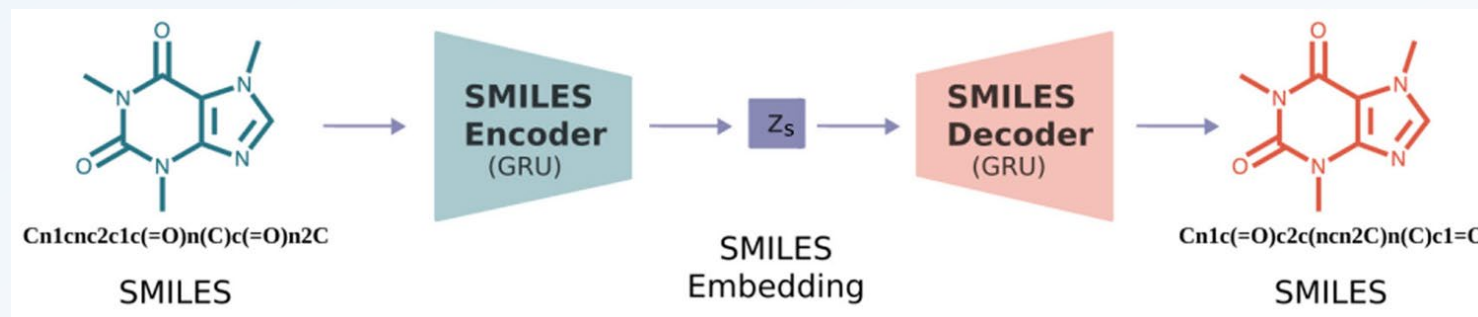
1 million spectrum-molecule pairs



135 million molecule

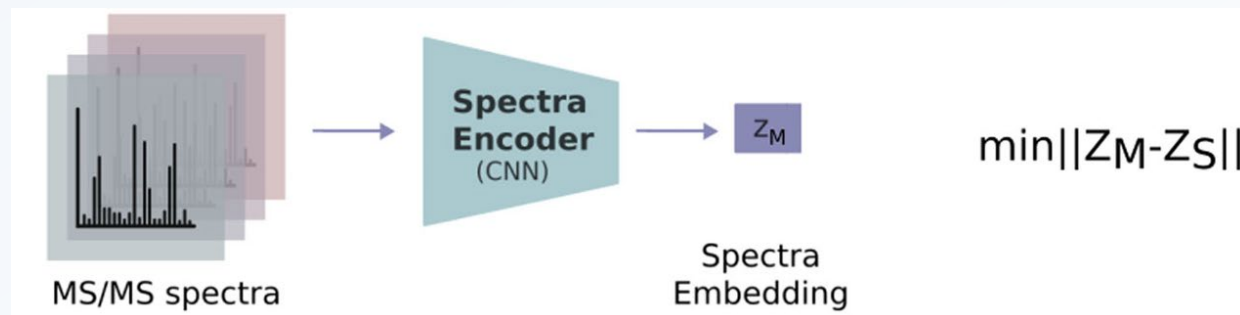


Step 1: learn how to generate a SMILES string



Learn the syntax of SMILES

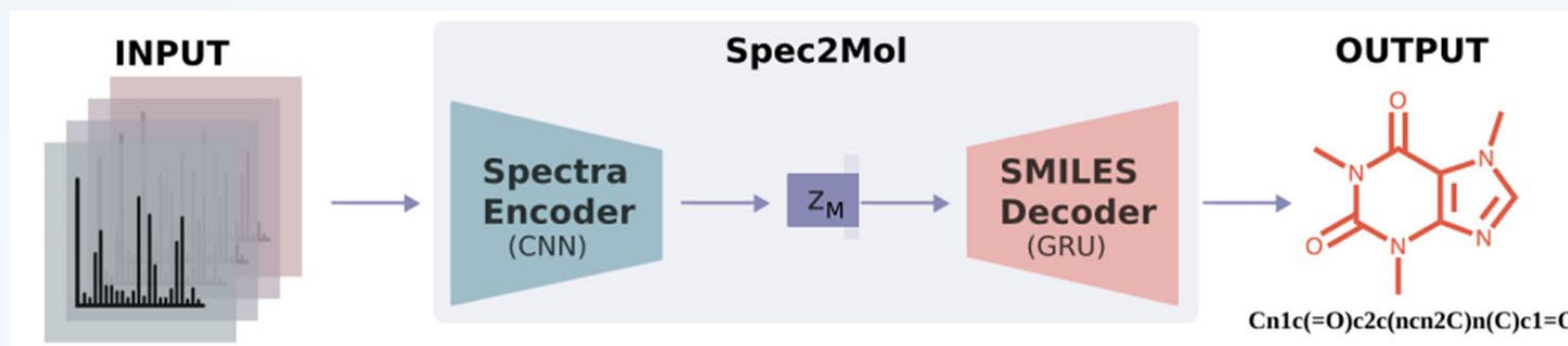
Step 2: learn how to extract spectral features



Force encoder extract features related to *de novo* generation.

Results

SIRIUS: fragmentation tree-based annotation



Method		Fngp _{cosine}
SIRIUS	Max	0.49
	Avg	0.33
Spec2Mol	Max	0.49
	Avg	0.34

$$\text{Fngp}_{\text{cosine}} = \cos \langle \text{Predict}, \text{True} \rangle$$

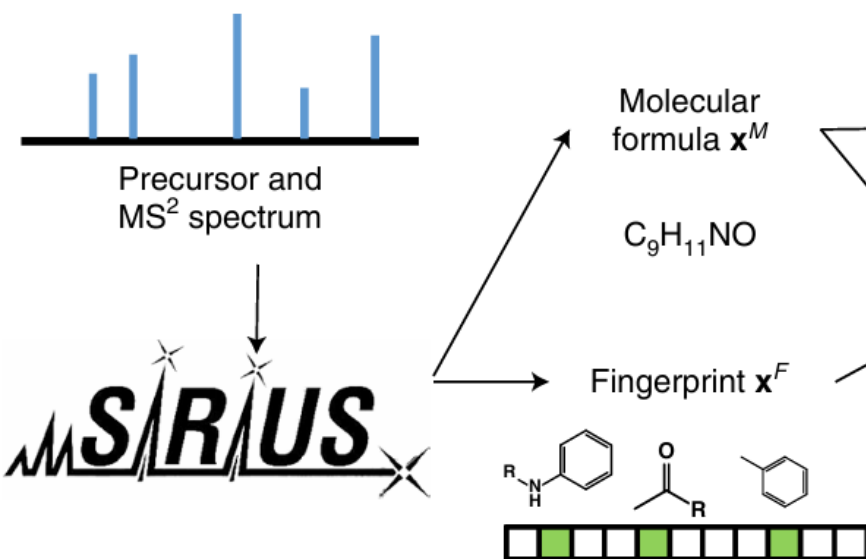
- On the test set of 947 cases, SIRIUS found the correct formula for about 98% of the test cases while it found the correct structure for about 67%.
- Spec2Mol only achieved **comparable performance when SIRIUS did not find an exact match.**

Incorporate chemical information

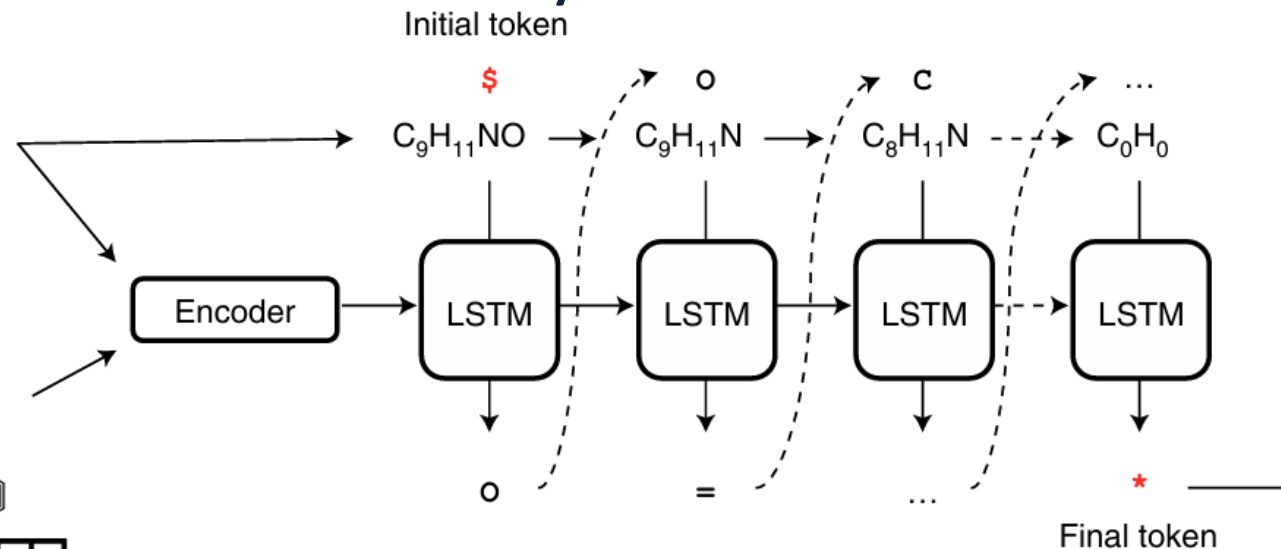
1 million spectrum-molecule pairs



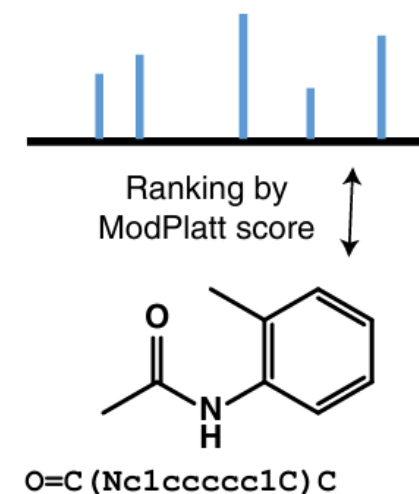
1. prepare initial state



2. decode initial state by RNN

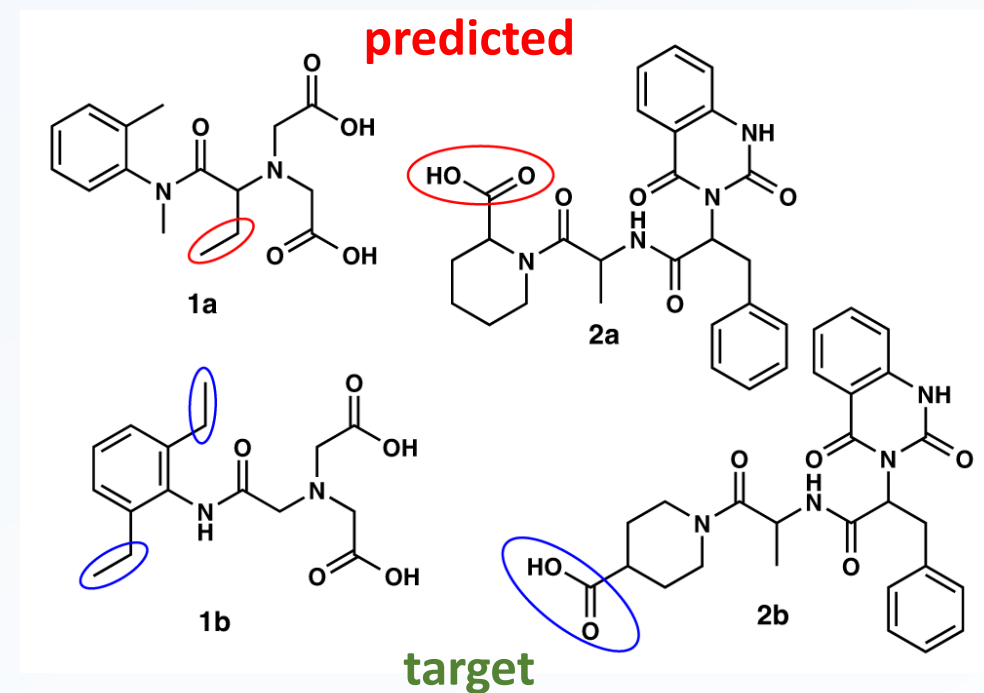
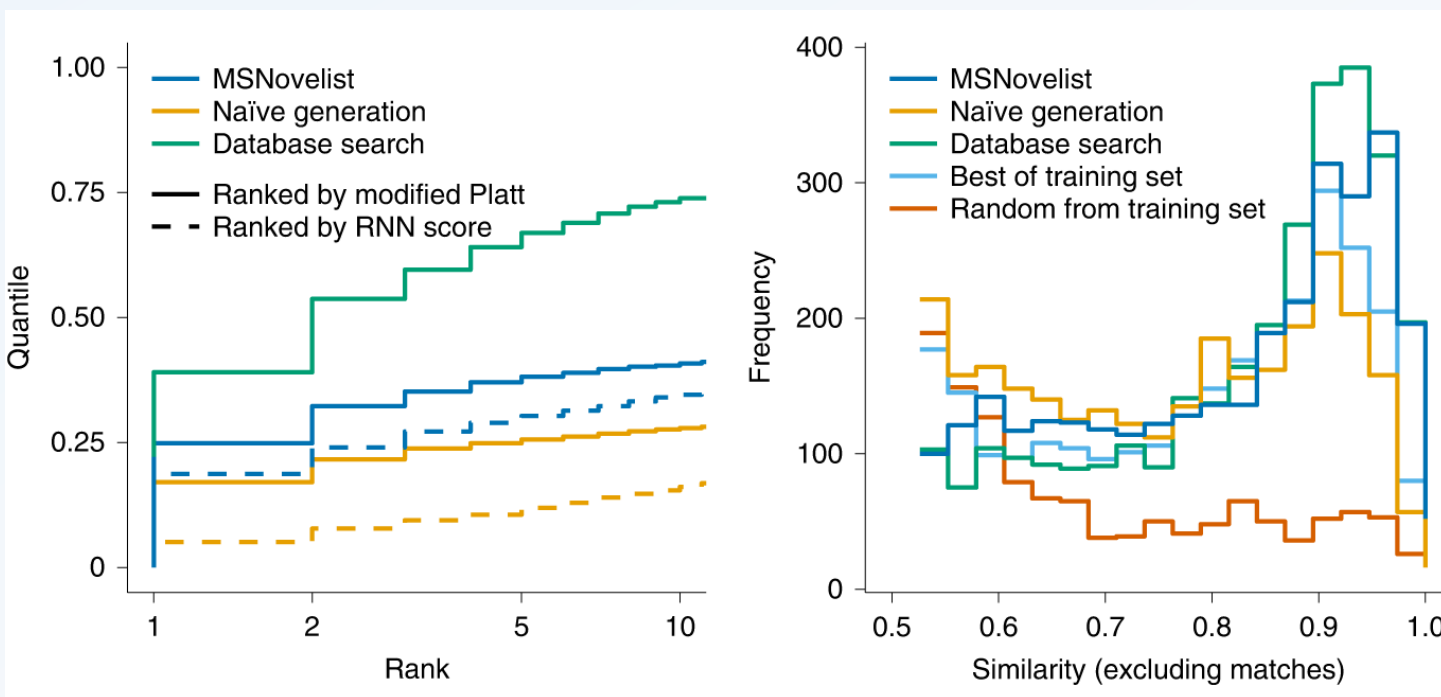


3. evaluation



independent of spectral libraries

Results



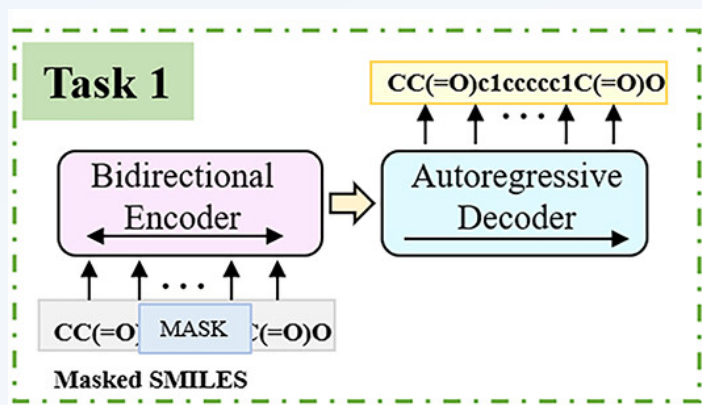
On the test set of 3863 samples, MSNovelist achieved Top-1 recall of 25%

More data, more information

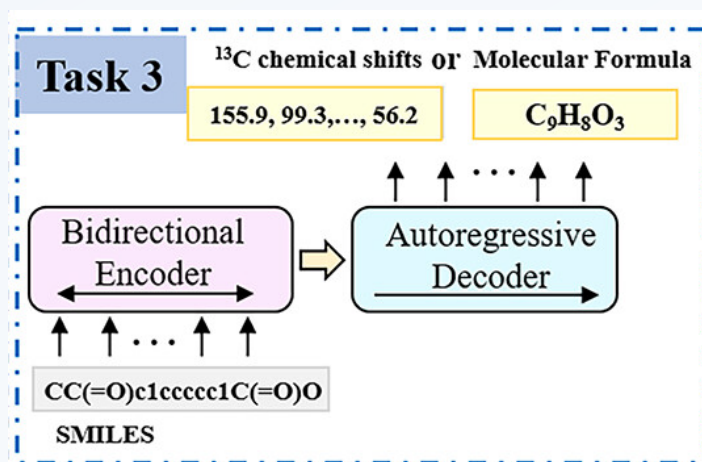


Pre-training with 360 million spectrum-molecule pairs

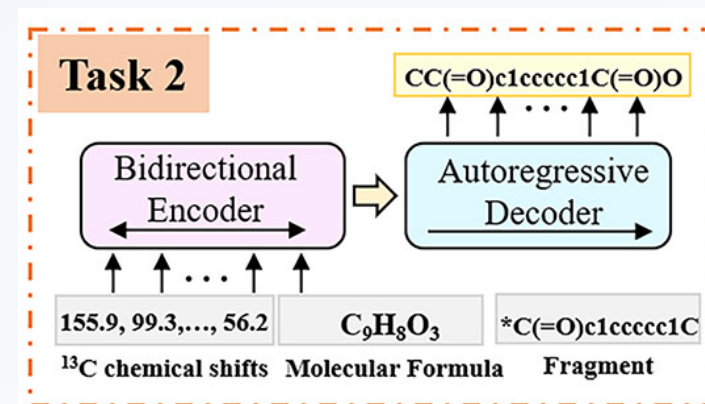
X. Wang et al. Anal. Chem. 2023, 95, 5393



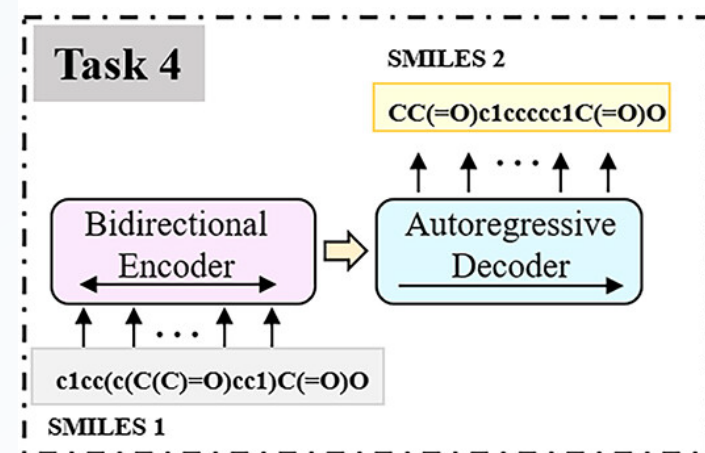
“Cloze test”



Predicting ¹³C NMR and formula by SMILES

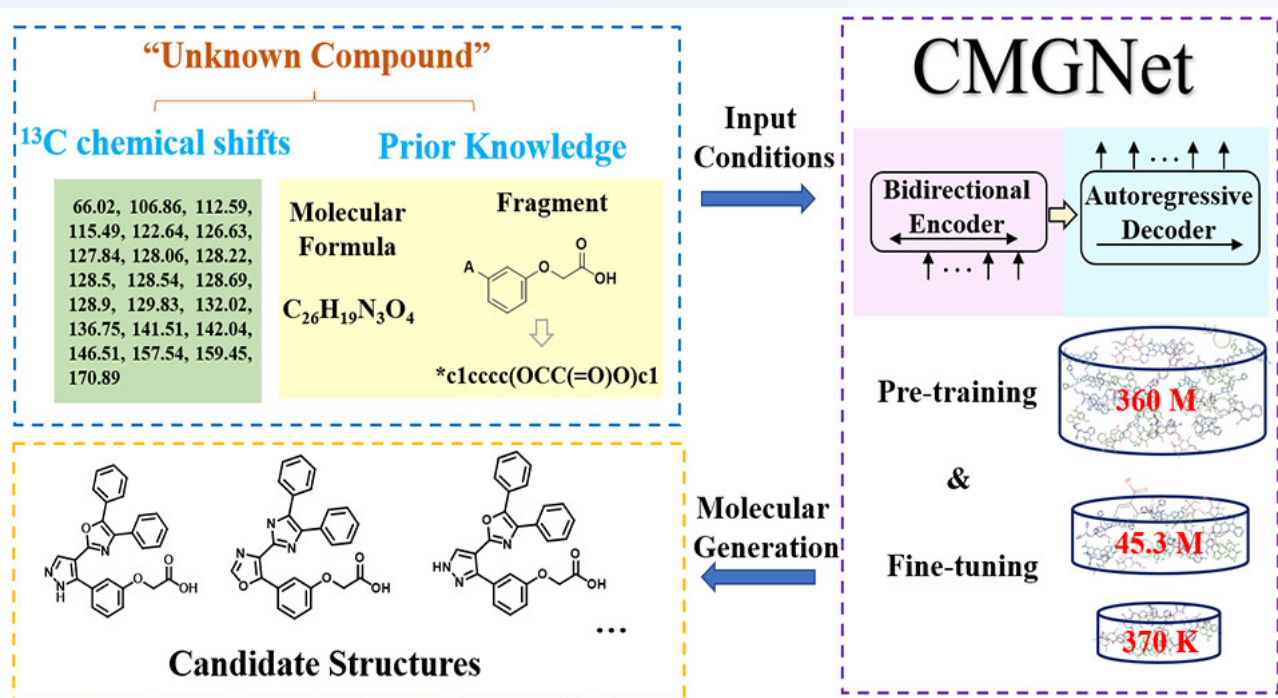


Predicting SMILES by ¹³C NMR, formula and fragment



Generating an equivalent SMILES representation

Results



More data and prior information brings higher accuracy

<https://drugflow.com/>

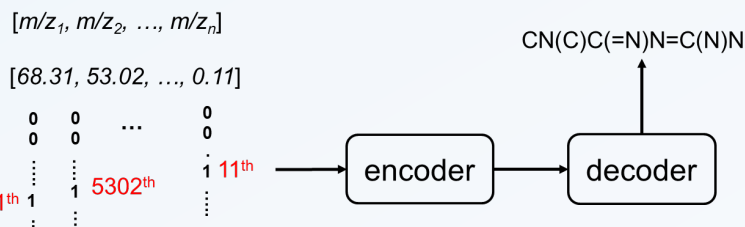


碳硅智慧
CarbonSilicon AI

conditions	validity	recovery (%)				
		top 1	top 5	top 10	top 50	top 100
NMR	98.66	32.91	42.77	48.29	60.78	65.01
NMR + MF	94.52	53.23	66.31	72.35	83.38	85.94
NMR + MF + MinFrag	93.85	57.60	69.82	75.39	85.61	87.84
NMR + MF + RandFrag	89.76	69.93	79.21	83.23	90.56	92.13
NMR + MF + MaxFrag	82.24	88.64	92.62	94.17	97.07	97.69

Summary

First publication about spectrum-to-structure



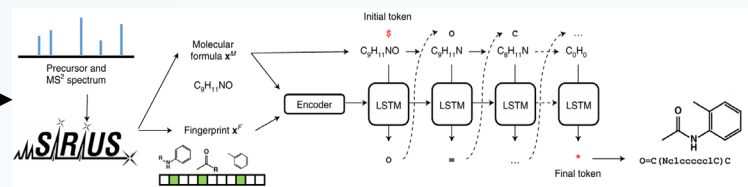
D. B. Kell, et al. Biomolecules 2021, 11, 1793.

More data



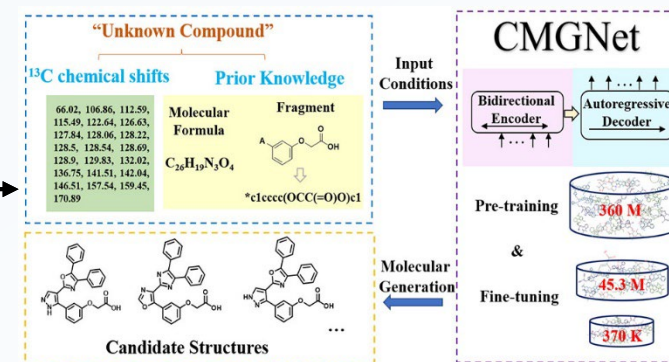
L. E. Kaviraki et al., Commun. Chem. 2023, 6, 132

More prior information



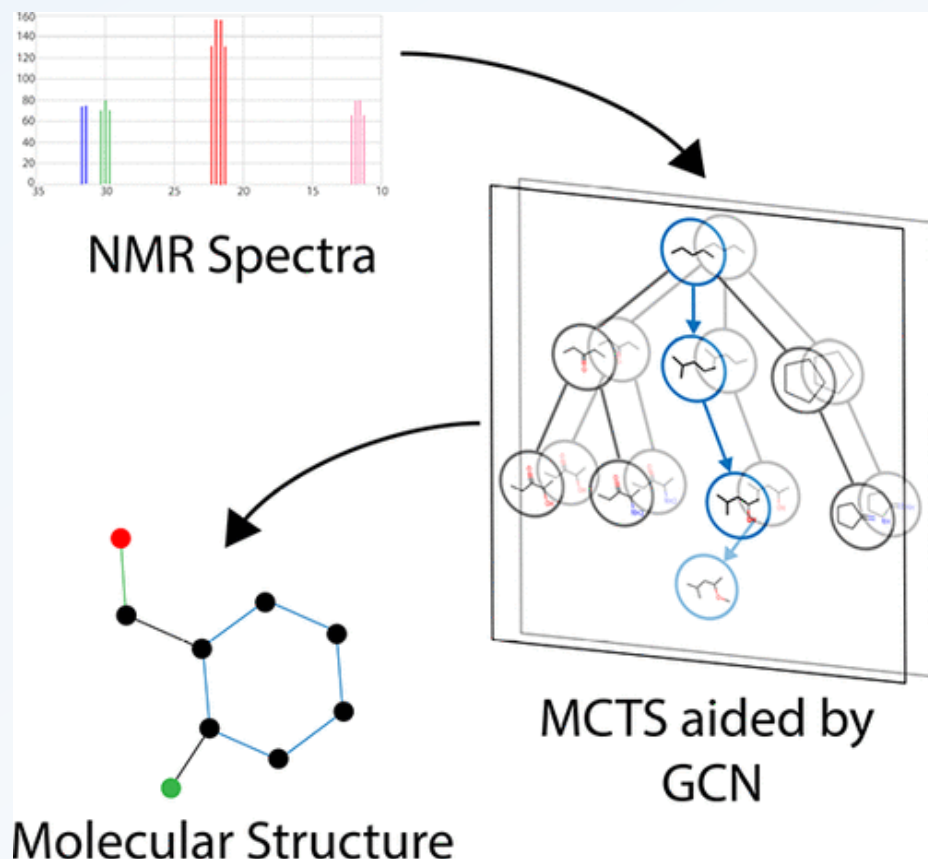
N. Zamboni et al. Nat Methods, 2022, 19, 865

Big data and rich information

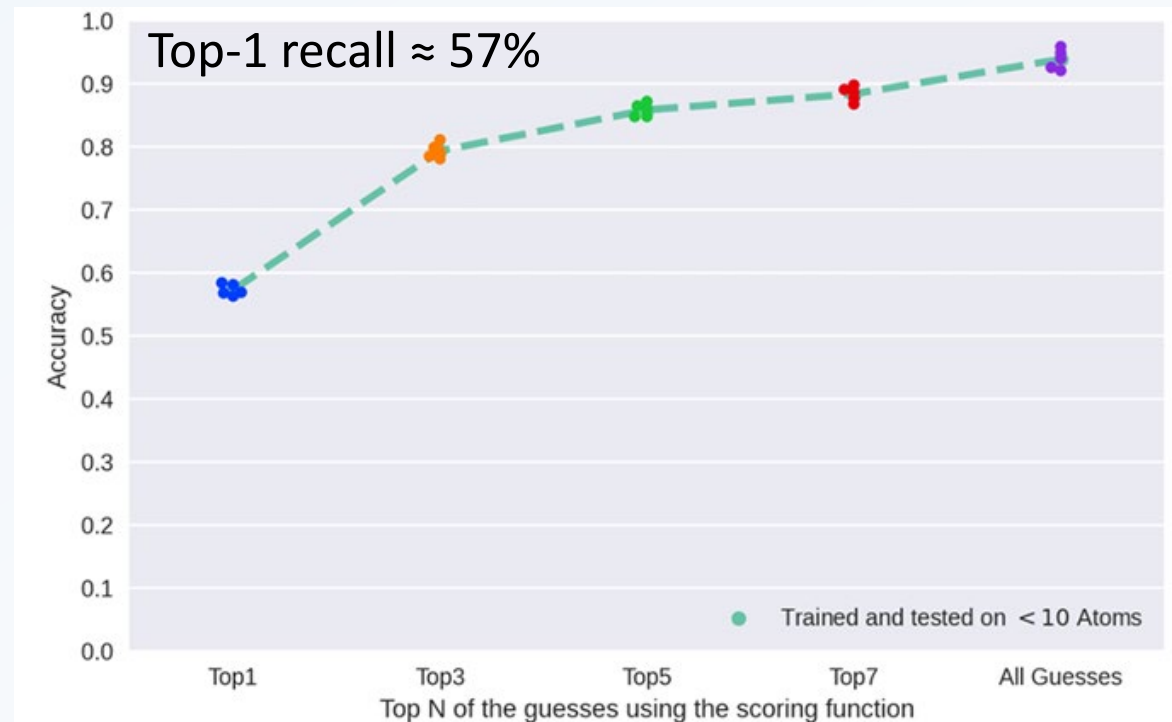


X. Wang et al. Anal. Chem. 2023, 95, 5393

Direct generation



J. Phys. Chem. Lett. 2022, 13, 22, 4924



Test set: nmrshiftdb2 (**<10 heavy atoms, only C, N and O**)

- Graph generation may overcome the limitation of SMILES and predict stereochemical structure.
- However, this method still needs further development.
(computational cost, dataset)

	MLP	CNN	RNN	Transformer	GNN
Encode spectra	√	√	√	√	×
Encode molecules	√ fingerprints	×	√ SMILES	√ SMILES	√ Graph
Generate molecules	√ fingerprints	×	√ SMILES	√ SMILES	×
characteristic	General change dim	Capture local feature	Describe & generate sequence	Describe & generate sequence	Describe molecular graph

2025-2026年，发表在Nature Machine Intelligence、Nature Communications、Nature Methods和AC等期刊，共计84篇。

文献详情分析									
A		B	C	D	E	F	G	H	I
1	Authors	Article Title	Source Title	Abstract	Publication Year	DOI Link	研究方向	解决问题	AI解决方法
2	Wu, CC; Tsantilas, KA; Park, J;	Enrichment of extracellular vesicles using Mag-Net for the	NATURE COMMUNICATIONS	Extracellular vesicles (EVs) in plasma are composed of exosomes.	2025	http://dx.doi.org/10.1038/s41467-025-60595-7	蛋白质组学	数据处理与特征提取	生成式模型 (GAN/Diffusion)
3	Bushuiev, R; Bushuiev, A; Samusevich, S;	Self-supervised learning of molecular representations from	NATURE BIOTECHNOLOGY	Characterizing biological and environmental samples at a molecular level	2025	http://dx.doi.org/10.1038/s41587-025-02663-3	代谢组学 (鉴定与注释)	复杂样本中分子的精准鉴定与注释	Transformer/注意力机制
4	Zhang, JY; Che, YL; Liu, RR; Wang, ZC;	Deep learning-driven multi-omics analysis: enhancing cancer	BRIEFINGS IN BIOINFORMATICS	Artificial intelligence (AI) excels at efficiently processing large volumes	2025	http://dx.doi.org/10.1093/bib/bbaf440	蛋白质组学	缺失数据预测或保留时间/谱图预测	深度学习
5	Movassaghi, CS; Sun, J; Jiang, YM;	Recent Advances in Mass Spectrometry-Based Bottom-Up Proteomics	ANALYTICAL CHEMISTRY	Mass spectrometry-based proteomics is about 35 years old, and recent	2025	http://dx.doi.org/10.1021/acs.analchem.4c06750	蛋白质组学	数据处理与特征提取	机器学习
6	Rosenberger, FA; Maedler, SC;	Deep Visual Proteomics maps proteotoxicity in a genetic liver disease	NATURE	Protein misfolding diseases, including alpha 1-antitrypsin deficiency	2025	http://dx.doi.org/10.1038/s41586-025-08885-4	蛋白质组学	高维空间分布数据的解析与可视化	机器学习
7	Eloff, K; Kalogeropoulos, K; Mabona, S; K; Mabona, X;	InstaNovo enables diffusion-powered de novo peptide sequencing	NATURE MACHINE INTELLIGENCE	Mass spectrometry-based proteomics focuses on identifying the peptide	2025	http://dx.doi.org/10.1038/s42256-025-01019-5	蛋白质组学 (从头测序)	提高肽段从头测序的准确率	Transformer/注意力机制
8	Zhang, X; Ling, TZ; Jin, Z; Xu,	π -PrimeNovo: an accurate and efficient non-autoregressive deep	NATURE COMMUNICATIONS	Peptide sequencing via tandem mass spectrometry (MS/MS) is essential in	2025	http://dx.doi.org/10.1038/s41467-024-55021-3	蛋白质组学 (从头测序)	提高肽段从头测序的准确率	Transformer/注意力机制
9	Guo, L; Xie, PS; Shen, XH; Lam, TKY;	Unraveling Spatial Heterogeneity in Mass Spectrometry Imaging	ADVANCED SCIENCE	Mass spectrometry imaging (MSI) provides valuable insights into metabolic	2025	http://dx.doi.org/10.1002/advs.202410840	代谢组学 (鉴定与注释)	复杂样本中分子的精准鉴定与注释	生成式模型 (GAN/Diffusion)
10	Samarah, LZ; Zheng, C; Xing, X; Lee,	Spatial metabolic gradients in the liver and small intestine	NATURE	The properties of mammalian cells depend on their location within	2025	http://dx.doi.org/10.1038/s41586-025-09616-5	代谢组学 (鉴定与注释)	复杂样本中分子的精准鉴定与注释	生成式模型 (GAN/Diffusion)
11	Brungs, C; Schmid, R; Heuckeroth,	MSnLib: efficient generation of open multi-stage	NATURE METHODS	Untargeted high-resolution mass spectrometry is a key tool in clinical	2025	http://dx.doi.org/10.1038/s41592-025-02813-0	代谢组学 (鉴定与注释)	复杂样本中分子的精准鉴定与注释	机器学习
12	Lu, Z; Qin, GM; Zheng, LL; Zhang,	The role of phytoplankton in structuring global	NATURE COMMUNICATIONS	Phytoplankton-derived dissolved organic carbon (DOC) is a major pathway	2025	http://dx.doi.org/10.1038/s41467-025-63105-x	其他	数据处理与特征提取	生成式模型 (GAN/Diffusion)
13	Koutarapu, S; Ge, JY; Dulewicz, M; Zhang, X;	Chemical imaging delineates A β plaque polymorphism across the Microfluidic Biochip-	NATURE COMMUNICATIONS	Amyloid-beta (A beta) plaque formation in Alzheimer's disease (AD)	2025	http://dx.doi.org/10.1038/s41467-025-59085-7	空间组学/质谱成像 (鉴定与注释)	复杂样本中分子的精准鉴定与注释	深度学习
			ADVANCED	Accurate early diagnosis	2025	http://dx.doi.org/10	蛋白质组学 (鉴定与注释)	复杂样本中分子的精准鉴定与注释	机器学习
文献详情分析 统计汇总报告									

数据量巨大：

现代质谱仪每秒产生数张谱图，一次实验产生的数据量可达 GB 甚至 TB 级别。

信噪比低：

复杂生物样本（如你研究的血液）中，背景噪声极多，传统算法难以提取微弱信号。

未知物质：

数据库里已知的分子不到 1%，剩下的 99% 需要 AI 通过从头测序 (De novo) 或生成式模型来“盲猜”结构。

	A	B	C	D	E	F	G	H
1	出版年	出版年_数量	出版物标题	物标题	研究方向	研究方向_数量	AI解决方法	AI解决方法_数量
2	2026	16	ANALYTICAL CHI	26	代谢组学（鉴定与注释）	18	机器学习	28
3	2025	68	NATURE COMMUN	25	蛋白质组学（鉴定与注释）	14	深度学习	23
4			ADVANCED SCIE	11	其他（鉴定与注释）	9	生成式模型（GAN/Diffus	15
5			BIOINFORMATICS	7	蛋白质组学	8	Transformer/注意力机制	8
6			BRIEFINGS IN I	7	代谢组学	5	传统机器学习	6
7			NATURE METHOD	3	蛋白质组学（从头测序）	4	卷积神经网络（CNN）	2
8			NATURE	2	其他	4	大语言模型（LLM）	1
9			NATURE BIOTECH	2	肽段分析（鉴定与注释）	4	图神经网络（GNN）	1
10			NATURE MACHIN	1	空间组学/质谱成像	3		
11					蛋白质组学（定量分析）	3		
12					脂质组学（鉴定与注释）	2		
13					药物研发（鉴定与注释）	2		
14					其他（定量分析）	2		
15					糖组学（鉴定与注释）	1		
16					代谢组学（定量分析）	1		
17					空间组学/质谱成像（定量分析）	1		
18					脂质组学（定量分析）	1		
19					空间组学/质谱成像（鉴定与注释）	1		
20					糖组学	1		
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文献详情分析统计汇总报告

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⋮

蛋白质组学：de novo
肽段测序、谱库预测等
问题。

代谢组学：分子注释、
代谢物鉴定。

空间组学与质谱成像：
图像重建、多模态对齐、

肽段测序

nature machine intelligence



Article

<https://doi.org/10.1038/s42256-025-01019-5>

InstaNovo enables diffusion-powered de novo peptide sequencing in large-scale proteomics experiments

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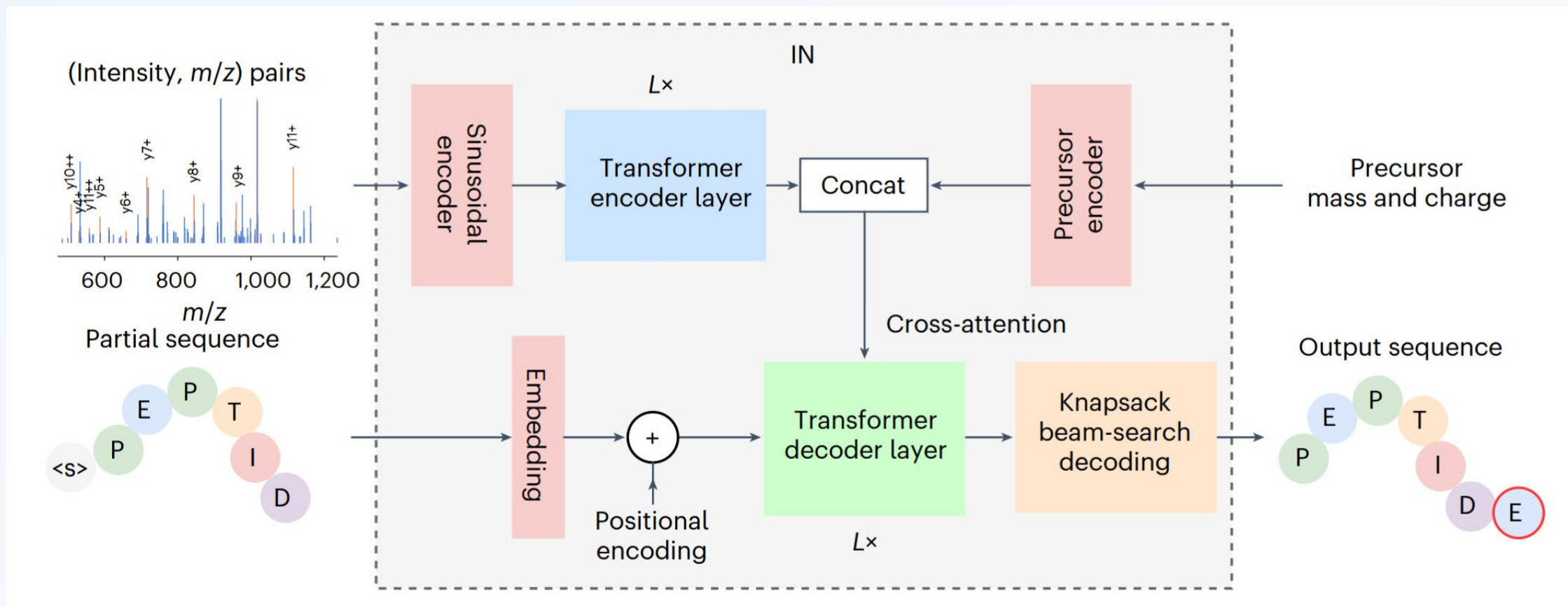
Published online: 31 March 2025

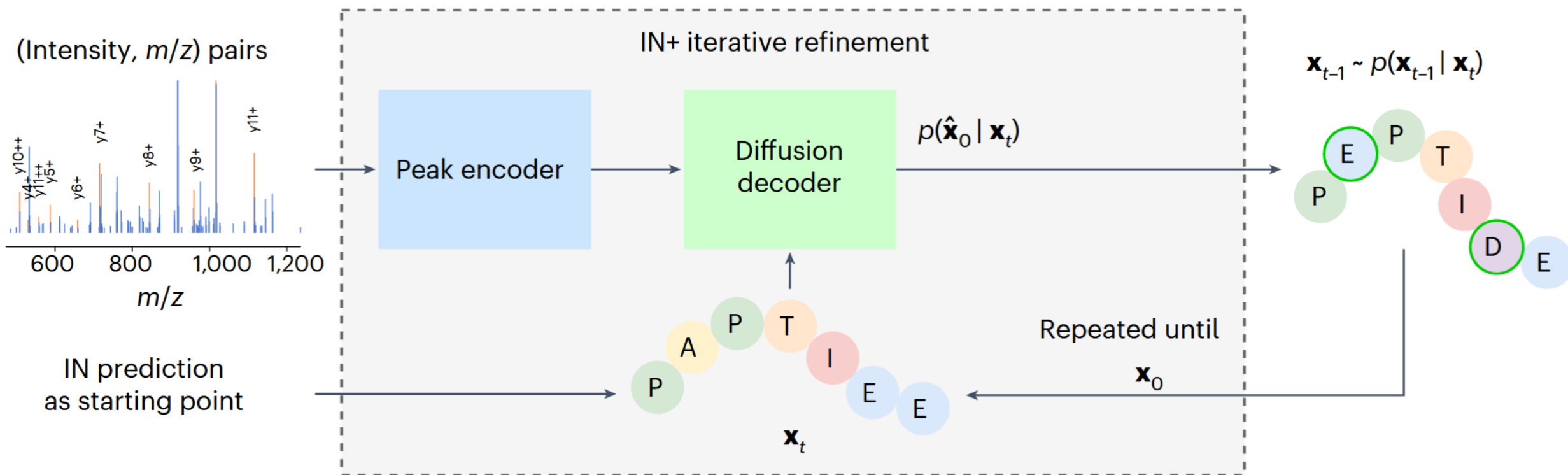
Check for updates

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研究背景：质谱蛋白质组学中，肽段识别主要依赖于将串联质谱与蛋白质数据库进行比对，但无法识别数据库中不存在的蛋白质序列，比如未知突变、新物种或非标准肽段。

局限：因此需要从肽段测序：不依赖先验知识，直接从质谱数据推断肽段序列。但现有从头测序算法准确性不足、计算成本高、假阳性率高，限制了其大规模应用。





代谢组学分子注释

nature biotechnology



Article

<https://doi.org/10.1038/s41587-025-02663-3>

Self-supervised learning of molecular representations from millions of tandem mass spectra using DreaMS

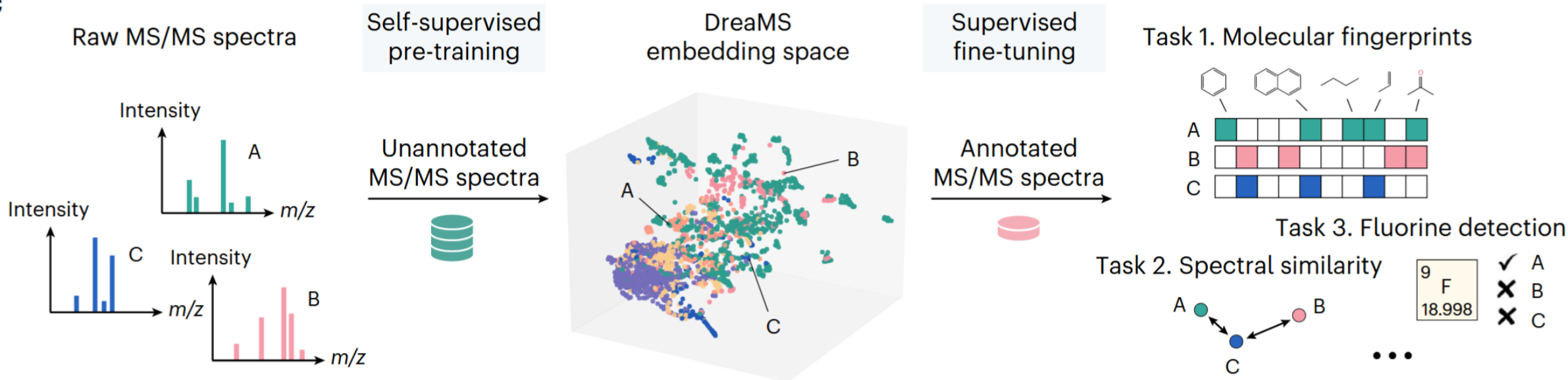
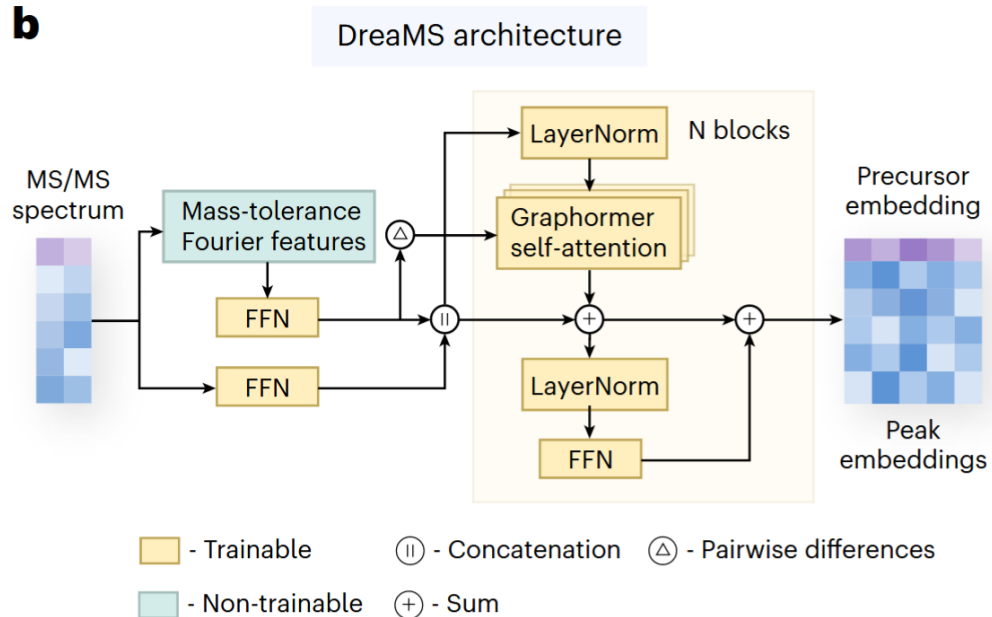
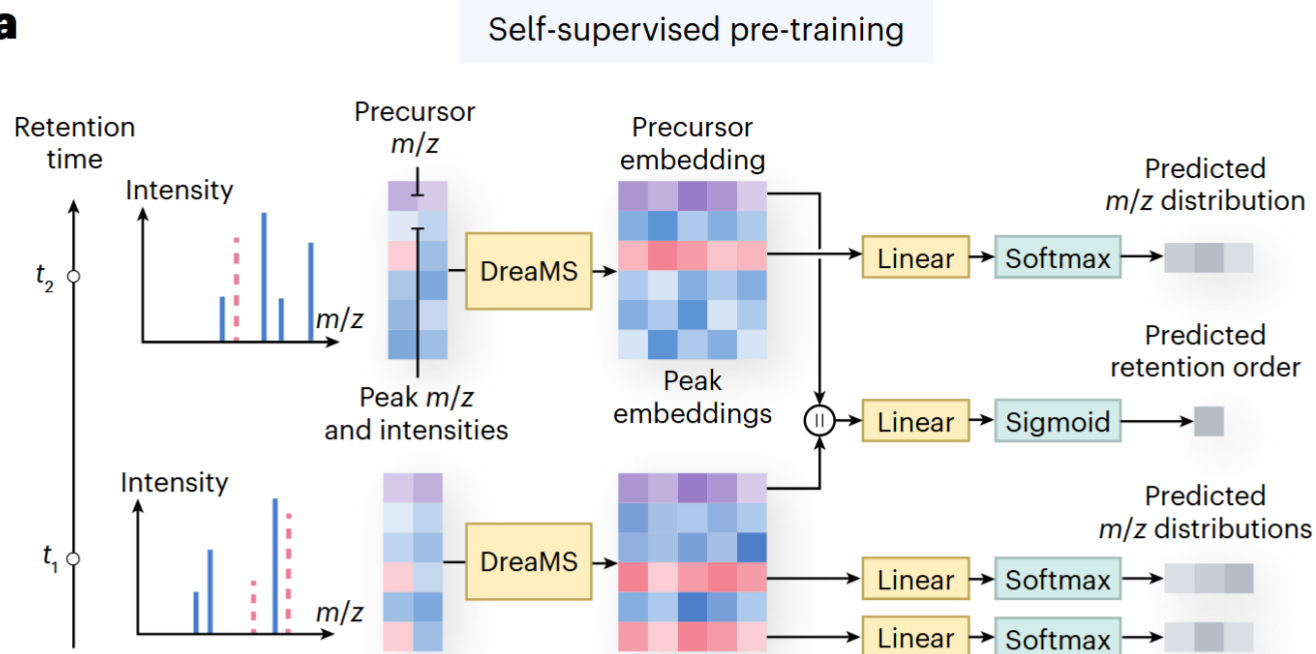
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研究背景：非靶向代谢组学中，绝大多数代谢物没有标准谱库可查。

局限：现有的AI模型依赖有标签的谱图，但这类数据极度稀缺，仅占总量的1%。

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Thinking: How to achieve Multimodal?

Homework 1

- **NIST database, scrap and train a model.**